Supplementary Material

Imputation of Missing Traffic Sensor Data: A Comparative Study of Classical, Graph-Based, and Deep Spatio-Temporal Methods

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This supplement contains the complete tabulated results accompanying the revised manuscript, using the completed road graphs and the five prescribed seeds. It reports test results from January 2023 and October–December 2022 in Darmstadt. Graph Laplacian, Graph Temporal and GRIN use graphs with 254 and 252 edges, respectively, with no isolated sensors after training-based qualification and connection through excluded sensors.

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- Tables S1–S24: complete primary results for all 13 masking scenarios, all 12 methods and MAE, RMSE and MRE.
- Tables S25–S96: corresponding results for sensors A023, A027 and A142.
- Tables A1–A31: persistent stream loss, graph and scaling controls, traffic-count and graph-degree strata, daily aggregation and the separate forecasting check.

Machine-readable summaries and individual-run results are provided in the accompanying `data/` directory. The `README.md` file maps the CSV files to the tables and describes how to compile this document. Training records and rounding comparisons are included as CSV files.

Every reported interval uses complete seed runs as the replication unit. The five-run intervals are conditional on fixed chronological splits, not estimates of between-city uncertainty. Natural-missingness realism has not been quantitatively validated. The two periods come from one city.

This document is self-contained and does not require the main manuscript's LaTeX class or bibliography.

Supplementary results: complete primary and selected-sensor comparisons

Tables S1–S24 report all 936 dataset–scenario–method–metric combinations in the main benchmark. Tables S25–S96 report all 2808 corresponding combinations for sensors A023, A027 and A142. Every estimate uses five seed runs on the same chronological split. Cells give mean ± sample SD and an unadjusted run-level 95% Student-t confidence interval. Negative lower bounds are retained when produced by this interval calculation; they do not imply negative error values. MAE and RMSE have units of vehicles per ten-minute interval; MRE is a dimensionless ratio evaluated on nonzero targets. Blocks use independent starts by sensor unless labelled synchronous. Bold identifies the lowest unrounded mean in a column, not a significant difference. The two KNN variants have the different scaling and k selection policies specified in Methods.

Table S1: MAE for January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 1$	Block $B = 2$	Block $B = 4$	Block $B = 8$
Linear	**23.69** ± 3.80 [18.97, 28.41]	23.65 ± 5.01 [17.43, 29.87]	26.68 ± 1.89 [24.34, 29.02]	28.62 ± 5.57 [21.69, 35.54]
Median	116.05 ± 6.82 [107.58, 124.52]	111.70 ± 15.42 [92.55, 130.85]	109.70 ± 10.91 [96.15, 123.24]	107.90 ± 13.95 [90.58, 125.22]
FFill + BFill	31.74 ± 5.34 [25.12, 38.37]	29.11 ± 7.49 [19.81, 38.41]	34.51 ± 2.36 [31.58, 37.44]	41.97 ± 8.25 [31.73, 52.21]
Rolling mean	31.37 ± 6.07 [23.82, 38.91]	29.46 ± 3.12 [25.58, 33.33]	34.20 ± 1.86 [31.89, 36.50]	33.21 ± 5.58 [26.27, 40.14]
KNN (scaled)	126.02 ± 7.26 [117.00, 135.03]	115.63 ± 18.70 [92.41, 138.84]	113.30 ± 11.34 [99.22, 127.38]	109.32 ± 13.93 [92.02, 126.62]
KNN (unscaled)	48.64 ± 10.50 [35.61, 61.68]	36.93 ± 6.76 [28.54, 45.32]	37.65 ± 4.03 [32.64, 42.65]	37.88 ± 10.21 [25.21, 50.56]
Historical	66.44 ± 11.50 [52.17, 80.72]	57.65 ± 12.70 [41.88, 73.42]	55.95 ± 5.15 [49.56, 62.35]	54.15 ± 8.55 [43.53, 64.77]
Graph Laplacian	67.48 ± 9.66 [55.49, 79.47]	67.64 ± 9.10 [56.34, 78.94]	62.54 ± 5.24 [56.03, 69.05]	66.61 ± 10.78 [53.23, 80.00]
Graph Temporal	52.98 ± 11.73 [38.41, 67.54]	61.59 ± 5.65 [54.57, 68.60]	60.43 ± 4.51 [54.84, 66.03]	66.70 ± 13.06 [50.48, 82.91]
Sage-Voronoi	33.38 ± 7.14 [24.51, 42.25]	28.06 ± 3.36 [23.89, 32.23]	30.61 ± 3.76 [25.93, 35.28]	36.01 ± 6.81 [27.55, 44.48]
GAIN	46.37 ± 9.23 [34.91, 57.84]	38.36 ± 6.85 [29.86, 46.86]	39.24 ± 5.42 [32.51, 45.98]	37.05 ± 8.69 [26.26, 47.84]
GRIN	25.69 ± 5.38 [19.00, 32.37]	**20.15** ± 5.01 [13.93, 26.37]	**23.43** ± 3.63 [18.92, 27.94]	**24.62** ± 8.01 [14.68, 34.56]

Table S2: MAE for January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 16$	Block $B = 32$	Block $B = 64$
Linear	34.94 ± 7.85 [25.19, 44.69]	49.84 ± 3.07 [46.02, 53.65]	75.75 ± 5.45 [68.97, 82.52]
Median	105.25 ± 6.61 [97.05, 113.46]	113.51 ± 8.80 [102.58, 124.44]	114.91 ± 7.02 [106.20, 123.62]
FFill + BFill	62.45 ± 11.08 [48.69, 76.20]	90.90 ± 7.37 [81.75, 100.05]	129.79 ± 15.96 [109.97, 149.60]
Rolling mean	38.48 ± 8.16 [28.35, 48.61]	50.79 ± 3.27 [46.72, 54.85]	75.82 ± 5.58 [68.89, 82.74]
KNN (scaled)	112.97 ± 7.51 [103.64, 122.30]	116.55 ± 8.55 [105.94, 127.16]	116.04 ± 11.38 [101.90, 130.17]
KNN (unscaled)	39.42 ± 6.23 [31.69, 47.15]	46.60 ± 12.23 [31.41, 61.79]	43.92 ± 6.69 [35.61, 52.23]
Historical	58.55 ± 4.66 [52.76, 64.33]	63.87 ± 11.37 [49.76, 77.98]	62.10 ± 7.54 [52.74, 71.47]
Graph Laplacian	57.27 ± 2.12 [54.63, 59.90]	67.90 ± 9.54 [56.06, 79.74]	69.89 ± 8.47 [59.37, 80.41]
Graph Temporal	61.16 ± 1.87 [58.83, 63.48]	70.88 ± 11.54 [56.55, 85.21]	76.18 ± 10.80 [62.77, 89.59]
Sage-Voronoi	45.26 ± 11.65 [30.80, 59.73]	52.78 ± 9.63 [40.83, 64.73]	58.89 ± 7.01 [50.19, 67.60]
GAIN	39.43 ± 6.95 [30.80, 48.06]	45.51 ± 10.92 [31.95, 59.08]	42.61 ± 6.41 [34.65, 50.57]
GRIN	**29.20** ± 4.49 [23.62, 34.78]	**40.71** ± 11.45 [26.49, 54.93]	**41.20** ± 6.65 [32.95, 49.46]

Table S3: MAE for January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 128$	Block $B = 144$
Linear	125.97 ± 6.87 [117.43, 134.51]	134.40 ± 5.56 [127.49, 141.30]
Median	111.59 ± 5.48 [104.79, 118.39]	114.06 ± 4.63 [108.31, 119.80]
FFill + BFill	142.98 ± 11.03 [129.29, 156.67]	141.63 ± 7.80 [131.95, 151.31]
Rolling mean	125.75 ± 6.98 [117.09, 134.41]	134.52 ± 5.57 [127.60, 141.44]
KNN (scaled)	110.90 ± 5.14 [104.53, 117.28]	113.60 ± 5.43 [106.85, 120.34]
KNN (unscaled)	41.29 ± 3.91 [36.44, 46.14]	42.70 ± 3.37 [38.52, 46.89]
Historical	59.00 ± 5.72 [51.89, 66.11]	59.26 ± 3.51 [54.91, 63.62]
Graph Laplacian	64.59 ± 4.94 [58.46, 70.72]	66.43 ± 5.07 [60.14, 72.72]
Graph Temporal	70.54 ± 5.96 [63.14, 77.93]	72.02 ± 4.78 [66.09, 77.96]
Sage-Voronoi	62.91 ± 10.29 [50.13, 75.69]	60.19 ± 3.87 [55.39, 64.99]
GAIN	38.32 ± 4.66 [32.54, 44.10]	40.35 ± 4.79 [34.40, 46.29]
GRIN	**38.19** ± 4.66 [32.40, 43.98]	**40.27** ± 4.54 [34.63, 45.92]

Table S4: MAE for January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	MCAR 10%	MCAR 30%	MCAR 50%	Synchronous $B = 144$
Linear	22.49 ± 0.38 [22.02, 22.96]	**23.45** ± 0.19 [23.21, 23.69]	**24.68** ± 0.19 [24.45, 24.91]	129.11 ± 28.07 [94.26, 163.96]
Median	111.81 ± 1.10 [110.44, 113.18]	112.11 ± 0.51 [111.48, 112.75]	112.04 ± 0.46 [111.47, 112.62]	114.94 ± 18.87 [91.51, 138.37]
FFill + BFill	27.85 ± 0.65 [27.04, 28.65]	29.78 ± 0.30 [29.41, 30.15]	32.15 ± 0.15 [31.96, 32.34]	128.06 ± 26.80 [94.78, 161.33]
Rolling mean	29.87 ± 0.53 [29.22, 30.52]	30.83 ± 0.29 [30.47, 31.19]	31.79 ± 0.10 [31.67, 31.92]	129.27 ± 27.80 [94.75, 163.79]
KNN (scaled)	111.50 ± 1.25 [109.95, 113.05]	104.12 ± 1.46 [102.30, 105.93]	94.51 ± 1.29 [92.91, 96.11]	117.73 ± 17.44 [96.08, 139.38]
KNN (unscaled)	41.89 ± 1.19 [40.41, 43.36]	43.13 ± 0.10 [43.00, 43.26]	44.49 ± 0.18 [44.27, 44.71]	117.73 ± 17.44 [96.08, 139.38]
Historical	58.60 ± 1.29 [57.00, 60.20]	59.35 ± 0.27 [59.02, 59.68]	59.69 ± 0.45 [59.13, 60.25]	**61.54** ± 5.85 [54.28, 68.80]
Graph Laplacian	65.70 ± 1.90 [63.34, 68.05]	66.42 ± 0.46 [65.84, 66.99]	67.64 ± 0.24 [67.34, 67.95]	117.73 ± 17.44 [96.08, 139.38]
Graph Temporal	51.95 ± 1.56 [50.01, 53.89]	53.79 ± 0.45 [53.23, 54.34]	56.96 ± 0.30 [56.59, 57.33]	161.98 ± 53.62 [95.40, 228.56]
Sage-Voronoi	29.08 ± 0.89 [27.98, 30.18]	31.08 ± 0.60 [30.33, 31.82]	34.28 ± 0.91 [33.15, 35.40]	126.29 ± 29.88 [89.19, 163.38]
GAIN	39.46 ± 0.66 [38.64, 40.27]	42.68 ± 1.05 [41.37, 43.99]	56.56 ± 1.85 [54.27, 58.86]	152.17 ± 10.18 [139.53, 164.81]
GRIN	**21.89** ± 0.87 [20.81, 22.97]	23.80 ± 0.44 [23.25, 24.36]	26.56 ± 0.34 [26.14, 26.97]	118.37 ± 21.18 [92.07, 144.67]

Table S5: RMSE for January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 1$	Block $B = 2$	Block $B = 4$	Block $B = 8$
Linear	**70.47** ± 20.29 [45.28, 95.66]	**73.00** ± 32.00 [33.27, 112.73]	**90.18** ± 15.90 [70.43, 109.93]	102.16 ± 43.66 [47.94, 156.37]
Median	237.40 ± 63.05 [159.11, 315.68]	197.12 ± 44.82 [141.46, 252.77]	194.44 ± 30.41 [156.68, 232.20]	187.63 ± 63.93 [108.25, 267.01]
FFill + BFill	107.72 ± 45.95 [50.67, 164.78]	91.46 ± 52.07 [26.81, 156.11]	111.16 ± 33.06 [70.10, 152.21]	110.70 ± 36.55 [65.31, 156.08]
Rolling mean	97.51 ± 43.90 [43.00, 152.01]	91.19 ± 32.58 [50.73, 131.65]	108.42 ± 30.19 [70.94, 145.91]	102.65 ± 33.45 [61.12, 144.18]
KNN (scaled)	264.65 ± 58.09 [192.51, 336.78]	224.98 ± 53.22 [158.90, 291.06]	217.79 ± 33.66 [175.99, 259.58]	207.70 ± 63.05 [129.41, 285.99]
KNN (unscaled)	195.02 ± 83.22 [91.69, 298.35]	128.02 ± 45.15 [71.96, 184.08]	131.66 ± 35.71 [87.32, 176.00]	125.07 ± 74.00 [33.18, 216.95]
Historical	203.84 ± 84.28 [99.19, 308.50]	152.95 ± 54.08 [85.80, 220.10]	154.36 ± 42.95 [101.02, 207.69]	130.99 ± 68.66 [45.74, 216.25]
Graph Laplacian	213.11 ± 79.85 [113.97, 312.25]	235.15 ± 66.73 [152.30, 318.01]	187.75 ± 42.60 [134.85, 240.65]	233.29 ± 72.04 [143.85, 322.73]
Graph Temporal	163.17 ± 66.48 [80.62, 245.72]	193.68 ± 25.20 [162.40, 224.96]	187.32 ± 21.22 [160.97, 213.66]	228.60 ± 52.96 [162.84, 294.35]
Sage-Voronoi	129.20 ± 72.98 [38.58, 219.82]	89.69 ± 23.27 [60.80, 118.58]	94.52 ± 29.23 [58.23, 130.81]	120.80 ± 48.17 [60.99, 180.61]
GAIN	185.71 ± 90.17 [73.76, 297.67]	124.79 ± 46.37 [67.21, 182.36]	134.96 ± 42.75 [81.88, 188.05]	119.36 ± 70.36 [31.99, 206.72]
GRIN	106.52 ± 62.14 [29.36, 183.68]	76.15 ± 47.05 [17.72, 134.57]	96.26 ± 31.96 [56.57, 135.94]	**85.09** ± 72.27 [−4.65, 174.83]

Table S6: RMSE for January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 16$	Block $B = 32$	Block $B = 64$
Linear	117.45 ± 56.62 [47.15, 187.76]	**140.47** ± 17.59 [118.62, 162.31]	**160.97** ± 25.02 [129.91, 192.04]
Median	186.19 ± 27.29 [152.31, 220.07]	232.15 ± 52.86 [166.51, 297.78]	227.39 ± 42.78 [174.27, 280.51]
FFill + BFill	185.62 ± 91.30 [72.26, 298.98]	220.66 ± 30.07 [183.32, 257.99]	253.82 ± 55.99 [184.29, 323.35]
Rolling mean	128.77 ± 65.04 [48.02, 209.53]	143.87 ± 16.75 [123.07, 164.67]	162.22 ± 24.63 [131.63, 192.81]
KNN (scaled)	212.30 ± 26.08 [179.92, 244.68]	250.05 ± 48.76 [189.50, 310.60]	244.24 ± 47.35 [185.45, 303.03]
KNN (unscaled)	132.30 ± 46.42 [74.65, 189.94]	187.95 ± 75.65 [94.02, 281.88]	175.93 ± 53.25 [109.82, 242.05]
Historical	161.28 ± 38.73 [113.19, 209.37]	204.91 ± 70.19 [117.76, 292.06]	187.97 ± 54.68 [120.08, 255.87]
Graph Laplacian	156.35 ± 26.22 [123.79, 188.91]	233.91 ± 46.78 [175.83, 291.99]	230.50 ± 51.56 [166.48, 294.52]
Graph Temporal	182.17 ± 20.91 [156.20, 208.14]	242.25 ± 53.27 [176.11, 308.40]	253.29 ± 51.05 [189.90, 316.68]
Sage-Voronoi	183.51 ± 116.16 [39.27, 327.75]	191.09 ± 37.95 [143.97, 238.21]	210.21 ± 44.80 [154.58, 265.84]
GAIN	137.08 ± 42.82 [83.92, 190.24]	191.34 ± 71.70 [102.31, 280.36]	173.80 ± 55.91 [104.38, 243.21]
GRIN	**105.59** ± 38.09 [58.29, 152.88]	176.13 ± 73.30 [85.11, 267.15]	170.73 ± 54.86 [102.60, 238.85]

Table S7: RMSE for January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 128$	Block $B = 144$
Linear	236.86 ± 25.90 [204.70, 269.01]	266.46 ± 21.85 [239.33, 293.59]
Median	209.61 ± 36.21 [164.65, 254.56]	218.97 ± 25.18 [187.71, 250.24]
FFill + BFill	265.15 ± 36.01 [220.44, 309.86]	297.06 ± 38.22 [249.60, 344.52]
Rolling mean	236.51 ± 25.89 [204.35, 268.66]	266.98 ± 22.11 [239.53, 294.43]
KNN (scaled)	227.42 ± 32.08 [187.59, 267.25]	237.08 ± 24.33 [206.87, 267.29]
KNN (unscaled)	161.62 ± 40.44 [111.41, 211.83]	163.75 ± 31.19 [125.02, 202.48]
Historical	177.85 ± 40.97 [126.98, 228.72]	180.08 ± 28.24 [145.01, 215.15]
Graph Laplacian	212.51 ± 39.39 [163.60, 261.41]	217.62 ± 34.26 [175.09, 260.16]
Graph Temporal	232.74 ± 34.71 [189.65, 275.83]	240.03 ± 26.62 [206.97, 273.08]
Sage-Voronoi	249.63 ± 122.27 [97.81, 401.45]	199.49 ± 31.27 [160.66, 238.33]
GAIN	154.00 ± 46.06 [96.81, 211.18]	162.37 ± 34.50 [119.54, 205.20]
GRIN	**153.85** ± 46.64 [95.93, 211.77]	**160.53** ± 34.04 [118.27, 202.80]

Table S8: RMSE for January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	MCAR 10%	MCAR 30%	MCAR 50%	Synchronous $B = 144$
Linear	**73.08** ± 2.79 [69.62, 76.55]	**77.99** ± 1.36 [76.30, 79.67]	**85.18** ± 1.47 [83.36, 87.01]	260.20 ± 69.44 [173.98, 346.42]
Median	214.16 ± 6.83 [205.68, 222.65]	217.15 ± 1.89 [214.80, 219.50]	217.70 ± 2.21 [214.95, 220.45]	227.38 ± 50.65 [164.50, 290.27]
FFill + BFill	93.34 ± 4.77 [87.41, 99.27]	101.05 ± 3.34 [96.89, 105.20]	109.34 ± 2.06 [106.78, 111.90]	257.26 ± 63.29 [178.67, 335.85]
Rolling mean	97.62 ± 5.34 [90.99, 104.24]	102.53 ± 3.71 [97.93, 107.13]	106.85 ± 1.51 [104.98, 108.73]	260.17 ± 69.00 [174.50, 345.85]
KNN (scaled)	231.20 ± 6.22 [223.48, 238.93]	227.19 ± 1.50 [225.33, 229.06]	220.11 ± 1.07 [218.78, 221.44]	231.67 ± 47.12 [173.17, 290.18]
KNN (unscaled)	164.99 ± 9.55 [153.13, 176.86]	170.92 ± 2.42 [167.92, 173.92]	173.74 ± 2.88 [170.17, 177.32]	231.67 ± 47.12 [173.17, 290.18]
Historical	178.95 ± 8.36 [168.57, 189.33]	184.79 ± 1.76 [182.60, 186.97]	186.38 ± 2.39 [183.41, 189.34]	**196.38** ± 45.90 [139.38, 253.37]
Graph Laplacian	219.24 ± 12.47 [203.76, 234.72]	222.58 ± 1.63 [220.56, 224.60]	223.94 ± 1.58 [221.98, 225.90]	231.67 ± 47.12 [173.17, 290.18]
Graph Temporal	164.59 ± 6.30 [156.77, 172.41]	173.76 ± 1.50 [171.90, 175.62]	185.57 ± 2.11 [182.95, 188.19]	318.46 ± 96.51 [198.62, 438.29]
Sage-Voronoi	106.03 ± 5.00 [99.83, 112.23]	115.38 ± 6.97 [106.72, 124.03]	126.29 ± 6.11 [118.71, 133.88]	267.60 ± 61.56 [191.16, 344.03]
GAIN	161.64 ± 9.57 [149.76, 173.52]	163.12 ± 3.90 [158.28, 167.96]	169.15 ± 3.01 [165.41, 172.89]	287.15 ± 29.25 [250.83, 323.47]
GRIN	94.67 ± 8.69 [83.87, 105.46]	106.46 ± 7.05 [97.71, 115.21]	118.22 ± 6.31 [110.38, 126.06]	241.27 ± 50.79 [178.21, 304.34]

Table S9: MRE for January 2023 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 1$	Block $B = 2$	Block $B = 4$	Block $B = 8$
Linear	0.232 ± 0.035 [0.188, 0.276]	0.292 ± 0.058 [0.220, 0.364]	0.302 ± 0.033 [0.261, 0.342]	0.332 ± 0.040 [0.282, 0.383]
Median	1.781 ± 0.317 [1.387, 2.175]	2.332 ± 0.465 [1.754, 2.910]	2.576 ± 0.364 [2.124, 3.028]	2.237 ± 0.424 [1.710, 2.764]
FFill + BFill	0.271 ± 0.063 [0.192, 0.350]	0.344 ± 0.050 [0.282, 0.405]	0.348 ± 0.049 [0.287, 0.409]	0.449 ± 0.058 [0.377, 0.521]
Rolling mean	0.287 ± 0.041 [0.236, 0.337]	0.351 ± 0.086 [0.245, 0.458]	0.351 ± 0.048 [0.292, 0.410]	0.360 ± 0.051 [0.297, 0.424]
KNN (scaled)	0.847 ± 0.123 [0.693, 1.000]	1.044 ± 0.159 [0.847, 1.242]	1.166 ± 0.146 [0.984, 1.348]	0.982 ± 0.173 [0.767, 1.197]
KNN (unscaled)	0.405 ± 0.093 [0.290, 0.520]	0.423 ± 0.048 [0.363, 0.483]	0.450 ± 0.069 [0.364, 0.536]	0.413 ± 0.069 [0.327, 0.499]
Historical	0.514 ± 0.089 [0.403, 0.625]	0.541 ± 0.097 [0.420, 0.661]	0.512 ± 0.080 [0.412, 0.611]	0.540 ± 0.048 [0.480, 0.600]
Graph Laplacian	0.727 ± 0.089 [0.616, 0.838]	0.838 ± 0.153 [0.648, 1.028]	1.078 ± 0.147 [0.895, 1.260]	0.949 ± 0.292 [0.588, 1.311]
Graph Temporal	0.675 ± 0.207 [0.418, 0.933]	0.805 ± 0.298 [0.435, 1.176]	1.362 ± 0.350 [0.927, 1.797]	1.087 ± 0.268 [0.754, 1.420]
Sage-Voronoi	0.255 ± 0.038 [0.208, 0.303]	0.311 ± 0.053 [0.245, 0.376]	0.364 ± 0.067 [0.281, 0.447]	0.377 ± 0.067 [0.293, 0.461]
GAIN	0.311 ± 0.073 [0.220, 0.402]	0.305 ± 0.031 [0.267, 0.343]	0.320 ± 0.032 [0.280, 0.359]	0.317 ± 0.030 [0.281, 0.354]
GRIN	**0.200** ± 0.013 [0.184, 0.217]	**0.214** ± 0.026 [0.181, 0.246]	**0.261** ± 0.052 [0.196, 0.326]	**0.292** ± 0.104 [0.163, 0.420]

Table S10: MRE for January 2023 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 16$	Block $B = 32$	Block $B = 64$
Linear	0.376 ± 0.060 [0.302, 0.450]	0.886 ± 0.094 [0.770, 1.003]	1.628 ± 0.127 [1.471, 1.786]
Median	2.015 ± 0.438 [1.472, 2.559]	2.383 ± 0.339 [1.962, 2.804]	2.272 ± 0.231 [1.985, 2.559]
FFill + BFill	0.619 ± 0.065 [0.538, 0.699]	1.136 ± 0.173 [0.920, 1.351]	2.281 ± 0.351 [1.845, 2.716]
Rolling mean	0.380 ± 0.049 [0.319, 0.440]	0.809 ± 0.095 [0.691, 0.928]	1.578 ± 0.124 [1.424, 1.733]
KNN (scaled)	0.940 ± 0.153 [0.749, 1.131]	1.034 ± 0.063 [0.955, 1.112]	1.034 ± 0.087 [0.926, 1.141]
KNN (unscaled)	0.420 ± 0.096 [0.300, 0.539]	0.445 ± 0.039 [0.396, 0.494]	0.430 ± 0.033 [0.389, 0.471]
Historical	0.506 ± 0.078 [0.410, 0.603]	0.547 ± 0.018 [0.525, 0.569]	0.568 ± 0.079 [0.470, 0.665]
Graph Laplacian	0.868 ± 0.316 [0.476, 1.261]	0.953 ± 0.157 [0.758, 1.148]	0.922 ± 0.159 [0.725, 1.119]
Graph Temporal	0.879 ± 0.495 [0.265, 1.493]	1.342 ± 0.638 [0.550, 2.134]	1.008 ± 0.423 [0.482, 1.533]
Sage-Voronoi	0.441 ± 0.100 [0.316, 0.565]	0.576 ± 0.095 [0.458, 0.694]	0.650 ± 0.106 [0.517, 0.782]
GAIN	0.305 ± 0.057 [0.234, 0.376]	**0.314** ± 0.036 [0.269, 0.359]	**0.311** ± 0.026 [0.279, 0.343]
GRIN	**0.290** ± 0.071 [0.202, 0.377]	0.463 ± 0.239 [0.166, 0.760]	0.416 ± 0.168 [0.206, 0.625]

Table S11: MRE for January 2023 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 128$	Block $B = 144$
Linear	2.858 ± 0.141 [2.683, 3.034]	2.936 ± 0.296 [2.568, 3.303]
Median	2.292 ± 0.093 [2.176, 2.408]	2.269 ± 0.066 [2.188, 2.351]
FFill + BFill	3.035 ± 0.088 [2.926, 3.144]	2.918 ± 0.233 [2.629, 3.206]
Rolling mean	2.847 ± 0.139 [2.674, 3.020]	2.935 ± 0.295 [2.568, 3.301]
KNN (scaled)	1.031 ± 0.053 [0.965, 1.098]	1.051 ± 0.068 [0.966, 1.136]
KNN (unscaled)	0.429 ± 0.024 [0.400, 0.459]	0.457 ± 0.035 [0.414, 0.501]
Historical	0.545 ± 0.036 [0.501, 0.590]	0.552 ± 0.018 [0.530, 0.574]
Graph Laplacian	0.901 ± 0.030 [0.864, 0.938]	0.900 ± 0.061 [0.824, 0.976]
Graph Temporal	1.222 ± 0.194 [0.981, 1.464]	1.050 ± 0.158 [0.853, 1.247]
Sage-Voronoi	0.787 ± 0.154 [0.596, 0.978]	0.796 ± 0.132 [0.632, 0.959]
GAIN	**0.325** ± 0.030 [0.287, 0.362]	**0.357** ± 0.038 [0.309, 0.404]
GRIN	0.397 ± 0.032 [0.357, 0.436]	0.440 ± 0.025 [0.410, 0.471]

Table S12: MRE for January 2023 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	MCAR 10%	MCAR 30%	MCAR 50%	Synchronous $B = 144$
Linear	0.258 ± 0.003 [0.254, 0.263]	0.266 ± 0.003 [0.262, 0.269]	0.277 ± 0.004 [0.273, 0.282]	4.287 ± 0.995 [3.051, 5.524]
Median	2.268 ± 0.038 [2.221, 2.315]	2.269 ± 0.020 [2.244, 2.294]	2.270 ± 0.010 [2.258, 2.283]	2.500 ± 0.369 [2.042, 2.958]
FFill + BFill	0.306 ± 0.004 [0.301, 0.311]	0.318 ± 0.004 [0.313, 0.323]	0.338 ± 0.003 [0.333, 0.342]	3.846 ± 1.724 [1.706, 5.987]
Rolling mean	0.305 ± 0.004 [0.300, 0.310]	0.316 ± 0.004 [0.311, 0.321]	0.328 ± 0.005 [0.322, 0.334]	4.284 ± 0.999 [3.044, 5.524]
KNN (scaled)	1.028 ± 0.027 [0.994, 1.062]	1.045 ± 0.024 [1.015, 1.074]	1.008 ± 0.047 [0.950, 1.067]	2.599 ± 0.414 [2.085, 3.113]
KNN (unscaled)	0.433 ± 0.005 [0.427, 0.439]	0.469 ± 0.014 [0.452, 0.486]	0.530 ± 0.016 [0.510, 0.550]	2.599 ± 0.414 [2.085, 3.113]
Historical	0.534 ± 0.007 [0.526, 0.542]	0.539 ± 0.006 [0.532, 0.546]	0.541 ± 0.002 [0.539, 0.544]	**0.590** ± 0.326 [0.186, 0.995]
Graph Laplacian	0.888 ± 0.016 [0.869, 0.907]	0.884 ± 0.017 [0.863, 0.906]	0.885 ± 0.029 [0.849, 0.921]	2.599 ± 0.414 [2.085, 3.113]
Graph Temporal	0.712 ± 0.051 [0.648, 0.776]	0.750 ± 0.031 [0.711, 0.788]	0.786 ± 0.029 [0.750, 0.822]	5.008 ± 1.311 [3.380, 6.635]
Sage-Voronoi	0.292 ± 0.011 [0.278, 0.306]	0.306 ± 0.011 [0.292, 0.320]	0.331 ± 0.014 [0.313, 0.348]	3.198 ± 1.499 [1.336, 5.059]
GAIN	0.333 ± 0.019 [0.309, 0.357]	0.528 ± 0.037 [0.481, 0.574]	1.000 ± 0.061 [0.925, 1.076]	4.976 ± 0.899 [3.860, 6.092]
GRIN	**0.227** ± 0.006 [0.220, 0.234]	**0.242** ± 0.007 [0.234, 0.251]	**0.274** ± 0.012 [0.259, 0.290]	1.902 ± 0.452 [1.341, 2.463]

Table S13: MAE for October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 1$	Block $B = 2$	Block $B = 4$	Block $B = 8$
Linear	19.81 ± 2.44 [16.78, 22.84]	20.40 ± 3.17 [16.47, 24.34]	19.29 ± 1.95 [16.87, 21.71]	24.73 ± 2.04 [22.20, 27.27]
Median	102.68 ± 8.81 [91.74, 113.62]	98.69 ± 6.38 [90.76, 106.62]	99.62 ± 4.84 [93.60, 105.63]	101.06 ± 2.28 [98.23, 103.89]
FFill + BFill	24.12 ± 1.84 [21.84, 26.41]	24.05 ± 3.15 [20.14, 27.97]	25.98 ± 2.06 [23.42, 28.53]	33.80 ± 2.86 [30.25, 37.34]
Rolling mean	25.79 ± 2.25 [23.01, 28.58]	28.69 ± 2.92 [25.07, 32.32]	26.10 ± 2.46 [23.05, 29.16]	27.79 ± 2.29 [24.94, 30.63]
KNN (scaled)	41.94 ± 6.22 [34.22, 49.66]	37.53 ± 3.09 [33.70, 41.37]	35.73 ± 7.47 [26.45, 45.01]	34.38 ± 1.87 [32.05, 36.71]
KNN (unscaled)	44.49 ± 8.13 [34.40, 54.58]	42.32 ± 3.99 [37.37, 47.27]	39.84 ± 6.45 [31.84, 47.85]	38.07 ± 2.23 [35.31, 40.84]
Historical	53.64 ± 8.97 [42.49, 64.78]	56.73 ± 4.86 [50.69, 62.76]	53.69 ± 6.27 [45.91, 61.48]	53.52 ± 3.51 [49.16, 57.88]
Graph Laplacian	47.62 ± 8.24 [37.39, 57.85]	40.26 ± 5.14 [33.88, 46.64]	40.84 ± 4.58 [35.15, 46.53]	41.82 ± 4.33 [36.44, 47.19]
Graph Temporal	36.23 ± 4.40 [30.77, 41.70]	34.03 ± 5.69 [26.96, 41.10]	38.15 ± 3.88 [33.33, 42.97]	40.25 ± 2.94 [36.60, 43.91]
Sage-Voronoi	22.43 ± 3.53 [18.05, 26.82]	22.24 ± 2.59 [19.03, 25.45]	21.08 ± 2.36 [18.14, 24.01]	24.73 ± 0.95 [23.56, 25.91]
GAIN	44.62 ± 8.77 [33.72, 55.51]	39.57 ± 4.64 [33.81, 45.33]	38.02 ± 5.42 [31.29, 44.75]	38.20 ± 1.73 [36.05, 40.35]
GRIN	**18.30** ± 3.21 [14.31, 22.29]	**17.70** ± 2.86 [14.15, 21.26]	**15.90** ± 2.27 [13.09, 18.72]	**18.45** ± 0.99 [17.22, 19.68]

Table S14: MAE for October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 16$	Block $B = 32$	Block $B = 64$
Linear	29.55 ± 2.19 [26.83, 32.28]	38.63 ± 4.86 [32.61, 44.66]	64.95 ± 3.00 [61.23, 68.68]
Median	110.37 ± 6.77 [101.96, 118.78]	102.83 ± 4.48 [97.27, 108.39]	104.40 ± 6.13 [96.79, 112.01]
FFill + BFill	49.46 ± 2.29 [46.62, 52.30]	73.11 ± 6.86 [64.60, 81.62]	108.15 ± 5.41 [101.43, 114.88]
Rolling mean	31.48 ± 1.67 [29.40, 33.55]	39.65 ± 4.86 [33.61, 45.68]	64.99 ± 3.27 [60.93, 69.05]
KNN (scaled)	40.64 ± 4.06 [35.60, 45.68]	34.90 ± 3.22 [30.90, 38.90]	39.09 ± 6.33 [31.24, 46.95]
KNN (unscaled)	43.65 ± 4.47 [38.10, 49.20]	38.40 ± 3.75 [33.74, 43.05]	42.98 ± 6.74 [34.62, 51.35]
Historical	59.68 ± 3.64 [55.17, 64.19]	50.95 ± 4.80 [44.99, 56.92]	55.25 ± 6.06 [47.72, 62.78]
Graph Laplacian	48.49 ± 5.12 [42.13, 54.85]	42.23 ± 4.01 [37.25, 47.20]	46.34 ± 8.24 [36.11, 56.57]
Graph Temporal	48.14 ± 5.03 [41.90, 54.38]	42.41 ± 5.12 [36.06, 48.76]	46.77 ± 6.80 [38.33, 55.21]
Sage-Voronoi	29.14 ± 1.16 [27.70, 30.58]	33.67 ± 3.43 [29.42, 37.93]	42.28 ± 7.28 [33.24, 51.32]
GAIN	46.75 ± 5.10 [40.42, 53.09]	38.15 ± 3.65 [33.62, 42.68]	42.50 ± 5.96 [35.10, 49.90]
GRIN	**23.92** ± 1.32 [22.28, 25.55]	**25.18** ± 2.95 [21.52, 28.84]	**32.65** ± 6.35 [24.77, 40.53]

Table S15: MAE for October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 128$	Block $B = 144$
Linear	114.24 ± 4.65 [108.46, 120.01]	117.12 ± 6.64 [108.88, 125.37]
Median	101.95 ± 2.80 [98.47, 105.43]	103.00 ± 3.04 [99.23, 106.78]
FFill + BFill	129.04 ± 4.19 [123.83, 134.24]	118.93 ± 7.12 [110.08, 127.77]
Rolling mean	114.13 ± 4.70 [108.30, 119.97]	117.13 ± 6.61 [108.93, 125.34]
KNN (scaled)	35.83 ± 0.94 [34.67, 37.00]	35.87 ± 1.81 [33.63, 38.11]
KNN (unscaled)	39.48 ± 1.41 [37.73, 41.23]	39.83 ± 2.04 [37.30, 42.36]
Historical	52.27 ± 2.41 [49.27, 55.26]	52.31 ± 2.66 [49.00, 55.61]
Graph Laplacian	43.96 ± 2.46 [40.91, 47.01]	44.07 ± 2.86 [40.52, 47.63]
Graph Temporal	44.68 ± 1.77 [42.47, 46.88]	44.67 ± 3.18 [40.72, 48.62]
Sage-Voronoi	42.32 ± 2.54 [39.16, 45.47]	42.62 ± 3.06 [38.82, 46.42]
GAIN	39.46 ± 2.29 [36.61, 42.30]	39.37 ± 3.60 [34.90, 43.85]
GRIN	**32.24** ± 1.78 [30.03, 34.45]	**31.50** ± 2.96 [27.83, 35.18]

Table S16: MAE for October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	MCAR 10%	MCAR 30%	MCAR 50%	Synchronous $B = 144$
Linear	20.26 ± 0.22 [19.98, 20.54]	20.88 ± 0.14 [20.71, 21.05]	21.88 ± 0.10 [21.75, 22.00]	107.07 ± 29.19 [70.82, 143.31]
Median	102.31 ± 0.40 [101.81, 102.81]	101.91 ± 0.34 [101.49, 102.34]	101.92 ± 0.13 [101.76, 102.08]	97.61 ± 10.68 [84.35, 110.87]
FFill + BFill	24.71 ± 0.31 [24.32, 25.09]	25.86 ± 0.19 [25.63, 26.09]	27.92 ± 0.17 [27.71, 28.13]	111.84 ± 35.75 [67.46, 156.23]
Rolling mean	25.58 ± 0.28 [25.23, 25.92]	26.34 ± 0.14 [26.16, 26.51]	27.28 ± 0.10 [27.15, 27.41]	107.24 ± 29.28 [70.88, 143.60]
KNN (scaled)	36.81 ± 0.23 [36.52, 37.09]	36.78 ± 0.19 [36.55, 37.02]	36.87 ± 0.10 [36.75, 36.99]	98.48 ± 11.67 [83.99, 112.97]
KNN (unscaled)	40.56 ± 0.35 [40.12, 40.99]	39.99 ± 0.11 [39.85, 40.13]	39.49 ± 0.06 [39.41, 39.56]	98.48 ± 11.67 [83.99, 112.97]
Historical	54.69 ± 0.15 [54.50, 54.87]	54.56 ± 0.23 [54.27, 54.85]	54.47 ± 0.10 [54.34, 54.59]	**58.25** ± 15.81 [38.63, 77.88]
Graph Laplacian	43.81 ± 0.57 [43.11, 44.52]	43.96 ± 0.20 [43.72, 44.21]	44.34 ± 0.20 [44.09, 44.58]	98.48 ± 11.67 [83.99, 112.97]
Graph Temporal	34.05 ± 0.39 [33.56, 34.54]	35.37 ± 0.16 [35.17, 35.57]	37.00 ± 0.05 [36.93, 37.06]	107.96 ± 29.90 [70.83, 145.09]
Sage-Voronoi	22.04 ± 0.14 [21.88, 22.21]	23.06 ± 0.12 [22.91, 23.21]	24.70 ± 0.23 [24.42, 24.98]	121.20 ± 37.88 [74.16, 168.23]
GAIN	37.68 ± 0.47 [37.10, 38.27]	41.58 ± 0.27 [41.24, 41.91]	62.44 ± 0.82 [61.42, 63.46]	205.60 ± 16.50 [185.11, 226.10]
GRIN	**16.92** ± 0.21 [16.65, 17.18]	**17.86** ± 0.12 [17.72, 18.01]	**19.50** ± 0.13 [19.34, 19.66]	86.52 ± 18.84 [63.12, 109.91]

Table S17: RMSE for October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 1$	Block $B = 2$	Block $B = 4$	Block $B = 8$
Linear	**48.70** ± 19.58 [24.39, 73.02]	64.12 ± 25.26 [32.75, 95.48]	43.97 ± 10.67 [30.72, 57.23]	81.48 ± 15.10 [62.73, 100.23]
Median	190.11 ± 33.16 [148.93, 231.28]	171.44 ± 6.46 [163.41, 179.46]	163.38 ± 10.42 [150.44, 176.33]	170.33 ± 13.77 [153.23, 187.42]
FFill + BFill	57.25 ± 8.04 [47.27, 67.23]	65.09 ± 21.81 [38.00, 92.17]	59.25 ± 14.69 [41.00, 77.49]	85.02 ± 12.83 [69.09, 100.95]
Rolling mean	62.85 ± 9.04 [51.62, 74.07]	90.44 ± 18.07 [68.00, 112.88]	58.83 ± 13.81 [41.69, 75.97]	80.48 ± 9.52 [68.66, 92.31]
KNN (scaled)	135.85 ± 30.64 [97.80, 173.90]	119.99 ± 23.49 [90.82, 149.16]	111.06 ± 29.17 [74.84, 147.29]	99.38 ± 5.94 [92.01, 106.76]
KNN (unscaled)	147.28 ± 44.29 [92.29, 202.27]	127.35 ± 24.35 [97.12, 157.58]	119.27 ± 29.31 [82.87, 155.67]	106.21 ± 7.92 [96.38, 116.05]
Historical	123.40 ± 30.73 [85.25, 161.55]	134.09 ± 20.77 [108.30, 159.88]	127.28 ± 28.96 [91.31, 163.24]	115.94 ± 14.26 [98.23, 133.65]
Graph Laplacian	152.68 ± 69.65 [66.20, 239.16]	113.05 ± 33.33 [71.67, 154.44]	105.54 ± 22.11 [78.09, 132.99]	104.37 ± 12.71 [88.58, 120.15]
Graph Temporal	98.75 ± 25.65 [66.90, 130.60]	97.39 ± 32.22 [57.38, 137.40]	94.64 ± 16.66 [73.95, 115.32]	96.85 ± 5.39 [90.16, 103.54]
Sage-Voronoi	61.17 ± 21.19 [34.86, 87.48]	70.71 ± 22.51 [42.77, 98.66]	46.06 ± 11.98 [31.19, 60.94]	64.42 ± 10.05 [51.94, 76.90]
GAIN	149.38 ± 58.47 [76.77, 221.98]	118.97 ± 33.12 [77.84, 160.10]	99.89 ± 23.60 [70.58, 129.19]	103.36 ± 12.19 [88.23, 118.50]
GRIN	57.08 ± 22.33 [29.35, 84.81]	**61.47** ± 26.31 [28.79, 94.14]	**39.72** ± 11.46 [25.49, 53.94]	**51.10** ± 11.92 [36.30, 65.91]

Table S18: RMSE for October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 16$	Block $B = 32$	Block $B = 64$
Linear	88.28 ± 17.81 [66.16, 110.39]	103.15 ± 30.10 [65.78, 140.52]	123.56 ± 5.17 [117.14, 129.99]
Median	214.65 ± 37.38 [168.23, 261.06]	171.87 ± 9.89 [159.59, 184.15]	193.37 ± 48.32 [133.38, 253.37]
FFill + BFill	115.93 ± 17.78 [93.86, 138.00]	148.75 ± 25.30 [117.34, 180.17]	192.64 ± 16.65 [171.97, 213.31]
Rolling mean	90.55 ± 15.04 [71.88, 109.22]	104.19 ± 29.10 [68.07, 140.32]	123.50 ± 5.63 [116.50, 130.49]
KNN (scaled)	148.81 ± 36.38 [103.64, 193.97]	102.49 ± 14.64 [84.31, 120.67]	141.63 ± 61.66 [65.07, 218.19]
KNN (unscaled)	152.10 ± 41.16 [100.99, 203.20]	107.12 ± 17.46 [85.43, 128.80]	145.63 ± 61.89 [68.78, 222.48]
Historical	147.75 ± 16.78 [126.91, 168.58]	112.92 ± 17.03 [91.78, 134.07]	141.88 ± 55.98 [72.37, 211.39]
Graph Laplacian	184.46 ± 60.45 [109.40, 259.52]	119.64 ± 30.13 [82.23, 157.05]	148.28 ± 59.60 [74.27, 222.29]
Graph Temporal	177.38 ± 55.64 [108.29, 246.47]	117.54 ± 31.61 [78.29, 156.79]	148.82 ± 57.78 [77.08, 220.56]
Sage-Voronoi	**80.56** ± 7.93 [70.71, 90.40]	98.89 ± 35.02 [55.41, 142.37]	139.53 ± 62.33 [62.14, 216.92]
GAIN	176.64 ± 49.86 [114.73, 238.55]	108.84 ± 20.77 [83.05, 134.62]	144.39 ± 63.17 [65.95, 222.83]
GRIN	81.76 ± 13.09 [65.50, 98.02]	**80.37** ± 25.40 [48.83, 111.91]	**119.40** ± 62.00 [42.41, 196.38]

Table S19: RMSE for October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 128$	Block $B = 144$
Linear	205.33 ± 14.53 [187.29, 223.37]	211.45 ± 23.51 [182.26, 240.64]
Median	182.06 ± 9.05 [170.83, 193.30]	178.53 ± 15.52 [159.26, 197.79]
FFill + BFill	229.57 ± 13.27 [213.09, 246.05]	213.10 ± 22.83 [184.74, 241.45]
Rolling mean	205.44 ± 14.63 [187.28, 223.60]	211.41 ± 23.40 [182.36, 240.46]
KNN (scaled)	**114.72** ± 4.15 [109.58, 119.87]	112.74 ± 6.58 [104.57, 120.92]
KNN (unscaled)	121.32 ± 6.58 [113.16, 129.48]	119.51 ± 8.30 [109.21, 129.81]
Historical	119.85 ± 6.95 [111.22, 128.49]	121.60 ± 10.58 [108.46, 134.74]
Graph Laplacian	143.95 ± 10.98 [130.32, 157.58]	134.49 ± 22.30 [106.80, 162.17]
Graph Temporal	144.26 ± 9.70 [132.22, 156.31]	133.66 ± 21.55 [106.90, 160.42]
Sage-Voronoi	139.61 ± 14.10 [122.11, 157.11]	129.11 ± 18.89 [105.66, 152.56]
GAIN	131.87 ± 13.45 [115.16, 148.57]	125.05 ± 22.19 [97.50, 152.60]
GRIN	115.58 ± 10.78 [102.19, 128.97]	**103.73** ± 18.67 [80.55, 126.91]

Table S20: RMSE for October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	MCAR 10%	MCAR 30%	MCAR 50%	Synchronous $B = 144$
Linear	57.36 ± 1.52 [55.47, 59.25]	60.94 ± 1.12 [59.55, 62.34]	66.26 ± 0.47 [65.68, 66.84]	182.74 ± 49.56 [121.20, 244.29]
Median	185.32 ± 1.82 [183.06, 187.59]	183.28 ± 0.95 [182.10, 184.46]	182.76 ± 0.41 [182.25, 183.27]	160.36 ± 15.33 [141.33, 179.40]
FFill + BFill	72.44 ± 1.71 [70.31, 74.57]	77.74 ± 1.59 [75.76, 79.71]	83.62 ± 0.93 [82.46, 84.78]	194.65 ± 55.75 [125.43, 263.87]
Rolling mean	74.51 ± 1.05 [73.20, 75.82]	78.12 ± 1.33 [76.47, 79.76]	81.68 ± 0.72 [80.79, 82.58]	182.98 ± 49.65 [121.33, 244.63]
KNN (scaled)	125.22 ± 3.16 [121.29, 129.15]	123.63 ± 1.28 [122.05, 125.22]	121.96 ± 0.88 [120.86, 123.05]	159.65 ± 15.04 [140.98, 178.32]
KNN (unscaled)	130.28 ± 2.95 [126.61, 133.94]	128.46 ± 0.92 [127.32, 129.61]	126.81 ± 0.74 [125.88, 127.73]	159.65 ± 15.04 [140.98, 178.32]
Historical	133.35 ± 2.19 [130.63, 136.06]	132.80 ± 1.33 [131.15, 134.45]	131.89 ± 0.62 [131.11, 132.66]	**124.80** ± 16.44 [104.40, 145.21]
Graph Laplacian	143.40 ± 4.69 [137.57, 149.22]	141.51 ± 0.84 [140.47, 142.55]	141.47 ± 0.37 [141.01, 141.92]	159.65 ± 15.04 [140.98, 178.32]
Graph Temporal	99.21 ± 2.01 [96.71, 101.71]	104.23 ± 1.62 [102.22, 106.23]	110.78 ± 0.76 [109.84, 111.72]	189.22 ± 49.38 [127.90, 250.54]
Sage-Voronoi	66.76 ± 0.62 [66.00, 67.53]	71.12 ± 1.10 [69.76, 72.48]	75.69 ± 0.97 [74.48, 76.90]	214.66 ± 56.69 [144.27, 285.05]
GAIN	132.67 ± 5.47 [125.88, 139.46]	127.18 ± 1.01 [125.92, 128.43]	141.50 ± 0.62 [140.73, 142.27]	360.95 ± 17.67 [339.01, 382.90]
GRIN	**54.40** ± 1.41 [52.65, 56.15]	**58.21** ± 1.23 [56.68, 59.74]	**62.90** ± 0.76 [61.96, 63.85]	147.33 ± 26.73 [114.13, 180.52]

Table S21: MRE for October–December 2022 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 1$	Block $B = 2$	Block $B = 4$	Block $B = 8$
Linear	0.226 ± 0.037 [0.181, 0.271]	0.240 ± 0.028 [0.206, 0.274]	0.244 ± 0.031 [0.206, 0.283]	0.298 ± 0.037 [0.252, 0.343]
Median	2.383 ± 0.463 [1.807, 2.958]	2.405 ± 0.422 [1.881, 2.929]	2.405 ± 0.471 [1.820, 2.990]	2.243 ± 0.256 [1.926, 2.561]
FFill + BFill	0.272 ± 0.048 [0.213, 0.332]	0.286 ± 0.029 [0.250, 0.322]	0.309 ± 0.051 [0.245, 0.373]	0.382 ± 0.044 [0.327, 0.436]
Rolling mean	0.258 ± 0.022 [0.230, 0.285]	0.272 ± 0.014 [0.254, 0.290]	0.303 ± 0.045 [0.247, 0.359]	0.310 ± 0.026 [0.277, 0.342]
KNN (scaled)	0.453 ± 0.303 [0.076, 0.829]	0.493 ± 0.222 [0.216, 0.769]	0.429 ± 0.225 [0.150, 0.709]	0.341 ± 0.083 [0.238, 0.443]
KNN (unscaled)	0.529 ± 0.309 [0.146, 0.912]	0.558 ± 0.193 [0.318, 0.798]	0.483 ± 0.189 [0.249, 0.718]	0.416 ± 0.073 [0.326, 0.507]
Historical	0.621 ± 0.101 [0.496, 0.746]	0.746 ± 0.133 [0.582, 0.911]	0.598 ± 0.132 [0.435, 0.762]	0.645 ± 0.050 [0.583, 0.707]
Graph Laplacian	0.578 ± 0.129 [0.418, 0.737]	0.480 ± 0.078 [0.384, 0.576]	0.524 ± 0.065 [0.444, 0.605]	0.572 ± 0.092 [0.457, 0.686]
Graph Temporal	0.471 ± 0.088 [0.361, 0.580]	0.419 ± 0.063 [0.342, 0.497]	0.488 ± 0.034 [0.446, 0.530]	0.581 ± 0.097 [0.460, 0.701]
Sage-Voronoi	0.226 ± 0.033 [0.186, 0.267]	0.225 ± 0.022 [0.197, 0.253]	0.255 ± 0.017 [0.234, 0.276]	0.277 ± 0.022 [0.249, 0.305]
GAIN	0.332 ± 0.061 [0.257, 0.408]	0.319 ± 0.034 [0.277, 0.360]	0.324 ± 0.050 [0.262, 0.385]	0.337 ± 0.066 [0.256, 0.419]
GRIN	**0.182** ± 0.015 [0.163, 0.201]	**0.184** ± 0.017 [0.163, 0.204]	**0.190** ± 0.030 [0.152, 0.227]	**0.232** ± 0.061 [0.157, 0.308]

Table S22: MRE for October–December 2022 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 16$	Block $B = 32$	Block $B = 64$
Linear	0.373 ± 0.052 [0.308, 0.438]	0.700 ± 0.129 [0.540, 0.860]	1.282 ± 0.121 [1.131, 1.432]
Median	2.531 ± 0.525 [1.880, 3.183]	2.695 ± 0.487 [2.090, 3.299]	2.311 ± 0.110 [2.175, 2.448]
FFill + BFill	0.560 ± 0.059 [0.487, 0.633]	1.068 ± 0.102 [0.941, 1.194]	1.840 ± 0.187 [1.607, 2.073]
Rolling mean	0.363 ± 0.047 [0.304, 0.421]	0.685 ± 0.108 [0.551, 0.819]	1.248 ± 0.126 [1.092, 1.404]
KNN (scaled)	0.636 ± 0.509 [0.004, 1.268]	0.628 ± 0.348 [0.197, 1.060]	0.398 ± 0.097 [0.278, 0.518]
KNN (unscaled)	0.796 ± 0.663 [−0.027, 1.618]	0.692 ± 0.369 [0.233, 1.150]	0.473 ± 0.100 [0.349, 0.598]
Historical	0.852 ± 0.318 [0.457, 1.246]	0.809 ± 0.171 [0.597, 1.021]	0.674 ± 0.119 [0.526, 0.822]
Graph Laplacian	0.544 ± 0.165 [0.340, 0.749]	0.661 ± 0.165 [0.456, 0.865]	0.583 ± 0.108 [0.448, 0.717]
Graph Temporal	0.552 ± 0.189 [0.317, 0.787]	0.684 ± 0.187 [0.451, 0.916]	0.593 ± 0.104 [0.464, 0.722]
Sage-Voronoi	0.294 ± 0.026 [0.261, 0.326]	0.410 ± 0.046 [0.354, 0.467]	0.416 ± 0.080 [0.318, 0.515]
GAIN	0.415 ± 0.166 [0.210, 0.621]	0.439 ± 0.121 [0.288, 0.589]	0.362 ± 0.097 [0.241, 0.483]
GRIN	**0.237** ± 0.074 [0.145, 0.329]	**0.350** ± 0.129 [0.189, 0.510]	**0.335** ± 0.098 [0.213, 0.457]

Table S23: MRE for October–December 2022 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 128$	Block $B = 144$
Linear	2.356 ± 0.113 [2.216, 2.496]	2.552 ± 0.112 [2.413, 2.691]
Median	2.399 ± 0.103 [2.270, 2.527]	2.449 ± 0.054 [2.382, 2.516]
FFill + BFill	2.561 ± 0.050 [2.499, 2.623]	2.499 ± 0.118 [2.351, 2.646]
Rolling mean	2.349 ± 0.110 [2.212, 2.486]	2.551 ± 0.112 [2.411, 2.690]
KNN (scaled)	0.516 ± 0.158 [0.319, 0.713]	0.491 ± 0.125 [0.335, 0.646]
KNN (unscaled)	0.591 ± 0.174 [0.376, 0.806]	0.588 ± 0.166 [0.382, 0.793]
Historical	0.735 ± 0.077 [0.640, 0.830]	0.749 ± 0.056 [0.680, 0.818]
Graph Laplacian	0.531 ± 0.059 [0.457, 0.605]	0.588 ± 0.069 [0.503, 0.674]
Graph Temporal	0.543 ± 0.052 [0.478, 0.608]	0.620 ± 0.089 [0.509, 0.731]
Sage-Voronoi	0.402 ± 0.048 [0.342, 0.462]	0.440 ± 0.040 [0.390, 0.491]
GAIN	0.353 ± 0.062 [0.276, 0.430]	0.441 ± 0.189 [0.206, 0.676]
GRIN	**0.299** ± 0.057 [0.228, 0.371]	**0.339** ± 0.058 [0.267, 0.411]

Table S24: MRE for October–December 2022 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	MCAR 10%	MCAR 30%	MCAR 50%	Synchronous $B = 144$
Linear	0.252 ± 0.003 [0.249, 0.256]	0.254 ± 0.001 [0.253, 0.256]	0.263 ± 0.002 [0.261, 0.265]	1.794 ± 1.318 [0.158, 3.430]
Median	2.408 ± 0.022 [2.381, 2.435]	2.386 ± 0.012 [2.370, 2.401]	2.387 ± 0.012 [2.372, 2.402]	2.003 ± 0.772 [1.044, 2.962]
FFill + BFill	0.298 ± 0.005 [0.292, 0.304]	0.301 ± 0.002 [0.298, 0.303]	0.318 ± 0.003 [0.315, 0.322]	1.857 ± 1.390 [0.131, 3.583]
Rolling mean	0.290 ± 0.004 [0.285, 0.295]	0.294 ± 0.002 [0.291, 0.297]	0.306 ± 0.002 [0.303, 0.308]	1.797 ± 1.319 [0.159, 3.434]
KNN (scaled)	0.460 ± 0.022 [0.433, 0.487]	0.457 ± 0.008 [0.447, 0.467]	0.444 ± 0.003 [0.440, 0.447]	2.032 ± 0.780 [1.063, 3.001]
KNN (unscaled)	0.527 ± 0.022 [0.500, 0.554]	0.519 ± 0.014 [0.501, 0.536]	0.492 ± 0.008 [0.482, 0.503]	2.032 ± 0.780 [1.063, 3.001]
Historical	0.714 ± 0.013 [0.698, 0.730]	0.714 ± 0.006 [0.707, 0.722]	0.714 ± 0.005 [0.708, 0.719]	**0.825** ± 0.480 [0.230, 1.421]
Graph Laplacian	0.551 ± 0.008 [0.542, 0.561]	0.549 ± 0.005 [0.543, 0.555]	0.551 ± 0.004 [0.546, 0.556]	2.032 ± 0.780 [1.063, 3.001]
Graph Temporal	0.448 ± 0.008 [0.438, 0.458]	0.457 ± 0.003 [0.454, 0.461]	0.473 ± 0.003 [0.469, 0.477]	1.852 ± 1.321 [0.212, 3.492]
Sage-Voronoi	0.252 ± 0.004 [0.247, 0.257]	0.255 ± 0.005 [0.249, 0.261]	0.270 ± 0.005 [0.263, 0.276]	1.477 ± 0.982 [0.257, 2.697]
GAIN	0.353 ± 0.011 [0.339, 0.367]	0.547 ± 0.019 [0.523, 0.571]	1.109 ± 0.033 [1.068, 1.150]	5.137 ± 1.664 [3.071, 7.203]
GRIN	**0.202** ± 0.002 [0.199, 0.204]	**0.212** ± 0.002 [0.209, 0.215]	**0.238** ± 0.005 [0.232, 0.245]	1.631 ± 0.801 [0.636, 2.625]

Table S25: MAE for sensor A023, January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 1$	Block $B = 2$	Block $B = 4$	Block $B = 8$
Linear	**9.50** ± 12.81 [−6.41, 25.41]	29.70 ± 18.44 [6.80, 52.60]	29.85 ± 17.04 [8.69, 51.01]	27.06 ± 16.68 [6.35, 47.76]
Median	86.70 ± 59.96 [12.25, 161.15]	129.10 ± 71.94 [39.77, 218.43]	122.30 ± 69.94 [35.45, 209.15]	129.53 ± 54.40 [61.97, 197.08]
FFill + BFill	17.60 ± 18.24 [−5.05, 40.25]	35.90 ± 28.29 [0.78, 71.02]	42.45 ± 25.54 [10.74, 74.16]	59.08 ± 35.84 [14.58, 103.57]
Rolling mean	28.95 ± 21.35 [2.44, 55.46]	23.75 ± 20.62 [−1.85, 49.35]	37.80 ± 20.39 [12.48, 63.12]	41.88 ± 24.47 [11.49, 72.27]
KNN (scaled)	122.14 ± 70.14 [35.05, 209.23]	126.73 ± 74.56 [34.15, 219.31]	60.38 ± 33.68 [18.56, 102.20]	116.92 ± 73.41 [25.76, 208.07]
KNN (unscaled)	23.12 ± 6.43 [15.13, 31.11]	21.52 ± 6.98 [12.86, 30.18]	17.41 ± 4.65 [11.63, 23.19]	**21.20** ± 5.65 [14.18, 28.21]
Historical	45.87 ± 20.58 [20.33, 71.42]	29.36 ± 43.43 [−24.57, 83.28]	53.68 ± 52.04 [−10.94, 118.29]	59.90 ± 41.33 [8.58, 111.22]
Graph Laplacian	45.59 ± 32.46 [5.29, 85.89]	58.69 ± 34.09 [16.36, 101.02]	27.78 ± 14.99 [9.16, 46.40]	48.25 ± 25.53 [16.55, 79.94]
Graph Temporal	19.84 ± 21.43 [−6.77, 46.44]	30.41 ± 16.66 [9.73, 51.10]	18.04 ± 9.95 [5.68, 30.39]	32.46 ± 18.74 [9.19, 55.73]
Sage-Voronoi	14.30 ± 8.44 [3.82, 24.78]	18.91 ± 11.52 [4.61, 33.22]	19.98 ± 11.08 [6.23, 33.73]	30.16 ± 16.90 [9.17, 51.14]
GAIN	15.09 ± 16.95 [−5.96, 36.13]	24.97 ± 15.22 [6.07, 43.87]	11.88 ± 2.73 [8.49, 15.27]	26.08 ± 11.69 [11.56, 40.60]
GRIN	10.05 ± 8.73 [−0.79, 20.89]	**18.90** ± 5.05 [12.63, 25.17]	**11.58** ± 3.81 [6.85, 16.31]	22.07 ± 8.18 [11.90, 32.23]

Table S26: MAE for sensor A023, January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 16$	Block $B = 32$	Block $B = 64$
Linear	29.37 ± 16.24 [9.21, 49.53]	27.57 ± 12.94 [11.50, 43.63]	92.53 ± 68.66 [7.27, 177.78]
Median	126.67 ± 36.86 [80.91, 172.44]	122.70 ± 6.91 [114.12, 131.28]	109.46 ± 48.15 [49.67, 169.25]
FFill + BFill	58.96 ± 42.35 [6.38, 111.55]	50.11 ± 31.77 [10.66, 89.56]	102.43 ± 56.51 [32.27, 172.60]
Rolling mean	19.89 ± 7.61 [10.44, 29.34]	28.08 ± 11.50 [13.81, 42.36]	90.26 ± 66.87 [7.24, 173.29]
KNN (scaled)	79.97 ± 49.21 [18.87, 141.07]	160.32 ± 62.91 [82.20, 238.44]	121.84 ± 45.88 [64.87, 178.81]
KNN (unscaled)	23.73 ± 8.99 [12.57, 34.89]	**24.12** ± 3.80 [19.40, 28.83]	24.36 ± 2.65 [21.07, 27.65]
Historical	31.11 ± 11.29 [17.09, 45.13]	62.08 ± 23.73 [32.61, 91.55]	48.88 ± 16.23 [28.73, 69.02]
Graph Laplacian	46.51 ± 19.99 [21.69, 71.33]	43.12 ± 12.15 [28.03, 58.22]	43.92 ± 11.97 [29.06, 58.78]
Graph Temporal	30.40 ± 9.28 [18.88, 41.92]	24.39 ± 3.36 [20.21, 28.56]	30.10 ± 3.38 [25.90, 34.29]
Sage-Voronoi	28.91 ± 10.83 [15.46, 42.36]	31.02 ± 4.74 [25.13, 36.91]	29.29 ± 8.87 [18.28, 40.31]
GAIN	20.05 ± 9.32 [8.48, 31.62]	24.24 ± 6.35 [16.35, 32.13]	24.32 ± 8.90 [13.27, 35.38]
GRIN	**16.68** ± 5.55 [9.79, 23.57]	24.81 ± 6.74 [16.44, 33.17]	**20.18** ± 5.75 [13.05, 27.32]

Table S27: MAE for sensor A023, January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 128$	Block $B = 144$
Linear	119.08 ± 41.15 [67.99, 170.17]	132.96 ± 19.36 [108.92, 157.00]
Median	108.20 ± 18.21 [85.59, 130.81]	120.96 ± 3.80 [116.24, 125.67]
FFill + BFill	141.16 ± 59.97 [66.69, 215.63]	130.77 ± 14.73 [112.48, 149.06]
Rolling mean	118.22 ± 42.29 [65.71, 170.74]	132.90 ± 19.10 [109.19, 156.61]
KNN (scaled)	100.37 ± 21.52 [73.65, 127.09]	110.16 ± 27.75 [75.71, 144.61]
KNN (unscaled)	24.21 ± 2.86 [20.66, 27.76]	25.92 ± 3.03 [22.16, 29.68]
Historical	46.06 ± 3.89 [41.24, 50.89]	45.33 ± 2.81 [41.84, 48.82]
Graph Laplacian	34.98 ± 6.38 [27.06, 42.90]	43.96 ± 7.67 [34.43, 53.49]
Graph Temporal	24.43 ± 3.90 [19.59, 29.26]	30.14 ± 1.15 [28.70, 31.57]
Sage-Voronoi	26.14 ± 5.25 [19.62, 32.65]	37.67 ± 5.62 [30.69, 44.64]
GAIN	22.36 ± 3.47 [18.05, 26.67]	23.04 ± 2.55 [19.88, 26.21]
GRIN	**21.06** ± 1.92 [18.68, 23.44]	**22.10** ± 1.35 [20.43, 23.77]

Table S28: MAE for sensor A023, January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	MCAR 10%	MCAR 30%	MCAR 50%	Synchronous $B = 144$
Linear	21.43 ± 1.58 [19.46, 23.39]	21.50 ± 1.41 [19.76, 23.25]	21.74 ± 0.70 [20.87, 22.60]	126.44 ± 10.34 [113.61, 139.27]
Median	114.44 ± 3.81 [109.71, 119.18]	112.57 ± 2.85 [109.03, 116.12]	111.77 ± 1.64 [109.73, 113.81]	117.44 ± 7.52 [108.10, 126.78]
FFill + BFill	25.13 ± 2.26 [22.33, 27.93]	25.97 ± 0.76 [25.02, 26.92]	28.74 ± 0.63 [27.96, 29.53]	135.15 ± 21.97 [107.88, 162.43]
Rolling mean	24.86 ± 2.27 [22.04, 27.68]	25.86 ± 1.21 [24.36, 27.36]	27.38 ± 1.00 [26.14, 28.62]	126.66 ± 10.65 [113.43, 139.88]
KNN (scaled)	108.96 ± 7.18 [100.04, 117.87]	101.72 ± 2.60 [98.49, 104.95]	90.09 ± 3.46 [85.79, 94.38]	121.75 ± 13.61 [104.86, 138.65]
KNN (unscaled)	23.81 ± 2.66 [20.51, 27.11]	26.88 ± 0.61 [26.13, 27.64]	27.55 ± 1.15 [26.13, 28.98]	121.75 ± 13.61 [104.86, 138.65]
Historical	45.48 ± 2.90 [41.88, 49.08]	49.09 ± 1.54 [47.18, 51.00]	47.91 ± 0.76 [46.97, 48.85]	**48.22** ± 8.25 [37.97, 58.46]
Graph Laplacian	43.48 ± 2.65 [40.18, 46.77]	45.79 ± 1.43 [44.01, 47.56]	47.59 ± 1.38 [45.87, 49.31]	121.75 ± 13.61 [104.86, 138.65]
Graph Temporal	25.64 ± 0.75 [24.72, 26.57]	27.46 ± 1.02 [26.18, 28.73]	27.59 ± 0.62 [26.82, 28.36]	159.28 ± 53.93 [92.31, 226.24]
Sage-Voronoi	20.95 ± 1.87 [18.63, 23.27]	22.74 ± 2.00 [20.25, 25.22]	23.89 ± 1.47 [22.06, 25.72]	119.28 ± 31.27 [80.45, 158.11]
GAIN	22.61 ± 1.89 [20.25, 24.96]	27.96 ± 2.40 [24.98, 30.94]	41.97 ± 3.84 [37.20, 46.73]	121.41 ± 12.95 [105.33, 137.48]
GRIN	**17.45** ± 0.28 [17.10, 17.80]	**18.54** ± 1.12 [17.14, 19.93]	**19.98** ± 1.18 [18.51, 21.44]	132.16 ± 25.48 [100.52, 163.80]

Table S29: RMSE for sensor A023, January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 1$	Block $B = 2$	Block $B = 4$	Block $B = 8$
Linear	**9.50** ± 12.81 [−6.41, 25.41]	32.82 ± 19.60 [8.48, 57.16]	33.28 ± 16.60 [12.67, 53.89]	32.30 ± 17.74 [10.26, 54.33]
Median	86.70 ± 59.96 [12.25, 161.15]	129.65 ± 71.95 [40.31, 218.99]	123.41 ± 69.61 [36.98, 209.84]	133.26 ± 51.40 [69.43, 197.09]
FFill + BFill	17.60 ± 18.24 [−5.05, 40.25]	38.47 ± 28.23 [3.42, 73.53]	45.21 ± 26.12 [12.78, 77.65]	65.28 ± 37.60 [18.60, 111.97]
Rolling mean	28.95 ± 21.35 [2.44, 55.46]	26.02 ± 22.16 [−1.50, 53.54]	41.46 ± 20.96 [15.44, 67.48]	48.04 ± 24.43 [17.70, 78.38]
KNN (scaled)	122.14 ± 70.14 [35.05, 209.23]	128.99 ± 72.94 [38.42, 219.56]	66.36 ± 33.09 [25.27, 107.45]	121.72 ± 73.05 [31.02, 212.42]
KNN (unscaled)	23.12 ± 6.43 [15.13, 31.11]	23.95 ± 7.51 [14.62, 33.27]	19.89 ± 5.89 [12.57, 27.20]	**25.36** ± 4.92 [19.25, 31.47]
Historical	45.87 ± 20.58 [20.33, 71.42]	30.97 ± 43.05 [−22.48, 84.43]	56.52 ± 50.84 [−6.60, 119.65]	64.46 ± 39.68 [15.19, 113.74]
Graph Laplacian	45.59 ± 32.46 [5.29, 85.89]	61.62 ± 32.29 [21.52, 101.72]	31.48 ± 13.78 [14.37, 48.58]	54.21 ± 26.82 [20.91, 87.50]
Graph Temporal	19.84 ± 21.43 [−6.77, 46.44]	33.39 ± 16.48 [12.92, 53.85]	20.12 ± 10.54 [7.04, 33.20]	35.35 ± 18.70 [12.13, 58.57]
Sage-Voronoi	14.30 ± 8.44 [3.82, 24.78]	**20.47** ± 12.33 [5.16, 35.78]	23.11 ± 11.09 [9.35, 36.88]	35.86 ± 20.11 [10.89, 60.84]
GAIN	15.09 ± 16.95 [−5.96, 36.13]	26.68 ± 14.55 [8.61, 44.75]	15.02 ± 3.82 [10.28, 19.76]	29.89 ± 11.95 [15.05, 44.73]
GRIN	10.05 ± 8.73 [−0.79, 20.89]	21.45 ± 7.70 [11.90, 31.01]	**14.53** ± 4.00 [9.56, 19.50]	25.74 ± 8.16 [15.61, 35.88]

Table S30: RMSE for sensor A023, January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 16$	Block $B = 32$	Block $B = 64$
Linear	35.74 ± 18.49 [12.79, 58.69]	33.39 ± 14.14 [15.83, 50.94]	106.03 ± 77.63 [9.64, 202.42]
Median	138.86 ± 36.31 [93.77, 183.94]	128.38 ± 11.46 [114.14, 142.61]	119.93 ± 49.56 [58.39, 181.47]
FFill + BFill	77.02 ± 56.40 [6.99, 147.06]	58.83 ± 37.00 [12.89, 104.78]	123.09 ± 65.66 [41.57, 204.61]
Rolling mean	23.81 ± 8.74 [12.96, 34.66]	34.07 ± 12.85 [18.11, 50.03]	103.73 ± 76.05 [9.30, 198.16]
KNN (scaled)	88.08 ± 46.40 [30.47, 145.70]	163.49 ± 61.56 [87.06, 239.92]	135.28 ± 40.58 [84.90, 185.67]
KNN (unscaled)	29.47 ± 12.66 [13.74, 45.19]	**30.04** ± 5.57 [23.12, 36.96]	31.02 ± 3.15 [27.11, 34.93]
Historical	38.85 ± 14.07 [21.38, 56.32]	68.20 ± 24.29 [38.04, 98.36]	59.70 ± 18.03 [37.32, 82.09]
Graph Laplacian	51.74 ± 17.71 [29.75, 73.72]	53.04 ± 13.70 [36.02, 70.05]	53.39 ± 14.74 [35.08, 71.69]
Graph Temporal	35.74 ± 10.94 [22.16, 49.32]	30.26 ± 4.66 [24.48, 36.04]	37.12 ± 3.15 [33.21, 41.03]
Sage-Voronoi	35.55 ± 13.82 [18.39, 52.71]	37.55 ± 5.52 [30.70, 44.40]	36.87 ± 9.67 [24.87, 48.87]
GAIN	25.41 ± 12.86 [9.43, 41.38]	30.12 ± 7.63 [20.65, 39.59]	31.20 ± 11.07 [17.45, 44.95]
GRIN	**20.74** ± 8.16 [10.60, 30.87]	30.68 ± 7.92 [20.85, 40.51]	**26.24** ± 6.69 [17.93, 34.54]

Table S31: RMSE for sensor A023, January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 128$	Block $B = 144$
Linear	135.78 ± 45.41 [79.39, 192.16]	160.31 ± 28.63 [124.76, 195.86]
Median	121.13 ± 17.71 [99.14, 143.12]	135.35 ± 3.97 [130.42, 140.29]
FFill + BFill	160.24 ± 65.48 [78.94, 241.55]	160.80 ± 22.56 [132.78, 188.81]
Rolling mean	134.86 ± 46.98 [76.53, 193.18]	160.20 ± 28.36 [124.98, 195.41]
KNN (scaled)	116.58 ± 20.32 [91.34, 141.81]	129.74 ± 24.98 [98.72, 160.76]
KNN (unscaled)	30.31 ± 4.30 [24.97, 35.65]	32.51 ± 4.26 [27.22, 37.80]
Historical	59.26 ± 7.01 [50.56, 67.96]	57.99 ± 2.09 [55.39, 60.58]
Graph Laplacian	42.80 ± 6.94 [34.19, 51.42]	53.33 ± 8.69 [42.54, 64.12]
Graph Temporal	30.90 ± 4.81 [24.93, 36.87]	37.05 ± 1.34 [35.39, 38.71]
Sage-Voronoi	33.37 ± 7.24 [24.38, 42.36]	45.83 ± 6.96 [37.19, 54.47]
GAIN	28.32 ± 3.74 [23.68, 32.96]	29.77 ± 2.73 [26.38, 33.17]
GRIN	**27.27** ± 2.43 [24.25, 30.30]	**28.67** ± 2.43 [25.65, 31.69]

Table S32: RMSE for sensor A023, January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	MCAR 10%	MCAR 30%	MCAR 50%	Synchronous $B = 144$
Linear	28.13 ± 2.28 [25.29, 30.96]	28.15 ± 1.80 [25.92, 30.38]	28.30 ± 0.78 [27.33, 29.27]	168.03 ± 23.27 [139.14, 196.93]
Median	129.95 ± 3.22 [125.95, 133.95]	128.63 ± 2.88 [125.06, 132.20]	127.51 ± 1.83 [125.23, 129.79]	133.91 ± 3.79 [129.21, 138.61]
FFill + BFill	32.17 ± 2.78 [28.71, 35.63]	34.07 ± 1.07 [32.74, 35.40]	37.80 ± 1.50 [35.94, 39.66]	179.50 ± 24.33 [149.29, 209.71]
Rolling mean	31.12 ± 2.66 [27.82, 34.42]	33.88 ± 1.58 [31.92, 35.83]	36.11 ± 1.50 [34.24, 37.97]	168.46 ± 23.12 [139.76, 197.16]
KNN (scaled)	129.01 ± 5.71 [121.92, 136.10]	124.30 ± 2.25 [121.50, 127.10]	116.01 ± 3.32 [111.89, 120.13]	134.87 ± 10.18 [122.22, 147.51]
KNN (unscaled)	31.01 ± 4.00 [26.05, 35.98]	34.52 ± 0.78 [33.55, 35.48]	37.21 ± 1.66 [35.14, 39.27]	134.87 ± 10.18 [122.22, 147.51]
Historical	60.59 ± 3.47 [56.29, 64.90]	63.32 ± 1.94 [60.91, 65.73]	61.47 ± 1.19 [59.99, 62.94]	**62.48** ± 12.94 [46.41, 78.55]
Graph Laplacian	53.51 ± 2.28 [50.68, 56.34]	57.43 ± 1.79 [55.22, 59.65]	60.84 ± 2.08 [58.26, 63.42]	134.87 ± 10.18 [122.22, 147.51]
Graph Temporal	32.24 ± 0.86 [31.17, 33.30]	34.33 ± 1.69 [32.24, 36.42]	34.78 ± 0.64 [33.99, 35.58]	200.07 ± 59.36 [126.37, 273.77]
Sage-Voronoi	27.28 ± 3.04 [23.50, 31.06]	30.12 ± 2.56 [26.93, 33.30]	31.55 ± 1.89 [29.20, 33.91]	157.21 ± 38.71 [109.15, 205.27]
GAIN	29.31 ± 2.09 [26.72, 31.90]	34.73 ± 2.77 [31.28, 38.17]	49.99 ± 4.26 [44.70, 55.28]	134.91 ± 9.11 [123.59, 146.23]
GRIN	**22.67** ± 1.08 [21.33, 24.01]	**24.27** ± 1.56 [22.34, 26.21]	**26.02** ± 1.40 [24.28, 27.76]	148.79 ± 26.15 [116.32, 181.26]

Table S33: MRE for sensor A023, January 2023 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 1$	Block $B = 2$	Block $B = 4$	Block $B = 8$
Linear	**0.031** ± 0.031 [−0.008, 0.070]	0.163 ± 0.089 [0.052, 0.274]	0.275 ± 0.159 [0.079, 0.472]	0.135 ± 0.056 [0.065, 0.204]
Median	0.385 ± 0.427 [−0.146, 0.915]	1.362 ± 1.660 [−0.699, 3.423]	1.552 ± 1.663 [−0.512, 3.617]	1.101 ± 1.372 [−0.602, 2.804]
FFill + BFill	0.056 ± 0.044 [0.002, 0.111]	0.170 ± 0.061 [0.095, 0.246]	0.339 ± 0.153 [0.149, 0.529]	0.339 ± 0.208 [0.081, 0.598]
Rolling mean	0.143 ± 0.161 [−0.057, 0.344]	0.169 ± 0.171 [−0.044, 0.381]	0.316 ± 0.115 [0.173, 0.459]	0.232 ± 0.126 [0.076, 0.388]
KNN (scaled)	0.400 ± 0.107 [0.268, 0.533]	0.755 ± 0.459 [0.185, 1.325]	0.733 ± 0.873 [−0.351, 1.817]	0.650 ± 0.576 [−0.066, 1.365]
KNN (unscaled)	0.092 ± 0.046 [0.034, 0.149]	0.217 ± 0.254 [−0.099, 0.533]	0.155 ± 0.081 [0.055, 0.256]	0.135 ± 0.133 [−0.030, 0.300]
Historical	0.166 ± 0.069 [0.080, 0.251]	**0.128** ± 0.096 [0.009, 0.248]	0.514 ± 0.452 [−0.048, 1.075]	0.283 ± 0.138 [0.112, 0.454]
Graph Laplacian	0.179 ± 0.139 [0.006, 0.352]	0.420 ± 0.434 [−0.120, 0.959]	0.364 ± 0.427 [−0.166, 0.893]	0.286 ± 0.264 [−0.041, 0.613]
Graph Temporal	0.074 ± 0.076 [−0.020, 0.168]	0.211 ± 0.225 [−0.068, 0.491]	0.190 ± 0.189 [−0.045, 0.424]	0.182 ± 0.118 [0.035, 0.329]
Sage-Voronoi	0.057 ± 0.040 [0.007, 0.107]	0.135 ± 0.128 [−0.024, 0.294]	0.192 ± 0.101 [0.066, 0.318]	0.144 ± 0.038 [0.096, 0.191]
GAIN	0.053 ± 0.045 [−0.003, 0.109]	0.150 ± 0.105 [0.019, 0.281]	**0.099** ± 0.027 [0.066, 0.132]	0.153 ± 0.115 [0.011, 0.296]
GRIN	0.036 ± 0.026 [0.004, 0.068]	0.159 ± 0.157 [−0.036, 0.354]	0.103 ± 0.039 [0.054, 0.151]	**0.102** ± 0.038 [0.055, 0.149]

Table S34: MRE for sensor A023, January 2023 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 16$	Block $B = 32$	Block $B = 64$
Linear	0.375 ± 0.361 [−0.073, 0.824]	0.094 ± 0.042 [0.042, 0.146]	1.270 ± 1.697 [−0.837, 3.377]
Median	1.697 ± 1.171 [0.242, 3.151]	0.642 ± 0.730 [−0.264, 1.548]	1.260 ± 1.403 [−0.482, 3.003]
FFill + BFill	0.390 ± 0.253 [0.076, 0.703]	0.396 ± 0.669 [−0.435, 1.227]	0.997 ± 1.319 [−0.641, 2.636]
Rolling mean	0.192 ± 0.117 [0.047, 0.336]	0.101 ± 0.056 [0.032, 0.170]	1.226 ± 1.660 [−0.835, 3.288]
KNN (scaled)	0.697 ± 0.377 [0.228, 1.166]	0.527 ± 0.074 [0.435, 0.620]	0.667 ± 0.368 [0.210, 1.125]
KNN (unscaled)	0.228 ± 0.159 [0.031, 0.425]	0.094 ± 0.059 [0.020, 0.167]	0.160 ± 0.115 [0.017, 0.303]
Historical	0.288 ± 0.186 [0.057, 0.519]	0.230 ± 0.122 [0.079, 0.382]	0.273 ± 0.171 [0.061, 0.485]
Graph Laplacian	0.477 ± 0.268 [0.144, 0.810]	0.192 ± 0.151 [0.005, 0.380]	0.368 ± 0.321 [−0.030, 0.766]
Graph Temporal	0.262 ± 0.139 [0.090, 0.435]	0.108 ± 0.094 [−0.008, 0.224]	0.228 ± 0.185 [−0.002, 0.457]
Sage-Voronoi	0.248 ± 0.137 [0.078, 0.418]	0.163 ± 0.188 [−0.071, 0.397]	0.219 ± 0.224 [−0.059, 0.498]
GAIN	0.149 ± 0.054 [0.081, 0.216]	0.092 ± 0.055 [0.024, 0.160]	0.121 ± 0.045 [0.065, 0.177]
GRIN	**0.129** ± 0.048 [0.069, 0.189]	**0.086** ± 0.033 [0.045, 0.126]	**0.094** ± 0.030 [0.056, 0.132]

Table S35: MRE for sensor A023, January 2023 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 128$	Block $B = 144$
Linear	0.644 ± 0.302 [0.268, 1.019]	1.642 ± 0.686 [0.791, 2.494]
Median	1.041 ± 0.464 [0.465, 1.617]	1.377 ± 0.187 [1.144, 1.610]
FFill + BFill	0.816 ± 0.227 [0.534, 1.098]	1.619 ± 0.616 [0.854, 2.384]
Rolling mean	0.635 ± 0.302 [0.260, 1.009]	1.640 ± 0.684 [0.791, 2.490]
KNN (scaled)	0.564 ± 0.120 [0.415, 0.713]	0.626 ± 0.165 [0.421, 0.830]
KNN (unscaled)	0.159 ± 0.026 [0.126, 0.191]	0.198 ± 0.037 [0.152, 0.244]
Historical	0.305 ± 0.083 [0.202, 0.408]	0.263 ± 0.045 [0.207, 0.319]
Graph Laplacian	0.293 ± 0.117 [0.148, 0.438]	0.384 ± 0.046 [0.327, 0.441]
Graph Temporal	0.172 ± 0.071 [0.084, 0.261]	0.238 ± 0.033 [0.197, 0.278]
Sage-Voronoi	0.178 ± 0.059 [0.105, 0.251]	0.351 ± 0.144 [0.172, 0.530]
GAIN	0.127 ± 0.014 [0.109, 0.145]	0.136 ± 0.021 [0.110, 0.163]
GRIN	**0.106** ± 0.007 [0.098, 0.115]	**0.110** ± 0.007 [0.101, 0.119]

Table S36: MRE for sensor A023, January 2023 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	MCAR 10%	MCAR 30%	MCAR 50%	Synchronous $B = 144$
Linear	0.126 ± 0.008 [0.117, 0.135]	0.128 ± 0.012 [0.113, 0.143]	0.122 ± 0.005 [0.115, 0.128]	1.782 ± 0.351 [1.346, 2.217]
Median	1.278 ± 0.113 [1.137, 1.418]	1.226 ± 0.142 [1.050, 1.401]	1.159 ± 0.090 [1.047, 1.270]	1.333 ± 0.250 [1.023, 1.644]
FFill + BFill	0.150 ± 0.017 [0.129, 0.171]	0.154 ± 0.012 [0.139, 0.169]	0.163 ± 0.008 [0.153, 0.173]	1.645 ± 0.629 [0.864, 2.427]
Rolling mean	0.152 ± 0.020 [0.127, 0.177]	0.156 ± 0.012 [0.141, 0.171]	0.156 ± 0.007 [0.147, 0.164]	1.783 ± 0.350 [1.349, 2.218]
KNN (scaled)	0.657 ± 0.045 [0.601, 0.712]	0.610 ± 0.059 [0.537, 0.683]	0.541 ± 0.019 [0.517, 0.564]	1.228 ± 0.209 [0.969, 1.488]
KNN (unscaled)	0.189 ± 0.030 [0.151, 0.226]	0.205 ± 0.021 [0.179, 0.232]	0.217 ± 0.020 [0.192, 0.241]	1.228 ± 0.209 [0.969, 1.488]
Historical	0.297 ± 0.016 [0.277, 0.317]	0.310 ± 0.019 [0.286, 0.334]	0.292 ± 0.020 [0.267, 0.317]	**0.309** ± 0.190 [0.073, 0.544]
Graph Laplacian	0.378 ± 0.033 [0.337, 0.419]	0.378 ± 0.043 [0.324, 0.432]	0.373 ± 0.021 [0.347, 0.399]	1.228 ± 0.209 [0.969, 1.488]
Graph Temporal	0.195 ± 0.018 [0.173, 0.218]	0.198 ± 0.020 [0.173, 0.224]	0.190 ± 0.017 [0.169, 0.211]	2.092 ± 0.586 [1.365, 2.819]
Sage-Voronoi	0.127 ± 0.009 [0.116, 0.138]	0.135 ± 0.017 [0.113, 0.156]	0.135 ± 0.012 [0.120, 0.150]	1.333 ± 0.453 [0.770, 1.896]
GAIN	0.143 ± 0.023 [0.115, 0.171]	0.203 ± 0.035 [0.160, 0.247]	0.322 ± 0.047 [0.264, 0.380]	1.241 ± 0.219 [0.969, 1.512]
GRIN	**0.099** ± 0.007 [0.090, 0.108]	**0.101** ± 0.007 [0.092, 0.109]	**0.104** ± 0.004 [0.098, 0.109]	0.999 ± 0.286 [0.643, 1.354]

Table S37: MAE for sensor A023, October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 1$	Block $B = 2$	Block $B = 4$	Block $B = 8$
Linear	21.80 ± 8.63 [11.09, 32.51]	21.73 ± 8.38 [11.33, 32.14]	19.57 ± 8.94 [8.47, 30.67]	22.59 ± 7.28 [13.56, 31.63]
Median	172.00 ± 73.24 [81.06, 262.94]	148.40 ± 73.41 [57.25, 239.55]	159.55 ± 86.97 [51.57, 267.53]	99.80 ± 71.80 [10.65, 188.95]
FFill + BFill	21.60 ± 15.22 [2.70, 40.50]	23.80 ± 17.28 [2.35, 45.25]	24.95 ± 17.47 [3.25, 46.65]	37.80 ± 18.47 [14.86, 60.74]
Rolling mean	25.45 ± 15.39 [6.34, 44.56]	16.13 ± 9.66 [4.14, 28.13]	30.97 ± 16.70 [10.23, 51.70]	23.99 ± 6.73 [15.63, 32.35]
KNN (scaled)	13.86 ± 15.13 [−4.93, 32.65]	16.78 ± 7.29 [7.73, 25.83]	16.12 ± 10.38 [3.22, 29.01]	19.75 ± 8.73 [8.92, 30.59]
KNN (unscaled)	23.64 ± 12.07 [8.65, 38.63]	26.76 ± 14.31 [8.99, 44.53]	17.07 ± 7.08 [8.27, 25.87]	26.84 ± 15.23 [7.94, 45.74]
Historical	27.08 ± 30.77 [−11.12, 65.29]	75.03 ± 66.96 [−8.10, 158.17]	29.15 ± 11.90 [14.37, 43.92]	49.80 ± 53.33 [−16.42, 116.02]
Graph Laplacian	29.83 ± 25.45 [−1.77, 61.43]	29.65 ± 9.76 [17.53, 41.77]	28.86 ± 15.60 [9.49, 48.22]	25.03 ± 8.83 [14.06, 36.00]
Graph Temporal	32.91 ± 16.68 [12.20, 53.61]	30.57 ± 8.37 [20.17, 40.97]	26.84 ± 13.27 [10.37, 43.32]	24.20 ± 7.41 [15.00, 33.39]
Sage-Voronoi	16.41 ± 15.03 [−2.25, 35.07]	18.46 ± 8.06 [8.46, 28.47]	19.89 ± 13.76 [2.80, 36.98]	19.66 ± 5.30 [13.08, 26.24]
GAIN	19.26 ± 18.06 [−3.16, 41.68]	23.33 ± 21.91 [−3.87, 50.53]	15.55 ± 5.11 [9.20, 21.90]	23.58 ± 4.88 [17.51, 29.64]
GRIN	**13.05** ± 9.83 [0.84, 25.26]	**14.42** ± 9.85 [2.19, 26.65]	**14.43** ± 8.96 [3.31, 25.56]	**18.16** ± 2.53 [15.01, 21.30]

Table S38: MAE for sensor A023, October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 16$	Block $B = 32$	Block $B = 64$
Linear	34.75 ± 7.28 [25.71, 43.79]	46.95 ± 21.33 [20.47, 73.43]	70.52 ± 45.49 [14.04, 127.00]
Median	72.59 ± 32.96 [31.67, 113.51]	115.01 ± 86.59 [7.49, 222.52]	108.52 ± 61.53 [32.12, 184.91]
FFill + BFill	55.36 ± 52.62 [−9.98, 120.70]	113.36 ± 65.95 [31.47, 195.24]	102.60 ± 91.58 [−11.11, 216.31]
Rolling mean	33.36 ± 12.09 [18.35, 48.37]	54.39 ± 25.62 [22.58, 86.20]	70.09 ± 48.93 [9.34, 130.84]
KNN (scaled)	21.35 ± 4.59 [15.65, 27.05]	22.79 ± 6.65 [14.53, 31.05]	28.29 ± 14.88 [9.82, 46.76]
KNN (unscaled)	25.54 ± 3.61 [21.05, 30.02]	26.78 ± 8.79 [15.87, 37.70]	34.25 ± 17.62 [12.37, 56.12]
Historical	22.26 ± 6.47 [14.22, 30.29]	59.16 ± 29.18 [22.93, 95.40]	67.80 ± 34.01 [25.57, 110.02]
Graph Laplacian	35.36 ± 4.70 [29.53, 41.19]	34.35 ± 13.96 [17.01, 51.69]	35.51 ± 9.10 [24.21, 46.81]
Graph Temporal	36.20 ± 11.04 [22.49, 49.91]	33.16 ± 13.46 [16.45, 49.88]	33.95 ± 8.60 [23.26, 44.63]
Sage-Voronoi	40.99 ± 17.79 [18.90, 63.08]	29.81 ± 13.46 [13.10, 46.52]	36.56 ± 5.21 [30.09, 43.02]
GAIN	26.63 ± 7.18 [17.72, 35.55]	26.69 ± 11.34 [12.61, 40.77]	**24.13** ± 10.09 [11.60, 36.66]
GRIN	**19.54** ± 5.21 [13.08, 26.01]	**19.90** ± 6.74 [11.53, 28.27]	25.21 ± 13.00 [9.07, 41.35]

Table S39: MAE for sensor A023, October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 128$	Block $B = 144$
Linear	182.17 ± 65.26 [101.14, 263.21]	126.79 ± 10.31 [113.99, 139.59]
Median	94.85 ± 15.95 [75.05, 114.65]	113.29 ± 7.10 [104.48, 122.10]
FFill + BFill	178.65 ± 76.79 [83.30, 274.00]	117.95 ± 8.44 [107.47, 128.43]
Rolling mean	181.80 ± 65.69 [100.23, 263.37]	126.61 ± 10.13 [114.04, 139.19]
KNN (scaled)	22.07 ± 2.37 [19.13, 25.01]	23.57 ± 2.25 [20.78, 26.37]
KNN (unscaled)	26.61 ± 0.45 [26.05, 27.17]	27.82 ± 2.01 [25.33, 30.31]
Historical	29.97 ± 7.52 [20.63, 39.31]	45.38 ± 20.47 [19.96, 70.80]
Graph Laplacian	34.23 ± 2.08 [31.65, 36.82]	35.72 ± 6.70 [27.40, 44.05]
Graph Temporal	35.15 ± 2.99 [31.44, 38.87]	33.87 ± 6.52 [25.78, 41.97]
Sage-Voronoi	34.79 ± 2.29 [31.95, 37.63]	32.62 ± 7.72 [23.03, 42.21]
GAIN	23.59 ± 3.55 [19.18, 27.99]	23.67 ± 2.92 [20.05, 27.30]
GRIN	**20.65** ± 0.66 [19.84, 21.47]	**21.72** ± 3.57 [17.28, 26.15]

Table S40: MAE for sensor A023, October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	MCAR 10%	MCAR 30%	MCAR 50%	Synchronous $B = 144$
Linear	22.87 ± 1.48 [21.03, 24.71]	23.53 ± 0.86 [22.46, 24.60]	23.94 ± 0.40 [23.44, 24.43]	124.28 ± 30.70 [86.16, 162.40]
Median	108.24 ± 2.26 [105.44, 111.04]	110.07 ± 1.82 [107.81, 112.34]	107.39 ± 0.89 [106.29, 108.49]	107.86 ± 13.79 [90.73, 124.98]
FFill + BFill	27.91 ± 1.29 [26.30, 29.51]	28.00 ± 0.89 [26.89, 29.10]	29.78 ± 0.86 [28.71, 30.86]	121.93 ± 27.77 [87.46, 156.41]
Rolling mean	26.87 ± 1.65 [24.83, 28.92]	27.40 ± 0.62 [26.63, 28.17]	28.59 ± 0.62 [27.83, 29.36]	124.48 ± 30.67 [86.41, 162.56]
KNN (scaled)	24.76 ± 1.20 [23.27, 26.26]	24.45 ± 0.94 [23.28, 25.61]	24.80 ± 0.45 [24.24, 25.36]	103.28 ± 14.90 [84.78, 121.77]
KNN (unscaled)	29.15 ± 2.09 [26.55, 31.75]	28.27 ± 0.85 [27.21, 29.33]	27.83 ± 0.29 [27.47, 28.19]	103.28 ± 14.90 [84.78, 121.77]
Historical	46.54 ± 2.48 [43.46, 49.62]	46.45 ± 1.89 [44.11, 48.80]	46.29 ± 0.56 [45.59, 46.99]	**54.86** ± 19.45 [30.71, 79.00]
Graph Laplacian	35.28 ± 0.74 [34.35, 36.20]	36.09 ± 1.07 [34.76, 37.41]	36.27 ± 0.30 [35.90, 36.64]	103.28 ± 14.90 [84.78, 121.77]
Graph Temporal	30.38 ± 1.12 [28.98, 31.77]	31.80 ± 0.41 [31.30, 32.31]	32.73 ± 0.36 [32.28, 33.17]	118.67 ± 24.66 [88.06, 149.29]
Sage-Voronoi	22.70 ± 1.32 [21.06, 24.35]	23.36 ± 0.66 [22.55, 24.18]	25.06 ± 0.52 [24.42, 25.71]	130.94 ± 42.92 [77.65, 184.24]
GAIN	24.53 ± 1.41 [22.78, 26.28]	28.33 ± 0.96 [27.13, 29.52]	39.49 ± 0.89 [38.38, 40.60]	104.91 ± 11.67 [90.42, 119.41]
GRIN	**19.46** ± 1.17 [18.01, 20.91]	**19.88** ± 0.79 [18.90, 20.87]	**20.51** ± 0.46 [19.94, 21.09]	89.28 ± 16.66 [68.60, 109.96]

Table S41: RMSE for sensor A023, October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 1$	Block $B = 2$	Block $B = 4$	Block $B = 8$
Linear	21.80 ± 8.63 [11.09, 32.51]	24.24 ± 7.45 [14.99, 33.49]	21.25 ± 9.30 [9.69, 32.80]	28.19 ± 9.50 [16.40, 39.99]
Median	172.00 ± 73.24 [81.06, 262.94]	149.35 ± 73.46 [58.14, 240.55]	160.87 ± 85.49 [54.72, 267.02]	104.64 ± 69.39 [18.48, 190.81]
FFill + BFill	21.60 ± 15.22 [2.70, 40.50]	24.61 ± 17.91 [2.37, 46.85]	28.47 ± 17.87 [6.28, 50.66]	42.99 ± 19.68 [18.56, 67.43]
Rolling mean	25.45 ± 15.39 [6.34, 44.56]	18.73 ± 9.90 [6.43, 31.03]	34.95 ± 16.52 [14.44, 55.46]	28.76 ± 8.02 [18.80, 38.73]
KNN (scaled)	13.86 ± 15.13 [−4.93, 32.65]	20.44 ± 9.27 [8.93, 31.95]	18.10 ± 11.05 [4.38, 31.82]	24.39 ± 8.71 [13.58, 35.21]
KNN (unscaled)	23.64 ± 12.07 [8.65, 38.63]	30.39 ± 14.95 [11.83, 48.96]	20.32 ± 8.77 [9.43, 31.21]	32.85 ± 14.00 [15.47, 50.24]
Historical	27.08 ± 30.77 [−11.12, 65.29]	78.58 ± 64.75 [−1.82, 158.98]	31.73 ± 11.85 [17.02, 46.44]	53.84 ± 51.67 [−10.32, 117.99]
Graph Laplacian	29.83 ± 25.45 [−1.77, 61.43]	33.48 ± 10.71 [20.18, 46.77]	31.73 ± 14.02 [14.32, 49.13]	29.82 ± 9.86 [17.58, 42.06]
Graph Temporal	32.91 ± 16.68 [12.20, 53.61]	35.46 ± 9.46 [23.72, 47.21]	29.52 ± 11.33 [15.46, 43.58]	29.18 ± 8.05 [19.18, 39.17]
Sage-Voronoi	16.41 ± 15.03 [−2.25, 35.07]	20.18 ± 7.59 [10.75, 29.61]	22.51 ± 13.76 [5.43, 39.59]	24.23 ± 5.71 [17.14, 31.32]
GAIN	19.26 ± 18.06 [−3.16, 41.68]	26.81 ± 23.25 [−2.06, 55.67]	19.03 ± 5.59 [12.10, 25.97]	29.65 ± 4.61 [23.92, 35.37]
GRIN	**13.05** ± 9.83 [0.84, 25.26]	**16.63** ± 10.94 [3.05, 30.21]	**16.34** ± 9.64 [4.37, 28.31]	**21.76** ± 3.40 [17.53, 25.98]

Table S42: RMSE for sensor A023, October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 16$	Block $B = 32$	Block $B = 64$
Linear	41.51 ± 5.68 [34.45, 48.56]	55.84 ± 27.60 [21.57, 90.12]	82.58 ± 49.65 [20.93, 144.23]
Median	84.20 ± 37.73 [37.34, 131.05]	124.45 ± 85.45 [18.35, 230.56]	122.37 ± 64.04 [42.86, 201.88]
FFill + BFill	67.48 ± 60.58 [−7.74, 142.70]	125.37 ± 69.46 [39.12, 211.62]	118.85 ± 90.20 [6.85, 230.85]
Rolling mean	39.87 ± 9.86 [27.63, 52.11]	63.45 ± 33.63 [21.69, 105.20]	82.26 ± 53.31 [16.07, 148.45]
KNN (scaled)	29.35 ± 6.17 [21.69, 37.01]	29.08 ± 9.08 [17.80, 40.35]	36.73 ± 21.40 [10.16, 63.30]
KNN (unscaled)	32.51 ± 4.38 [27.08, 37.95]	33.13 ± 10.37 [20.25, 46.00]	44.29 ± 22.86 [15.90, 72.68]
Historical	29.92 ± 7.25 [20.92, 38.92]	64.77 ± 31.16 [26.09, 103.46]	81.56 ± 41.90 [29.53, 133.59]
Graph Laplacian	43.76 ± 6.76 [35.37, 52.15]	40.72 ± 13.13 [24.42, 57.02]	44.93 ± 11.94 [30.11, 59.75]
Graph Temporal	45.31 ± 14.34 [27.50, 63.12]	38.56 ± 13.05 [22.35, 54.77]	42.80 ± 13.24 [26.37, 59.24]
Sage-Voronoi	48.57 ± 18.47 [25.63, 71.50]	35.33 ± 14.56 [17.26, 53.41]	46.25 ± 10.71 [32.95, 59.55]
GAIN	36.95 ± 8.83 [25.98, 47.92]	33.35 ± 13.12 [17.06, 49.65]	**33.07** ± 17.86 [10.90, 55.25]
GRIN	**27.72** ± 6.93 [19.12, 36.33]	**24.77** ± 7.43 [15.55, 33.99]	33.81 ± 20.02 [8.96, 58.66]

Table S43: RMSE for sensor A023, October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 128$	Block $B = 144$
Linear	213.48 ± 55.88 [144.10, 282.87]	164.90 ± 21.60 [138.07, 191.72]
Median	117.24 ± 17.04 [96.08, 138.40]	138.75 ± 10.01 [126.32, 151.18]
FFill + BFill	207.25 ± 71.18 [118.88, 295.63]	155.89 ± 22.97 [127.37, 184.42]
Rolling mean	213.14 ± 56.41 [143.10, 283.18]	164.72 ± 21.40 [138.15, 191.29]
KNN (scaled)	28.36 ± 2.58 [25.16, 31.56]	32.36 ± 6.24 [24.61, 40.11]
KNN (unscaled)	33.78 ± 1.07 [32.45, 35.11]	36.43 ± 5.70 [29.35, 43.51]
Historical	38.69 ± 8.63 [27.97, 49.41]	60.50 ± 26.27 [27.88, 93.12]
Graph Laplacian	41.93 ± 2.66 [38.62, 45.23]	43.75 ± 9.81 [31.57, 55.93]
Graph Temporal	42.52 ± 3.43 [38.26, 46.78]	42.31 ± 10.47 [29.30, 55.32]
Sage-Voronoi	44.24 ± 3.44 [39.96, 48.51]	42.97 ± 11.76 [28.36, 57.57]
GAIN	30.13 ± 4.93 [24.01, 36.24]	32.49 ± 7.06 [23.73, 41.26]
GRIN	**26.97** ± 0.46 [26.39, 27.54]	**29.85** ± 8.66 [19.10, 40.60]

Table S44: RMSE for sensor A023, October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	MCAR 10%	MCAR 30%	MCAR 50%	Synchronous $B = 144$
Linear	31.01 ± 2.90 [27.41, 34.62]	32.66 ± 1.96 [30.23, 35.10]	33.36 ± 0.60 [32.61, 34.10]	148.03 ± 39.90 [98.49, 197.57]
Median	133.32 ± 2.13 [130.67, 135.96]	135.44 ± 1.73 [133.29, 137.60]	132.83 ± 0.83 [131.79, 133.87]	129.88 ± 15.38 [110.78, 148.97]
FFill + BFill	37.14 ± 3.18 [33.20, 41.09]	38.61 ± 2.31 [35.74, 41.48]	41.34 ± 1.03 [40.06, 42.61]	146.10 ± 38.02 [98.90, 193.30]
Rolling mean	36.84 ± 3.02 [33.10, 40.59]	37.72 ± 1.30 [36.10, 39.34]	39.56 ± 0.63 [38.77, 40.34]	148.02 ± 39.88 [98.50, 197.53]
KNN (scaled)	33.73 ± 2.16 [31.04, 36.41]	33.83 ± 1.94 [31.42, 36.25]	34.72 ± 1.20 [33.24, 36.21]	118.27 ± 14.45 [100.32, 136.22]
KNN (unscaled)	38.36 ± 3.38 [34.17, 42.56]	38.03 ± 1.77 [35.82, 40.23]	38.20 ± 0.88 [37.10, 39.30]	118.27 ± 14.45 [100.32, 136.22]
Historical	65.14 ± 3.60 [60.68, 69.61]	65.48 ± 2.66 [62.18, 68.78]	65.08 ± 0.73 [64.17, 65.98]	**71.49** ± 23.31 [42.55, 100.44]
Graph Laplacian	44.52 ± 1.70 [42.42, 46.63]	45.34 ± 1.89 [42.99, 47.69]	45.90 ± 0.76 [44.96, 46.84]	118.27 ± 14.45 [100.32, 136.22]
Graph Temporal	38.39 ± 2.47 [35.33, 41.45]	40.48 ± 1.24 [38.94, 42.03]	41.98 ± 0.91 [40.84, 43.11]	141.58 ± 34.27 [99.03, 184.14]
Sage-Voronoi	31.35 ± 3.27 [27.30, 35.41]	32.52 ± 1.56 [30.59, 34.46]	34.98 ± 1.30 [33.36, 36.60]	163.35 ± 51.87 [98.95, 227.75]
GAIN	33.49 ± 2.65 [30.20, 36.78]	37.40 ± 2.29 [34.56, 40.24]	48.69 ± 1.07 [47.36, 50.01]	124.06 ± 14.27 [106.34, 141.77]
GRIN	**27.24** ± 2.71 [23.87, 30.61]	**28.33** ± 2.03 [25.80, 30.85]	**29.47** ± 1.05 [28.17, 30.77]	107.69 ± 24.63 [77.11, 138.27]

Table S45: MRE for sensor A023, October–December 2022 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 1$	Block $B = 2$	Block $B = 4$	Block $B = 8$
Linear	0.208 ± 0.122 [0.057, 0.359]	0.215 ± 0.163 [0.012, 0.417]	0.173 ± 0.061 [0.097, 0.249]	0.131 ± 0.108 [−0.003, 0.266]
Median	2.227 ± 1.950 [−0.194, 4.649]	1.613 ± 1.437 [−0.171, 3.397]	2.182 ± 2.185 [−0.531, 4.895]	0.882 ± 1.174 [−0.576, 2.341]
FFill + BFill	0.212 ± 0.176 [−0.006, 0.430]	0.195 ± 0.146 [0.013, 0.376]	0.205 ± 0.099 [0.082, 0.328]	0.239 ± 0.245 [−0.066, 0.543]
Rolling mean	0.290 ± 0.210 [0.029, 0.551]	0.172 ± 0.167 [−0.035, 0.379]	0.268 ± 0.108 [0.134, 0.403]	0.146 ± 0.122 [−0.005, 0.298]
KNN (scaled)	0.139 ± 0.157 [−0.056, 0.334]	0.148 ± 0.100 [0.024, 0.272]	0.152 ± 0.097 [0.031, 0.273]	0.110 ± 0.073 [0.020, 0.201]
KNN (unscaled)	0.202 ± 0.104 [0.073, 0.332]	0.196 ± 0.125 [0.040, 0.351]	0.172 ± 0.147 [−0.010, 0.355]	0.159 ± 0.130 [−0.003, 0.321]
Historical	0.305 ± 0.404 [−0.196, 0.807]	0.519 ± 0.492 [−0.093, 1.130]	0.293 ± 0.179 [0.071, 0.515]	0.287 ± 0.368 [−0.170, 0.743]
Graph Laplacian	0.446 ± 0.463 [−0.128, 1.021]	0.321 ± 0.284 [−0.031, 0.673]	0.419 ± 0.489 [−0.187, 1.026]	0.195 ± 0.231 [−0.092, 0.482]
Graph Temporal	0.408 ± 0.363 [−0.042, 0.859]	0.297 ± 0.227 [0.014, 0.579]	0.381 ± 0.431 [−0.154, 0.916]	0.174 ± 0.186 [−0.058, 0.405]
Sage-Voronoi	0.201 ± 0.201 [−0.049, 0.450]	0.174 ± 0.149 [−0.010, 0.359]	0.182 ± 0.100 [0.058, 0.306]	0.109 ± 0.065 [0.028, 0.190]
GAIN	0.175 ± 0.196 [−0.069, 0.418]	0.166 ± 0.118 [0.019, 0.313]	0.143 ± 0.072 [0.053, 0.233]	0.125 ± 0.059 [0.052, 0.199]
GRIN	**0.137** ± 0.129 [−0.023, 0.298]	**0.144** ± 0.134 [−0.022, 0.310]	**0.114** ± 0.049 [0.053, 0.174]	**0.102** ± 0.057 [0.032, 0.173]

Table S46: MRE for sensor A023, October–December 2022 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 16$	Block $B = 32$	Block $B = 64$
Linear	0.121 ± 0.013 [0.104, 0.137]	0.462 ± 0.541 [−0.210, 1.134]	0.517 ± 0.539 [−0.152, 1.186]
Median	0.436 ± 0.523 [−0.214, 1.085]	1.889 ± 2.543 [−1.268, 5.047]	1.200 ± 1.379 [−0.512, 2.912]
FFill + BFill	0.250 ± 0.298 [−0.121, 0.620]	0.988 ± 0.820 [−0.031, 2.006]	0.795 ± 0.831 [−0.237, 1.826]
Rolling mean	0.140 ± 0.106 [0.008, 0.272]	0.505 ± 0.535 [−0.159, 1.169]	0.506 ± 0.503 [−0.119, 1.131]
KNN (scaled)	0.080 ± 0.026 [0.048, 0.112]	0.162 ± 0.090 [0.050, 0.274]	0.162 ± 0.098 [0.041, 0.284]
KNN (unscaled)	0.103 ± 0.033 [0.062, 0.143]	0.191 ± 0.093 [0.075, 0.307]	0.201 ± 0.128 [0.043, 0.360]
Historical	0.087 ± 0.037 [0.041, 0.133]	0.408 ± 0.176 [0.189, 0.626]	0.522 ± 0.652 [−0.288, 1.332]
Graph Laplacian	0.151 ± 0.083 [0.048, 0.254]	0.456 ± 0.564 [−0.244, 1.157]	0.324 ± 0.291 [−0.037, 0.685]
Graph Temporal	0.146 ± 0.067 [0.062, 0.230]	0.446 ± 0.546 [−0.233, 1.124]	0.307 ± 0.260 [−0.015, 0.630]
Sage-Voronoi	0.168 ± 0.131 [0.006, 0.330]	0.281 ± 0.290 [−0.079, 0.640]	0.264 ± 0.170 [0.053, 0.475]
GAIN	0.100 ± 0.040 [0.051, 0.150]	0.166 ± 0.063 [0.088, 0.244]	**0.133** ± 0.062 [0.057, 0.210]
GRIN	**0.074** ± 0.028 [0.040, 0.109]	**0.134** ± 0.056 [0.064, 0.203]	0.140 ± 0.058 [0.068, 0.212]

Table S47: MRE for sensor A023, October–December 2022 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 128$	Block $B = 144$
Linear	1.073 ± 0.596 [0.333, 1.814]	1.489 ± 0.753 [0.554, 2.424]
Median	0.901 ± 0.395 [0.411, 1.392]	1.508 ± 0.339 [1.087, 1.930]
FFill + BFill	1.080 ± 0.533 [0.418, 1.743]	1.428 ± 0.736 [0.514, 2.342]
Rolling mean	1.072 ± 0.597 [0.332, 1.813]	1.488 ± 0.750 [0.556, 2.419]
KNN (scaled)	0.107 ± 0.022 [0.079, 0.134]	0.168 ± 0.029 [0.133, 0.204]
KNN (unscaled)	0.140 ± 0.032 [0.100, 0.180]	0.199 ± 0.041 [0.148, 0.250]
Historical	0.143 ± 0.062 [0.066, 0.220]	0.364 ± 0.279 [0.018, 0.710]
Graph Laplacian	0.243 ± 0.085 [0.137, 0.349]	0.384 ± 0.101 [0.259, 0.510]
Graph Temporal	0.232 ± 0.083 [0.130, 0.335]	0.358 ± 0.086 [0.251, 0.465]
Sage-Voronoi	0.166 ± 0.040 [0.116, 0.216]	0.238 ± 0.073 [0.147, 0.328]
GAIN	0.106 ± 0.023 [0.077, 0.134]	0.155 ± 0.023 [0.126, 0.183]
GRIN	**0.092** ± 0.013 [0.076, 0.109]	**0.145** ± 0.017 [0.123, 0.166]

Table S48: MRE for sensor A023, October–December 2022 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	MCAR 10%	MCAR 30%	MCAR 50%	Synchronous $B = 144$
Linear	0.131 ± 0.012 [0.117, 0.145]	0.136 ± 0.005 [0.130, 0.142]	0.135 ± 0.003 [0.131, 0.140]	0.895 ± 0.518 [0.251, 1.538]
Median	1.254 ± 0.034 [1.212, 1.296]	1.283 ± 0.041 [1.232, 1.335]	1.232 ± 0.027 [1.198, 1.266]	1.054 ± 0.426 [0.525, 1.582]
FFill + BFill	0.160 ± 0.015 [0.142, 0.178]	0.161 ± 0.005 [0.155, 0.167]	0.168 ± 0.006 [0.160, 0.176]	0.890 ± 0.492 [0.278, 1.501]
Rolling mean	0.158 ± 0.014 [0.140, 0.175]	0.160 ± 0.006 [0.152, 0.167]	0.163 ± 0.005 [0.157, 0.169]	0.896 ± 0.517 [0.254, 1.538]
KNN (scaled)	0.146 ± 0.009 [0.135, 0.158]	0.146 ± 0.003 [0.142, 0.150]	0.140 ± 0.003 [0.136, 0.144]	0.895 ± 0.354 [0.456, 1.335]
KNN (unscaled)	0.180 ± 0.014 [0.163, 0.198]	0.174 ± 0.003 [0.171, 0.177]	0.163 ± 0.004 [0.158, 0.167]	0.895 ± 0.354 [0.456, 1.335]
Historical	0.324 ± 0.041 [0.273, 0.375]	0.327 ± 0.016 [0.307, 0.347]	0.317 ± 0.009 [0.306, 0.328]	**0.374** ± 0.212 [0.112, 0.637]
Graph Laplacian	0.319 ± 0.009 [0.308, 0.331]	0.328 ± 0.010 [0.316, 0.340]	0.321 ± 0.010 [0.308, 0.334]	0.895 ± 0.354 [0.456, 1.335]
Graph Temporal	0.266 ± 0.014 [0.249, 0.283]	0.280 ± 0.007 [0.272, 0.288]	0.282 ± 0.008 [0.272, 0.293]	0.890 ± 0.474 [0.301, 1.478]
Sage-Voronoi	0.134 ± 0.016 [0.114, 0.154]	0.138 ± 0.006 [0.131, 0.145]	0.145 ± 0.005 [0.140, 0.151]	0.690 ± 0.369 [0.231, 1.149]
GAIN	0.139 ± 0.014 [0.122, 0.157]	0.190 ± 0.004 [0.186, 0.195]	0.313 ± 0.012 [0.299, 0.327]	0.992 ± 0.403 [0.492, 1.493]
GRIN	**0.111** ± 0.013 [0.095, 0.127]	**0.115** ± 0.004 [0.110, 0.121]	**0.116** ± 0.005 [0.109, 0.122]	0.842 ± 0.481 [0.244, 1.439]

Table S49: MAE for sensor A027, January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 1$	Block $B = 2$	Block $B = 4$	Block $B = 8$
Linear	36.90 ± 29.76 [−0.05, 73.85]	**24.20** ± 6.26 [16.43, 31.97]	26.34 ± 9.33 [14.76, 37.92]	28.88 ± 13.73 [11.83, 45.92]
Median	108.40 ± 55.79 [39.12, 177.68]	176.70 ± 89.13 [66.03, 287.37]	143.60 ± 68.41 [58.66, 228.54]	138.40 ± 70.16 [51.28, 225.52]
FFill + BFill	36.80 ± 45.89 [−20.18, 93.78]	34.10 ± 24.30 [3.93, 64.27]	48.40 ± 16.42 [28.01, 68.79]	65.80 ± 44.55 [10.49, 121.11]
Rolling mean	29.00 ± 40.14 [−20.84, 78.84]	48.40 ± 17.82 [26.27, 70.53]	59.62 ± 20.49 [34.17, 85.06]	44.60 ± 30.92 [6.21, 82.99]
KNN (scaled)	149.44 ± 81.67 [48.04, 250.84]	236.62 ± 113.13 [96.15, 377.09]	182.10 ± 127.10 [24.28, 339.92]	165.99 ± 91.72 [52.10, 279.87]
KNN (unscaled)	40.08 ± 39.37 [−8.80, 88.96]	32.08 ± 25.35 [0.60, 63.56]	34.36 ± 11.18 [20.48, 48.24]	31.85 ± 12.99 [15.72, 47.98]
Historical	31.48 ± 26.95 [−1.98, 64.94]	139.17 ± 78.89 [41.21, 237.12]	98.45 ± 60.37 [23.49, 173.41]	60.59 ± 32.58 [20.13, 101.05]
Graph Laplacian	85.13 ± 58.04 [13.07, 157.19]	85.38 ± 63.78 [6.19, 164.57]	81.41 ± 36.48 [36.11, 126.70]	64.88 ± 36.06 [20.10, 109.66]
Graph Temporal	61.79 ± 38.00 [14.60, 108.98]	133.32 ± 137.35 [−37.23, 303.86]	54.54 ± 33.47 [12.98, 96.09]	68.84 ± 50.37 [6.30, 131.39]
Sage-Voronoi	26.51 ± 32.63 [−14.00, 67.02]	27.90 ± 14.74 [9.59, 46.20]	38.77 ± 21.77 [11.73, 65.80]	34.61 ± 15.93 [14.83, 54.40]
GAIN	39.88 ± 23.44 [10.77, 68.99]	47.66 ± 33.55 [6.01, 89.31]	31.25 ± 11.94 [16.43, 46.07]	35.18 ± 14.65 [16.98, 53.37]
GRIN	**19.32** ± 18.21 [−3.29, 41.94]	28.89 ± 19.55 [4.62, 53.16]	**25.91** ± 13.07 [9.68, 42.14]	**27.49** ± 6.48 [19.45, 35.53]

Table S50: MAE for sensor A027, January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 16$	Block $B = 32$	Block $B = 64$
Linear	35.96 ± 23.03 [7.37, 64.55]	93.15 ± 57.76 [21.43, 164.87]	80.00 ± 23.17 [51.22, 108.77]
Median	130.84 ± 55.12 [62.39, 199.28]	128.14 ± 49.14 [67.12, 189.15]	146.73 ± 19.25 [122.83, 170.63]
FFill + BFill	54.59 ± 30.60 [16.60, 92.58]	191.38 ± 113.75 [50.14, 332.61]	213.74 ± 71.91 [124.45, 303.03]
Rolling mean	35.28 ± 23.17 [6.51, 64.05]	104.27 ± 65.50 [22.94, 185.60]	83.67 ± 25.37 [52.16, 115.17]
KNN (scaled)	101.38 ± 70.62 [13.69, 189.07]	127.07 ± 60.81 [51.56, 202.58]	162.68 ± 48.07 [102.99, 222.37]
KNN (unscaled)	30.11 ± 15.72 [10.60, 49.63]	34.86 ± 13.37 [18.26, 51.46]	35.89 ± 10.92 [22.33, 49.45]
Historical	29.43 ± 6.49 [21.37, 37.49]	93.26 ± 41.60 [41.61, 144.91]	57.38 ± 14.98 [38.78, 75.98]
Graph Laplacian	69.86 ± 25.20 [38.56, 101.15]	78.16 ± 8.74 [67.31, 89.02]	92.70 ± 16.67 [72.00, 113.39]
Graph Temporal	95.79 ± 78.06 [−1.14, 192.71]	124.26 ± 97.35 [3.38, 245.13]	105.24 ± 43.06 [51.78, 158.71]
Sage-Voronoi	44.51 ± 24.97 [13.50, 75.52]	58.25 ± 14.81 [39.87, 76.64]	52.44 ± 11.70 [37.91, 66.97]
GAIN	**23.79** ± 11.00 [10.14, 37.45]	**24.07** ± 9.11 [12.76, 35.38]	**24.12** ± 3.70 [19.53, 28.71]
GRIN	33.98 ± 29.59 [−2.76, 70.72]	55.19 ± 50.14 [−7.07, 117.45]	44.54 ± 26.34 [11.84, 77.25]

Table S51: MAE for sensor A027, January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 128$	Block $B = 144$
Linear	208.26 ± 78.89 [110.31, 306.22]	203.13 ± 49.62 [141.51, 264.74]
Median	147.52 ± 10.27 [134.77, 160.27]	148.14 ± 14.22 [130.49, 165.79]
FFill + BFill	227.00 ± 56.32 [157.06, 296.93]	204.96 ± 58.51 [132.31, 277.61]
Rolling mean	208.30 ± 78.59 [110.72, 305.89]	203.08 ± 49.92 [141.09, 265.06]
KNN (scaled)	164.57 ± 15.29 [145.59, 183.55]	148.46 ± 31.91 [108.84, 188.08]
KNN (unscaled)	37.44 ± 3.25 [33.40, 41.47]	26.87 ± 4.77 [20.95, 32.80]
Historical	52.21 ± 6.89 [43.65, 60.77]	57.47 ± 4.69 [51.65, 63.29]
Graph Laplacian	98.97 ± 13.68 [81.98, 115.96]	82.90 ± 14.34 [65.10, 100.70]
Graph Temporal	119.39 ± 60.94 [43.72, 195.06]	90.10 ± 17.24 [68.69, 111.51]
Sage-Voronoi	66.60 ± 10.10 [54.07, 79.14]	51.60 ± 7.48 [42.31, 60.89]
GAIN	**27.03** ± 1.71 [24.90, 29.15]	**24.24** ± 2.28 [21.41, 27.07]
GRIN	46.45 ± 32.45 [6.16, 86.74]	36.72 ± 13.75 [19.64, 53.80]

Table S52: MAE for sensor A027, January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	MCAR 10%	MCAR 30%	MCAR 50%	Synchronous $B = 144$
Linear	22.26 ± 1.78 [20.06, 24.47]	23.17 ± 0.81 [22.16, 24.17]	**22.98** ± 0.82 [21.96, 23.99]	154.13 ± 16.06 [134.19, 174.07]
Median	146.12 ± 6.35 [138.24, 154.00]	145.01 ± 1.76 [142.82, 147.19]	146.42 ± 1.09 [145.06, 147.77]	147.67 ± 16.56 [127.10, 168.23]
FFill + BFill	27.47 ± 3.40 [23.26, 31.69]	29.66 ± 1.01 [28.40, 30.91]	32.05 ± 2.25 [29.25, 34.85]	163.17 ± 25.71 [131.25, 195.09]
Rolling mean	30.40 ± 2.05 [27.86, 32.94]	30.89 ± 0.49 [30.29, 31.50]	31.00 ± 1.08 [29.66, 32.34]	154.86 ± 16.23 [134.71, 175.02]
KNN (scaled)	136.86 ± 2.86 [133.30, 140.41]	130.16 ± 5.39 [123.47, 136.85]	111.84 ± 3.09 [108.01, 115.68]	150.19 ± 19.41 [126.08, 174.29]
KNN (unscaled)	30.40 ± 2.89 [26.81, 33.98]	33.19 ± 0.98 [31.97, 34.40]	34.29 ± 0.93 [33.13, 35.44]	150.19 ± 19.41 [126.08, 174.29]
Historical	57.92 ± 4.11 [52.82, 63.02]	57.88 ± 1.80 [55.64, 60.11]	56.35 ± 0.99 [55.11, 57.58]	**57.54** ± 3.63 [53.04, 62.05]
Graph Laplacian	80.28 ± 3.80 [75.57, 85.00]	78.64 ± 1.50 [76.78, 80.51]	76.90 ± 2.03 [74.37, 79.42]	150.19 ± 19.41 [126.08, 174.29]
Graph Temporal	80.22 ± 9.91 [67.93, 92.52]	87.26 ± 6.19 [79.57, 94.95]	94.08 ± 6.15 [86.44, 101.71]	198.23 ± 63.63 [119.22, 277.24]
Sage-Voronoi	25.56 ± 2.23 [22.79, 28.32]	27.57 ± 1.09 [26.22, 28.92]	29.96 ± 1.74 [27.80, 32.13]	156.16 ± 37.34 [109.80, 202.53]
GAIN	26.06 ± 2.17 [23.36, 28.75]	34.48 ± 0.92 [33.34, 35.62]	48.33 ± 2.32 [45.45, 51.21]	143.79 ± 7.86 [134.03, 153.55]
GRIN	**20.05** ± 1.62 [18.03, 22.07]	**22.18** ± 1.09 [20.83, 23.53]	24.42 ± 3.10 [20.57, 28.27]	154.93 ± 23.91 [125.23, 184.62]

Table S53: RMSE for sensor A027, January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 1$	Block $B = 2$	Block $B = 4$	Block $B = 8$
Linear	36.90 ± 29.76 [−0.05, 73.85]	**26.81** ± 7.80 [17.13, 36.49]	**30.35** ± 10.35 [17.50, 43.20]	33.76 ± 14.90 [15.25, 52.26]
Median	108.40 ± 55.79 [39.12, 177.68]	177.07 ± 89.33 [66.15, 287.99]	147.83 ± 66.93 [64.73, 230.93]	145.78 ± 65.12 [64.92, 226.64]
FFill + BFill	36.80 ± 45.89 [−20.18, 93.78]	36.41 ± 24.25 [6.29, 66.52]	57.32 ± 18.06 [34.90, 79.74]	73.94 ± 49.31 [12.71, 135.17]
Rolling mean	29.00 ± 40.14 [−20.84, 78.84]	50.05 ± 17.92 [27.80, 72.30]	65.22 ± 21.38 [38.67, 91.77]	50.29 ± 32.16 [10.36, 90.21]
KNN (scaled)	149.44 ± 81.67 [48.04, 250.84]	238.51 ± 111.10 [100.56, 376.45]	186.54 ± 124.52 [31.93, 341.15]	175.87 ± 88.61 [65.85, 285.89]
KNN (unscaled)	40.08 ± 39.37 [−8.80, 88.96]	32.72 ± 25.16 [1.48, 63.96]	44.19 ± 11.70 [29.66, 58.72]	38.05 ± 15.70 [18.56, 57.54]
Historical	31.48 ± 26.95 [−1.98, 64.94]	139.64 ± 78.92 [41.64, 237.64]	105.50 ± 54.52 [37.81, 173.19]	68.77 ± 32.81 [28.03, 109.52]
Graph Laplacian	85.13 ± 58.04 [13.07, 157.19]	88.03 ± 64.52 [7.91, 168.14]	94.59 ± 39.90 [45.05, 144.12]	70.70 ± 38.46 [22.95, 118.46]
Graph Temporal	61.79 ± 38.00 [14.60, 108.98]	136.31 ± 135.63 [−32.10, 304.72]	59.81 ± 33.19 [18.60, 101.02]	77.94 ± 53.55 [11.45, 144.42]
Sage-Voronoi	26.51 ± 32.63 [−14.00, 67.02]	29.16 ± 15.59 [9.81, 48.51]	44.12 ± 24.40 [13.83, 74.42]	40.45 ± 17.37 [18.88, 62.01]
GAIN	39.88 ± 23.44 [10.77, 68.99]	49.33 ± 32.75 [8.67, 89.99]	38.16 ± 14.79 [19.79, 56.52]	42.94 ± 17.25 [21.53, 64.35]
GRIN	**19.32** ± 18.21 [−3.29, 41.94]	31.27 ± 18.72 [8.03, 54.51]	30.38 ± 14.02 [12.97, 47.78]	**32.38** ± 6.99 [23.70, 41.05]

Table S54: RMSE for sensor A027, January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 16$	Block $B = 32$	Block $B = 64$
Linear	44.33 ± 28.56 [8.87, 79.79]	106.68 ± 66.68 [23.89, 189.47]	98.93 ± 32.01 [59.19, 138.68]
Median	138.83 ± 51.70 [74.64, 203.02]	140.45 ± 45.84 [83.53, 197.37]	159.83 ± 17.61 [137.97, 181.69]
FFill + BFill	65.15 ± 36.74 [19.52, 110.77]	208.16 ± 118.66 [60.83, 355.49]	244.09 ± 69.76 [157.47, 330.70]
Rolling mean	42.79 ± 30.11 [5.40, 80.18]	117.86 ± 73.44 [26.67, 209.06]	102.64 ± 35.11 [59.04, 146.24]
KNN (scaled)	112.99 ± 73.71 [21.47, 204.51]	147.28 ± 67.81 [63.08, 231.47]	185.23 ± 39.78 [135.84, 234.62]
KNN (unscaled)	39.77 ± 21.72 [12.80, 66.73]	43.08 ± 14.83 [24.66, 61.50]	44.31 ± 11.76 [29.70, 58.91]
Historical	38.15 ± 15.44 [18.98, 57.31]	106.55 ± 47.68 [47.35, 165.75]	71.34 ± 19.44 [47.20, 95.49]
Graph Laplacian	76.52 ± 28.71 [40.88, 112.17]	91.04 ± 8.16 [80.91, 101.16]	104.18 ± 17.59 [82.35, 126.02]
Graph Temporal	110.03 ± 101.54 [−16.05, 236.11]	162.83 ± 131.88 [−0.92, 326.58]	140.75 ± 79.19 [42.42, 239.08]
Sage-Voronoi	53.60 ± 27.03 [20.03, 87.17]	67.72 ± 13.05 [51.51, 83.92]	62.23 ± 15.29 [43.24, 81.22]
GAIN	**31.70** ± 18.05 [9.29, 54.11]	**29.64** ± 11.28 [15.64, 43.65]	**30.82** ± 5.15 [24.42, 37.22]
GRIN	38.85 ± 31.49 [−0.26, 77.95]	63.20 ± 52.13 [−1.52, 127.92]	61.72 ± 45.33 [5.44, 118.01]

Table S55: RMSE for sensor A027, January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 128$	Block $B = 144$
Linear	249.50 ± 86.08 [142.62, 356.39]	238.85 ± 57.05 [168.01, 309.69]
Median	162.86 ± 10.56 [149.75, 175.97]	164.31 ± 10.95 [150.71, 177.91]
FFill + BFill	272.99 ± 45.27 [216.78, 329.21]	238.79 ± 68.29 [154.00, 323.58]
Rolling mean	249.55 ± 85.92 [142.86, 356.23]	238.84 ± 57.34 [167.64, 310.04]
KNN (scaled)	188.23 ± 15.25 [169.29, 207.17]	176.00 ± 37.61 [129.30, 222.69]
KNN (unscaled)	48.20 ± 4.53 [42.58, 53.83]	35.23 ± 6.02 [27.75, 42.71]
Historical	68.27 ± 9.04 [57.04, 79.50]	77.21 ± 7.57 [67.82, 86.61]
Graph Laplacian	109.77 ± 13.50 [93.00, 126.54]	96.55 ± 15.54 [77.26, 115.84]
Graph Temporal	151.15 ± 82.68 [48.49, 253.80]	120.52 ± 32.93 [79.63, 161.41]
Sage-Voronoi	80.44 ± 12.80 [64.55, 96.33]	64.33 ± 8.10 [54.27, 74.39]
GAIN	**34.70** ± 2.78 [31.25, 38.15]	**32.38** ± 2.82 [28.87, 35.88]
GRIN	61.33 ± 46.56 [3.52, 119.14]	48.23 ± 21.74 [21.23, 75.22]

Table S56: RMSE for sensor A027, January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	MCAR 10%	MCAR 30%	MCAR 50%	Synchronous $B = 144$
Linear	29.08 ± 2.47 [26.01, 32.14]	31.39 ± 0.78 [30.42, 32.37]	**31.21** ± 1.40 [29.47, 32.96]	196.01 ± 29.13 [159.84, 232.18]
Median	163.15 ± 5.12 [156.79, 169.51]	161.73 ± 1.00 [160.49, 162.97]	162.69 ± 1.02 [161.43, 163.95]	163.71 ± 12.85 [147.75, 179.66]
FFill + BFill	36.05 ± 3.87 [31.24, 40.85]	40.26 ± 1.86 [37.95, 42.58]	43.70 ± 3.99 [38.75, 48.66]	208.73 ± 37.68 [161.94, 255.52]
Rolling mean	40.39 ± 2.64 [37.11, 43.67]	41.60 ± 0.83 [40.57, 42.63]	41.48 ± 1.17 [40.03, 42.94]	196.58 ± 28.88 [160.72, 232.45]
KNN (scaled)	170.94 ± 2.63 [167.68, 174.21]	164.83 ± 5.36 [158.18, 171.49]	150.16 ± 2.87 [146.60, 153.72]	165.25 ± 17.24 [143.84, 186.65]
KNN (unscaled)	40.43 ± 2.95 [36.77, 44.09]	44.69 ± 1.66 [42.63, 46.75]	48.59 ± 1.43 [46.81, 50.37]	165.25 ± 17.24 [143.84, 186.65]
Historical	78.80 ± 4.61 [73.08, 84.52]	77.07 ± 3.02 [73.31, 80.82]	75.25 ± 1.32 [73.62, 76.89]	**75.51** ± 8.10 [65.45, 85.57]
Graph Laplacian	97.07 ± 3.14 [93.17, 100.96]	98.69 ± 2.96 [95.02, 102.36]	101.39 ± 2.62 [98.14, 104.64]	165.25 ± 17.24 [143.84, 186.65]
Graph Temporal	123.35 ± 15.36 [104.28, 142.41]	133.59 ± 14.87 [115.12, 152.06]	137.97 ± 9.71 [125.91, 150.03]	243.72 ± 72.89 [153.22, 334.22]
Sage-Voronoi	33.40 ± 2.85 [29.86, 36.94]	36.82 ± 2.65 [33.53, 40.11]	39.33 ± 2.91 [35.71, 42.94]	199.33 ± 48.74 [138.81, 259.84]
GAIN	36.28 ± 3.30 [32.18, 40.38]	44.56 ± 1.66 [42.50, 46.62]	58.88 ± 3.19 [54.91, 62.84]	177.46 ± 10.70 [164.17, 190.75]
GRIN	**26.99** ± 2.26 [24.19, 29.79]	**29.80** ± 1.68 [27.72, 31.89]	33.49 ± 5.37 [26.82, 40.17]	175.21 ± 22.86 [146.82, 203.59]

Table S57: MRE for sensor A027, January 2023 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 1$	Block $B = 2$	Block $B = 4$	Block $B = 8$
Linear	0.170 ± 0.146 [−0.011, 0.352]	**0.059** ± 0.017 [0.037, 0.080]	**0.097** ± 0.051 [0.034, 0.161]	**0.104** ± 0.035 [0.061, 0.147]
Median	0.656 ± 0.894 [−0.454, 1.766]	0.393 ± 0.123 [0.241, 0.546]	0.527 ± 0.386 [0.047, 1.007]	0.998 ± 1.555 [−0.933, 2.929]
FFill + BFill	0.125 ± 0.137 [−0.046, 0.295]	0.080 ± 0.062 [0.004, 0.157]	0.204 ± 0.204 [−0.050, 0.457]	0.234 ± 0.107 [0.101, 0.366]
Rolling mean	0.106 ± 0.130 [−0.055, 0.267]	0.113 ± 0.031 [0.074, 0.152]	0.257 ± 0.235 [−0.034, 0.549]	0.161 ± 0.078 [0.064, 0.258]
KNN (scaled)	0.495 ± 0.099 [0.371, 0.618]	0.537 ± 0.166 [0.332, 0.743]	0.471 ± 0.209 [0.211, 0.731]	0.649 ± 0.381 [0.175, 1.122]
KNN (unscaled)	0.162 ± 0.135 [−0.006, 0.329]	0.070 ± 0.040 [0.020, 0.120]	0.130 ± 0.070 [0.043, 0.217]	0.122 ± 0.052 [0.057, 0.187]
Historical	0.095 ± 0.065 [0.014, 0.176]	0.306 ± 0.124 [0.152, 0.461]	0.285 ± 0.096 [0.166, 0.404]	0.236 ± 0.137 [0.067, 0.406]
Graph Laplacian	0.367 ± 0.267 [0.036, 0.698]	0.189 ± 0.112 [0.049, 0.328]	0.280 ± 0.096 [0.161, 0.399]	0.434 ± 0.597 [−0.307, 1.174]
Graph Temporal	0.215 ± 0.089 [0.105, 0.325]	0.300 ± 0.322 [−0.100, 0.699]	0.168 ± 0.046 [0.112, 0.225]	0.528 ± 0.826 [−0.498, 1.554]
Sage-Voronoi	0.112 ± 0.114 [−0.030, 0.254]	0.064 ± 0.030 [0.027, 0.101]	0.139 ± 0.096 [0.019, 0.258]	0.136 ± 0.059 [0.063, 0.209]
GAIN	0.175 ± 0.122 [0.024, 0.326]	0.110 ± 0.068 [0.026, 0.195]	0.109 ± 0.056 [0.040, 0.179]	0.126 ± 0.039 [0.077, 0.174]
GRIN	**0.082** ± 0.069 [−0.003, 0.168]	0.067 ± 0.046 [0.010, 0.125]	0.098 ± 0.056 [0.029, 0.168]	0.162 ± 0.181 [−0.063, 0.387]

Table S58: MRE for sensor A027, January 2023 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 16$	Block $B = 32$	Block $B = 64$
Linear	0.183 ± 0.048 [0.123, 0.243]	0.453 ± 0.350 [0.018, 0.888]	0.965 ± 1.430 [−0.811, 2.740]
Median	2.556 ± 4.359 [−2.857, 7.969]	1.176 ± 1.698 [−0.932, 3.285]	2.158 ± 2.686 [−1.178, 5.493]
FFill + BFill	0.431 ± 0.245 [0.126, 0.736]	0.694 ± 0.230 [0.409, 0.979]	1.894 ± 1.499 [0.032, 3.755]
Rolling mean	0.176 ± 0.034 [0.135, 0.218]	0.466 ± 0.259 [0.144, 0.787]	0.916 ± 1.253 [−0.640, 2.471]
KNN (scaled)	0.896 ± 1.064 [−0.425, 2.217]	0.620 ± 0.372 [0.157, 1.082]	1.307 ± 1.498 [−0.552, 3.167]
KNN (unscaled)	0.196 ± 0.102 [0.069, 0.323]	0.209 ± 0.104 [0.079, 0.338]	0.453 ± 0.668 [−0.377, 1.283]
Historical	0.374 ± 0.492 [−0.237, 0.984]	0.744 ± 1.096 [−0.617, 2.105]	0.338 ± 0.261 [0.013, 0.662]
Graph Laplacian	0.799 ± 1.021 [−0.469, 2.067]	0.679 ± 0.847 [−0.374, 1.731]	1.023 ± 1.017 [−0.241, 2.286]
Graph Temporal	0.849 ± 0.881 [−0.245, 1.944]	1.673 ± 2.905 [−1.934, 5.281]	1.249 ± 1.191 [−0.230, 2.727]
Sage-Voronoi	0.412 ± 0.283 [0.061, 0.764]	0.532 ± 0.694 [−0.330, 1.394]	0.840 ± 0.867 [−0.237, 1.917]
GAIN	**0.159** ± 0.103 [0.031, 0.288]	**0.126** ± 0.061 [0.050, 0.201]	**0.166** ± 0.143 [−0.012, 0.344]
GRIN	0.248 ± 0.209 [−0.012, 0.508]	0.772 ± 1.414 [−0.983, 2.528]	0.520 ± 0.679 [−0.322, 1.363]

Table S59: MRE for sensor A027, January 2023 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 128$	Block $B = 144$
Linear	3.428 ± 2.936 [−0.217, 7.074]	1.222 ± 0.666 [0.395, 2.050]
Median	2.604 ± 1.339 [0.941, 4.266]	2.741 ± 0.142 [2.565, 2.917]
FFill + BFill	4.137 ± 3.071 [0.324, 7.950]	1.291 ± 0.617 [0.524, 2.058]
Rolling mean	3.429 ± 2.931 [−0.210, 7.069]	1.221 ± 0.668 [0.392, 2.050]
KNN (scaled)	1.605 ± 0.766 [0.653, 2.556]	1.146 ± 0.091 [1.034, 1.259]
KNN (unscaled)	0.527 ± 0.334 [0.112, 0.943]	0.280 ± 0.089 [0.169, 0.391]
Historical	0.356 ± 0.126 [0.199, 0.513]	0.479 ± 0.249 [0.169, 0.789]
Graph Laplacian	1.282 ± 0.657 [0.467, 2.098]	0.995 ± 0.102 [0.868, 1.121]
Graph Temporal	1.631 ± 1.811 [−0.617, 3.880]	1.054 ± 0.438 [0.510, 1.597]
Sage-Voronoi	1.364 ± 0.963 [0.169, 2.560]	0.734 ± 0.354 [0.295, 1.173]
GAIN	**0.278** ± 0.139 [0.106, 0.450]	**0.204** ± 0.037 [0.159, 0.250]
GRIN	0.930 ± 1.487 [−0.916, 2.776]	0.420 ± 0.311 [0.034, 0.806]

Table S60: MRE for sensor A027, January 2023 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	MCAR 10%	MCAR 30%	MCAR 50%	Synchronous $B = 144$
Linear	**0.170** ± 0.023 [0.141, 0.199]	**0.173** ± 0.015 [0.154, 0.192]	**0.167** ± 0.019 [0.143, 0.190]	4.371 ± 1.179 [2.908, 5.835]
Median	2.888 ± 0.675 [2.050, 3.726]	2.666 ± 0.149 [2.481, 2.850]	2.679 ± 0.225 [2.400, 2.959]	3.030 ± 0.245 [2.726, 3.335]
FFill + BFill	0.198 ± 0.039 [0.149, 0.246]	0.210 ± 0.025 [0.179, 0.240]	0.230 ± 0.036 [0.185, 0.274]	4.084 ± 2.135 [1.433, 6.735]
Rolling mean	0.211 ± 0.044 [0.156, 0.266]	0.217 ± 0.009 [0.206, 0.228]	0.224 ± 0.030 [0.187, 0.262]	4.373 ± 1.183 [2.904, 5.842]
KNN (scaled)	1.139 ± 0.200 [0.891, 1.388]	1.082 ± 0.124 [0.928, 1.236]	1.037 ± 0.083 [0.934, 1.140]	2.866 ± 0.236 [2.573, 3.158]
KNN (unscaled)	0.371 ± 0.091 [0.258, 0.485]	0.382 ± 0.024 [0.353, 0.412]	0.450 ± 0.038 [0.403, 0.497]	2.866 ± 0.236 [2.573, 3.158]
Historical	0.472 ± 0.082 [0.370, 0.574]	0.466 ± 0.013 [0.449, 0.482]	0.452 ± 0.044 [0.397, 0.507]	**0.496** ± 0.189 [0.261, 0.731]
Graph Laplacian	1.126 ± 0.182 [0.899, 1.352]	1.028 ± 0.071 [0.940, 1.115]	0.995 ± 0.084 [0.890, 1.099]	2.866 ± 0.236 [2.573, 3.158]
Graph Temporal	1.165 ± 0.263 [0.838, 1.491]	1.260 ± 0.199 [1.012, 1.507]	1.238 ± 0.159 [1.040, 1.436]	5.431 ± 1.843 [3.142, 7.720]
Sage-Voronoi	0.222 ± 0.049 [0.161, 0.282]	0.238 ± 0.022 [0.211, 0.265]	0.273 ± 0.056 [0.203, 0.343]	3.460 ± 1.981 [1.000, 5.920]
GAIN	0.216 ± 0.022 [0.188, 0.244]	0.400 ± 0.061 [0.325, 0.476]	0.766 ± 0.060 [0.691, 0.841]	4.017 ± 0.291 [3.655, 4.378]
GRIN	0.181 ± 0.049 [0.120, 0.241]	0.222 ± 0.025 [0.190, 0.253]	0.267 ± 0.087 [0.159, 0.375]	2.257 ± 0.332 [1.844, 2.669]

Table S61: MAE for sensor A027, October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 1$	Block $B = 2$	Block $B = 4$	Block $B = 8$
Linear	29.80 ± 21.88 [2.64, 56.96]	12.87 ± 4.27 [7.56, 18.17]	27.03 ± 9.60 [15.11, 38.95]	25.30 ± 11.29 [11.29, 39.32]
Median	134.30 ± 86.02 [27.49, 241.11]	175.20 ± 53.00 [109.40, 241.00]	119.50 ± 115.99 [−24.51, 263.51]	131.32 ± 64.61 [51.10, 211.55]
FFill + BFill	31.20 ± 29.99 [−6.04, 68.44]	20.50 ± 19.04 [−3.14, 44.14]	32.25 ± 9.68 [20.23, 44.27]	37.67 ± 17.21 [16.30, 59.05]
Rolling mean	**15.20** ± 21.39 [−11.36, 41.76]	37.72 ± 31.22 [−1.04, 76.48]	36.00 ± 13.37 [19.40, 52.60]	30.87 ± 9.92 [18.55, 43.19]
KNN (scaled)	55.58 ± 30.94 [17.16, 94.00]	15.56 ± 10.43 [2.60, 28.52]	20.36 ± 8.96 [9.24, 31.49]	22.97 ± 10.30 [10.19, 35.75]
KNN (unscaled)	38.92 ± 30.67 [0.84, 77.00]	16.58 ± 14.40 [−1.30, 34.46]	29.36 ± 10.46 [16.37, 42.35]	23.86 ± 9.86 [11.61, 36.11]
Historical	40.26 ± 26.94 [6.81, 73.71]	24.96 ± 10.44 [12.00, 37.92]	31.31 ± 32.59 [−9.15, 71.77]	34.18 ± 18.57 [11.13, 57.24]
Graph Laplacian	41.96 ± 32.67 [1.39, 82.53]	24.20 ± 18.50 [1.23, 47.17]	45.10 ± 12.15 [30.01, 60.19]	35.69 ± 14.41 [17.81, 53.58]
Graph Temporal	49.44 ± 29.39 [12.95, 85.94]	23.55 ± 17.11 [2.30, 44.80]	44.02 ± 13.07 [27.79, 60.26]	56.58 ± 14.65 [38.39, 74.77]
Sage-Voronoi	19.32 ± 27.22 [−14.48, 53.12]	22.90 ± 16.33 [2.63, 43.18]	16.87 ± 8.95 [5.76, 27.99]	25.74 ± 10.21 [13.07, 38.42]
GAIN	40.31 ± 40.83 [−10.38, 91.01]	20.06 ± 12.08 [5.06, 35.06]	29.14 ± 18.95 [5.61, 52.67]	23.22 ± 6.60 [15.03, 31.41]
GRIN	25.17 ± 18.90 [1.70, 48.63]	**8.72** ± 4.77 [2.80, 14.65]	**13.91** ± 7.79 [4.24, 23.58]	**21.70** ± 8.43 [11.23, 32.17]

Table S62: MAE for sensor A027, October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 16$	Block $B = 32$	Block $B = 64$
Linear	27.11 ± 10.93 [13.53, 40.69]	34.28 ± 16.69 [13.56, 55.00]	85.91 ± 60.05 [11.35, 160.47]
Median	167.45 ± 61.40 [91.21, 243.69]	200.53 ± 56.11 [130.87, 270.20]	133.16 ± 68.01 [48.72, 217.60]
FFill + BFill	60.08 ± 32.36 [19.90, 100.25]	91.77 ± 46.52 [34.01, 149.53]	159.95 ± 40.23 [110.00, 209.90]
Rolling mean	27.56 ± 11.32 [13.50, 41.62]	40.70 ± 12.90 [24.68, 56.72]	85.80 ± 57.42 [14.51, 157.10]
KNN (scaled)	19.07 ± 3.27 [15.01, 23.14]	24.23 ± 15.08 [5.51, 42.95]	26.10 ± 4.26 [20.82, 31.39]
KNN (unscaled)	21.36 ± 4.76 [15.45, 27.26]	25.83 ± 9.43 [14.12, 37.54]	32.38 ± 2.81 [28.90, 35.87]
Historical	76.20 ± 102.75 [−51.38, 203.79]	28.78 ± 12.27 [13.54, 44.01]	48.99 ± 25.68 [17.10, 80.87]
Graph Laplacian	29.18 ± 13.21 [12.78, 45.59]	42.52 ± 20.18 [17.46, 67.57]	44.44 ± 10.65 [31.22, 57.66]
Graph Temporal	38.14 ± 17.20 [16.78, 59.49]	42.71 ± 10.31 [29.91, 55.51]	52.33 ± 4.72 [46.47, 58.19]
Sage-Voronoi	22.66 ± 4.92 [16.56, 28.77]	22.88 ± 5.89 [15.56, 30.20]	40.36 ± 13.85 [23.16, 57.55]
GAIN	30.35 ± 12.52 [14.80, 45.89]	22.87 ± 7.12 [14.03, 31.71]	30.65 ± 14.02 [13.24, 48.06]
GRIN	**15.97** ± 4.61 [10.26, 21.69]	**14.81** ± 6.30 [6.99, 22.63]	**25.31** ± 7.76 [15.67, 34.95]

Table S63: MAE for sensor A027, October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 128$	Block $B = 144$
Linear	121.38 ± 14.72 [103.10, 139.66]	139.12 ± 10.89 [125.60, 152.64]
Median	134.35 ± 10.90 [120.82, 147.88]	143.14 ± 8.25 [132.91, 153.38]
FFill + BFill	134.49 ± 16.46 [114.05, 154.93]	137.51 ± 17.52 [115.75, 159.26]
Rolling mean	120.90 ± 14.67 [102.69, 139.11]	139.23 ± 10.82 [125.79, 152.67]
KNN (scaled)	**25.84** ± 2.60 [22.61, 29.07]	**24.54** ± 3.10 [20.69, 28.39]
KNN (unscaled)	30.27 ± 3.81 [25.54, 34.99]	29.66 ± 3.25 [25.62, 33.69]
Historical	48.95 ± 9.61 [37.01, 60.89]	54.64 ± 27.10 [20.99, 88.29]
Graph Laplacian	39.45 ± 4.04 [34.42, 44.47]	40.72 ± 7.43 [31.49, 49.95]
Graph Temporal	54.07 ± 12.61 [38.42, 69.73]	54.26 ± 12.06 [39.28, 69.23]
Sage-Voronoi	39.45 ± 11.89 [24.68, 54.21]	35.46 ± 5.17 [29.04, 41.88]
GAIN	27.56 ± 4.89 [21.48, 33.64]	27.02 ± 2.63 [23.75, 30.29]
GRIN	31.64 ± 8.88 [20.62, 42.66]	31.17 ± 18.38 [8.35, 54.00]

Table S64: MAE for sensor A027, October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	MCAR 10%	MCAR 30%	MCAR 50%	Synchronous $B = 144$
Linear	22.89 ± 1.11 [21.51, 24.27]	23.10 ± 0.31 [22.72, 23.48]	23.69 ± 0.51 [23.06, 24.32]	146.03 ± 25.85 [113.93, 178.13]
Median	135.48 ± 1.51 [133.61, 137.36]	137.23 ± 1.66 [135.16, 139.29]	137.24 ± 2.41 [134.25, 140.23]	136.19 ± 10.56 [123.08, 149.30]
FFill + BFill	27.77 ± 1.45 [25.96, 29.57]	28.87 ± 1.22 [27.36, 30.38]	31.16 ± 0.69 [30.30, 32.02]	143.93 ± 24.08 [114.03, 173.83]
Rolling mean	28.71 ± 1.42 [26.95, 30.47]	28.99 ± 0.85 [27.94, 30.05]	30.18 ± 0.71 [29.30, 31.05]	146.04 ± 25.40 [114.51, 177.58]
KNN (scaled)	24.70 ± 1.43 [22.93, 26.47]	24.87 ± 0.75 [23.94, 25.80]	25.11 ± 0.44 [24.57, 25.65]	129.34 ± 16.24 [109.18, 149.51]
KNN (unscaled)	28.40 ± 1.27 [26.83, 29.97]	27.80 ± 0.90 [26.68, 28.92]	27.61 ± 0.19 [27.37, 27.84]	129.34 ± 16.24 [109.18, 149.51]
Historical	54.04 ± 1.94 [51.62, 56.45]	50.60 ± 1.12 [49.21, 51.99]	53.32 ± 1.53 [51.42, 55.22]	**61.57** ± 21.84 [34.45, 88.69]
Graph Laplacian	37.80 ± 5.59 [30.87, 44.74]	37.82 ± 3.26 [33.77, 41.87]	37.49 ± 3.16 [33.57, 41.42]	129.34 ± 16.24 [109.18, 149.51]
Graph Temporal	39.19 ± 1.64 [37.16, 41.23]	40.38 ± 0.63 [39.60, 41.15]	42.39 ± 0.66 [41.57, 43.21]	145.32 ± 27.82 [110.78, 179.87]
Sage-Voronoi	23.12 ± 1.40 [21.38, 24.86]	23.06 ± 0.47 [22.48, 23.65]	24.40 ± 0.36 [23.95, 24.85]	162.10 ± 58.68 [89.23, 234.96]
GAIN	25.68 ± 2.05 [23.14, 28.22]	29.73 ± 2.51 [26.61, 32.85]	47.18 ± 4.89 [41.11, 53.25]	179.17 ± 29.69 [142.31, 216.04]
GRIN	**17.49** ± 1.38 [15.78, 19.20]	**18.72** ± 0.49 [18.11, 19.33]	**21.26** ± 0.27 [20.93, 21.59]	115.51 ± 19.32 [91.53, 139.50]

Table S65: RMSE for sensor A027, October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 1$	Block $B = 2$	Block $B = 4$	Block $B = 8$
Linear	29.80 ± 21.88 [2.64, 56.96]	15.53 ± 6.11 [7.94, 23.11]	30.19 ± 12.69 [14.43, 45.95]	29.59 ± 13.67 [12.62, 46.56]
Median	134.30 ± 86.02 [27.49, 241.11]	175.36 ± 53.15 [109.36, 241.36]	121.71 ± 114.60 [−20.58, 264.01]	134.84 ± 62.93 [56.70, 212.99]
FFill + BFill	31.20 ± 29.99 [−6.04, 68.44]	22.54 ± 18.40 [−0.31, 45.39]	36.64 ± 9.94 [24.30, 48.98]	43.39 ± 16.93 [22.37, 64.41]
Rolling mean	**15.20** ± 21.39 [−11.36, 41.76]	38.96 ± 30.56 [1.02, 76.90]	40.22 ± 13.05 [24.01, 56.43]	36.76 ± 11.71 [22.22, 51.31]
KNN (scaled)	55.58 ± 30.94 [17.16, 94.00]	17.58 ± 11.69 [3.06, 32.09]	24.82 ± 12.15 [9.73, 39.90]	**27.83** ± 12.68 [12.08, 43.58]
KNN (unscaled)	38.92 ± 30.67 [0.84, 77.00]	19.17 ± 15.35 [0.11, 38.24]	35.78 ± 13.72 [18.75, 52.82]	30.88 ± 14.58 [12.77, 48.99]
Historical	40.26 ± 26.94 [6.81, 73.71]	26.93 ± 10.80 [13.52, 40.33]	35.18 ± 33.41 [−6.31, 76.67]	38.48 ± 18.21 [15.87, 61.10]
Graph Laplacian	41.96 ± 32.67 [1.39, 82.53]	25.67 ± 19.51 [1.45, 49.90]	46.05 ± 12.14 [30.98, 61.12]	41.61 ± 16.27 [21.41, 61.82]
Graph Temporal	49.44 ± 29.39 [12.95, 85.94]	24.36 ± 17.73 [2.36, 46.37]	44.86 ± 12.70 [29.08, 60.63]	62.86 ± 13.81 [45.72, 80.01]
Sage-Voronoi	19.32 ± 27.22 [−14.48, 53.12]	24.92 ± 15.55 [5.61, 44.23]	20.32 ± 9.88 [8.05, 32.58]	30.90 ± 12.30 [15.62, 46.17]
GAIN	40.31 ± 40.83 [−10.38, 91.01]	21.90 ± 12.54 [6.33, 37.48]	33.34 ± 19.07 [9.66, 57.03]	29.01 ± 9.51 [17.20, 40.82]
GRIN	25.17 ± 18.90 [1.70, 48.63]	**9.37** ± 4.60 [3.66, 15.08]	**16.55** ± 9.31 [5.00, 28.11]	28.39 ± 11.46 [14.17, 42.62]

Table S66: RMSE for sensor A027, October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 16$	Block $B = 32$	Block $B = 64$
Linear	31.50 ± 11.51 [17.21, 45.78]	39.14 ± 16.19 [19.03, 59.25]	101.71 ± 65.40 [20.50, 182.92]
Median	171.98 ± 60.32 [97.09, 246.88]	206.51 ± 54.52 [138.82, 274.21]	152.69 ± 71.75 [63.60, 241.78]
FFill + BFill	69.89 ± 31.80 [30.40, 109.37]	101.01 ± 50.73 [38.03, 164.00]	190.86 ± 52.48 [125.69, 256.02]
Rolling mean	33.64 ± 12.07 [18.65, 48.62]	45.92 ± 12.70 [30.15, 61.69]	101.07 ± 63.38 [22.38, 179.77]
KNN (scaled)	23.75 ± 4.27 [18.44, 29.05]	31.66 ± 19.51 [7.43, 55.89]	40.47 ± 10.78 [27.09, 53.85]
KNN (unscaled)	25.91 ± 5.63 [18.92, 32.90]	32.45 ± 11.47 [18.21, 46.68]	46.96 ± 8.78 [36.06, 57.86]
Historical	80.46 ± 101.78 [−45.92, 206.84]	33.98 ± 14.11 [16.46, 51.50]	62.21 ± 25.44 [30.62, 93.80]
Graph Laplacian	33.74 ± 13.26 [17.27, 50.21]	47.04 ± 18.76 [23.75, 70.33]	54.51 ± 7.96 [44.63, 64.40]
Graph Temporal	42.74 ± 16.75 [21.95, 63.54]	47.39 ± 9.30 [35.85, 58.93]	62.02 ± 5.40 [55.31, 68.72]
Sage-Voronoi	27.88 ± 6.69 [19.57, 36.19]	27.32 ± 6.88 [18.78, 35.87]	56.32 ± 13.21 [39.92, 72.72]
GAIN	34.91 ± 12.39 [19.52, 50.30]	28.61 ± 8.15 [18.49, 38.72]	42.30 ± 17.67 [20.36, 64.24]
GRIN	**19.66** ± 6.33 [11.81, 27.52]	**19.06** ± 7.20 [10.12, 28.00]	**38.77** ± 13.92 [21.48, 56.05]

Table S67: RMSE for sensor A027, October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 128$	Block $B = 144$
Linear	142.53 ± 23.56 [113.27, 171.79]	171.31 ± 20.85 [145.42, 197.19]
Median	163.70 ± 3.34 [159.55, 167.85]	169.09 ± 7.75 [159.47, 178.71]
FFill + BFill	151.59 ± 12.99 [135.46, 167.73]	169.46 ± 28.38 [134.22, 204.70]
Rolling mean	141.93 ± 23.48 [112.78, 171.08]	171.28 ± 20.88 [145.36, 197.21]
KNN (scaled)	**36.51** ± 8.05 [26.52, 46.50]	**32.90** ± 4.06 [27.86, 37.93]
KNN (unscaled)	41.39 ± 8.28 [31.12, 51.67]	38.84 ± 3.45 [34.56, 43.12]
Historical	66.53 ± 14.66 [48.33, 84.73]	72.35 ± 33.95 [30.20, 114.50]
Graph Laplacian	49.03 ± 4.82 [43.05, 55.01]	48.05 ± 7.33 [38.95, 57.15]
Graph Temporal	66.83 ± 16.92 [45.82, 87.84]	63.69 ± 17.60 [41.83, 85.55]
Sage-Voronoi	50.17 ± 12.55 [34.59, 65.75]	46.89 ± 7.96 [37.01, 56.78]
GAIN	38.28 ± 9.22 [26.83, 49.73]	35.20 ± 3.09 [31.37, 39.04]
GRIN	44.95 ± 11.31 [30.91, 58.98]	42.09 ± 23.23 [13.24, 70.94]

Table S68: RMSE for sensor A027, October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	MCAR 10%	MCAR 30%	MCAR 50%	Synchronous $B = 144$
Linear	31.82 ± 2.36 [28.89, 34.75]	32.69 ± 0.90 [31.57, 33.81]	33.72 ± 0.77 [32.77, 34.68]	179.75 ± 37.36 [133.35, 226.14]
Median	162.30 ± 1.99 [159.82, 164.77]	163.87 ± 1.33 [162.22, 165.52]	164.06 ± 2.19 [161.34, 166.78]	162.68 ± 11.16 [148.83, 176.54]
FFill + BFill	39.73 ± 2.55 [36.57, 42.89]	41.27 ± 2.26 [38.46, 44.08]	44.55 ± 1.32 [42.91, 46.20]	177.11 ± 36.14 [132.24, 221.99]
Rolling mean	39.35 ± 2.26 [36.53, 42.16]	40.26 ± 1.73 [38.11, 42.42]	42.02 ± 0.83 [40.99, 43.06]	179.70 ± 37.07 [133.67, 225.72]
KNN (scaled)	34.49 ± 3.48 [30.17, 38.81]	35.21 ± 1.43 [33.44, 36.98]	36.14 ± 0.91 [35.01, 37.26]	147.05 ± 13.97 [129.70, 164.39]
KNN (unscaled)	39.04 ± 2.86 [35.49, 42.59]	38.56 ± 1.28 [36.97, 40.15]	38.82 ± 0.58 [38.10, 39.53]	147.05 ± 13.97 [129.70, 164.39]
Historical	78.04 ± 2.61 [74.80, 81.27]	72.87 ± 1.77 [70.67, 75.07]	76.57 ± 2.10 [73.96, 79.18]	**83.70** ± 27.63 [49.40, 118.01]
Graph Laplacian	46.84 ± 6.55 [38.70, 54.97]	47.36 ± 4.01 [42.38, 52.34]	47.73 ± 3.87 [42.93, 52.54]	147.05 ± 13.97 [129.70, 164.39]
Graph Temporal	47.97 ± 2.51 [44.85, 51.09]	49.90 ± 0.55 [49.22, 50.59]	52.74 ± 0.73 [51.83, 53.65]	173.23 ± 40.34 [123.13, 223.32]
Sage-Voronoi	32.68 ± 2.83 [29.16, 36.20]	32.38 ± 1.28 [30.79, 33.97]	34.44 ± 0.53 [33.77, 35.10]	204.61 ± 64.41 [124.63, 284.59]
GAIN	36.13 ± 3.60 [31.66, 40.59]	39.07 ± 2.77 [35.62, 42.51]	56.56 ± 5.48 [49.76, 63.36]	220.70 ± 19.73 [196.20, 245.19]
GRIN	**25.92** ± 3.67 [21.37, 30.47]	**28.10** ± 1.71 [25.98, 30.22]	**31.98** ± 0.69 [31.13, 32.83]	134.51 ± 24.83 [103.68, 165.35]

Table S69: MRE for sensor A027, October–December 2022 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 1$	Block $B = 2$	Block $B = 4$	Block $B = 8$
Linear	0.112 ± 0.081 [0.012, 0.212]	0.127 ± 0.117 [−0.018, 0.273]	0.266 ± 0.294 [−0.099, 0.632]	0.095 ± 0.057 [0.024, 0.166]
Median	1.284 ± 2.341 [−1.622, 4.191]	1.639 ± 1.461 [−0.175, 3.454]	1.909 ± 2.568 [−1.280, 5.097]	1.021 ± 1.737 [−1.135, 3.177]
FFill + BFill	0.094 ± 0.058 [0.021, 0.166]	0.184 ± 0.162 [−0.018, 0.385]	0.235 ± 0.152 [0.047, 0.423]	0.223 ± 0.298 [−0.146, 0.593]
Rolling mean	0.063 ± 0.072 [−0.027, 0.152]	0.315 ± 0.254 [−0.001, 0.630]	0.248 ± 0.169 [0.038, 0.458]	0.148 ± 0.154 [−0.044, 0.339]
KNN (scaled)	0.197 ± 0.101 [0.072, 0.322]	0.123 ± 0.067 [0.040, 0.207]	0.142 ± 0.107 [0.009, 0.274]	0.088 ± 0.061 [0.012, 0.163]
KNN (unscaled)	0.100 ± 0.066 [0.019, 0.182]	0.115 ± 0.095 [−0.003, 0.233]	0.205 ± 0.112 [0.066, 0.344]	0.089 ± 0.060 [0.014, 0.163]
Historical	0.297 ± 0.473 [−0.291, 0.884]	0.210 ± 0.150 [0.023, 0.396]	0.163 ± 0.111 [0.025, 0.300]	0.133 ± 0.101 [0.007, 0.258]
Graph Laplacian	0.285 ± 0.434 [−0.254, 0.825]	0.280 ± 0.349 [−0.154, 0.714]	0.502 ± 0.573 [−0.209, 1.213]	0.215 ± 0.304 [−0.163, 0.592]
Graph Temporal	0.252 ± 0.296 [−0.116, 0.619]	0.247 ± 0.316 [−0.145, 0.639]	0.431 ± 0.452 [−0.130, 0.992]	0.265 ± 0.280 [−0.082, 0.612]
Sage-Voronoi	**0.048** ± 0.051 [−0.016, 0.111]	0.182 ± 0.123 [0.030, 0.335]	0.125 ± 0.079 [0.027, 0.223]	0.106 ± 0.085 [0.000, 0.211]
GAIN	0.106 ± 0.089 [−0.005, 0.217]	0.167 ± 0.110 [0.031, 0.303]	0.157 ± 0.044 [0.102, 0.212]	0.087 ± 0.058 [0.016, 0.159]
GRIN	0.099 ± 0.078 [0.002, 0.196]	**0.078** ± 0.058 [0.006, 0.150]	**0.086** ± 0.035 [0.042, 0.130]	**0.071** ± 0.028 [0.037, 0.106]

Table S70: MRE for sensor A027, October–December 2022 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 16$	Block $B = 32$	Block $B = 64$
Linear	0.237 ± 0.202 [−0.014, 0.487]	0.820 ± 1.203 [−0.674, 2.314]	2.885 ± 4.530 [−2.740, 8.509]
Median	2.057 ± 2.006 [−0.434, 4.548]	4.214 ± 4.022 [−0.780, 9.208]	4.248 ± 4.683 [−1.566, 10.063]
FFill + BFill	0.568 ± 0.446 [0.015, 1.121]	1.882 ± 1.107 [0.507, 3.257]	2.763 ± 3.125 [−1.117, 6.644]
Rolling mean	0.222 ± 0.127 [0.064, 0.379]	0.885 ± 1.174 [−0.573, 2.343]	2.793 ± 4.454 [−2.738, 8.323]
KNN (scaled)	0.157 ± 0.099 [0.035, 0.280]	0.246 ± 0.155 [0.053, 0.438]	0.270 ± 0.307 [−0.111, 0.650]
KNN (unscaled)	0.167 ± 0.094 [0.050, 0.284]	0.312 ± 0.205 [0.058, 0.567]	0.424 ± 0.438 [−0.120, 0.968]
Historical	1.207 ± 2.167 [−1.483, 3.897]	0.341 ± 0.182 [0.114, 0.567]	0.445 ± 0.338 [0.025, 0.865]
Graph Laplacian	0.407 ± 0.415 [−0.108, 0.923]	1.053 ± 1.263 [−0.515, 2.621]	1.133 ± 1.278 [−0.454, 2.720]
Graph Temporal	0.407 ± 0.372 [−0.055, 0.870]	0.940 ± 1.008 [−0.311, 2.192]	1.101 ± 1.172 [−0.354, 2.556]
Sage-Voronoi	0.224 ± 0.190 [−0.012, 0.459]	0.337 ± 0.161 [0.137, 0.536]	0.575 ± 0.775 [−0.387, 1.537]
GAIN	0.228 ± 0.140 [0.054, 0.402]	0.243 ± 0.120 [0.095, 0.392]	0.229 ± 0.129 [0.068, 0.389]
GRIN	**0.118** ± 0.061 [0.042, 0.193]	**0.151** ± 0.078 [0.054, 0.247]	**0.191** ± 0.160 [−0.008, 0.390]

Table S71: MRE for sensor A027, October–December 2022 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 128$	Block $B = 144$
Linear	2.251 ± 1.435 [0.470, 4.033]	2.865 ± 1.162 [1.422, 4.308]
Median	2.739 ± 1.062 [1.421, 4.057]	3.042 ± 0.759 [2.099, 3.984]
FFill + BFill	2.093 ± 1.206 [0.596, 3.590]	2.903 ± 1.126 [1.505, 4.300]
Rolling mean	2.237 ± 1.430 [0.461, 4.013]	2.867 ± 1.158 [1.429, 4.305]
KNN (scaled)	0.177 ± 0.063 [0.099, 0.255]	**0.171** ± 0.030 [0.134, 0.208]
KNN (unscaled)	0.230 ± 0.077 [0.135, 0.326]	0.241 ± 0.054 [0.174, 0.308]
Historical	0.347 ± 0.126 [0.190, 0.503]	0.519 ± 0.525 [−0.133, 1.171]
Graph Laplacian	0.670 ± 0.332 [0.259, 1.082]	0.735 ± 0.277 [0.391, 1.079]
Graph Temporal	0.830 ± 0.542 [0.157, 1.503]	0.909 ± 0.414 [0.396, 1.423]
Sage-Voronoi	0.463 ± 0.221 [0.189, 0.737]	0.421 ± 0.153 [0.231, 0.611]
GAIN	**0.172** ± 0.037 [0.127, 0.218]	0.179 ± 0.046 [0.122, 0.237]
GRIN	0.382 ± 0.407 [−0.124, 0.887]	0.367 ± 0.529 [−0.289, 1.023]

Table S72: MRE for sensor A027, October–December 2022 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	MCAR 10%	MCAR 30%	MCAR 50%	Synchronous $B = 144$
Linear	0.161 ± 0.021 [0.134, 0.187]	0.165 ± 0.007 [0.157, 0.173]	0.171 ± 0.009 [0.160, 0.182]	1.814 ± 1.486 [−0.032, 3.659]
Median	2.703 ± 0.437 [2.160, 3.245]	2.919 ± 0.165 [2.714, 3.124]	2.977 ± 0.153 [2.787, 3.167]	2.184 ± 0.934 [1.024, 3.343]
FFill + BFill	0.188 ± 0.029 [0.152, 0.224]	0.200 ± 0.010 [0.188, 0.212]	0.225 ± 0.011 [0.211, 0.238]	1.741 ± 1.460 [−0.073, 3.554]
Rolling mean	0.194 ± 0.020 [0.169, 0.219]	0.209 ± 0.008 [0.199, 0.218]	0.218 ± 0.010 [0.206, 0.230]	1.813 ± 1.483 [−0.029, 3.655]
KNN (scaled)	0.174 ± 0.023 [0.145, 0.202]	0.179 ± 0.017 [0.158, 0.200]	0.178 ± 0.013 [0.162, 0.195]	1.846 ± 0.799 [0.854, 2.838]
KNN (unscaled)	0.228 ± 0.042 [0.176, 0.280]	0.222 ± 0.022 [0.195, 0.249]	0.215 ± 0.019 [0.191, 0.239]	1.846 ± 0.799 [0.854, 2.838]
Historical	0.467 ± 0.036 [0.422, 0.513]	0.433 ± 0.022 [0.406, 0.461]	0.472 ± 0.033 [0.432, 0.513]	**0.559** ± 0.351 [0.124, 0.995]
Graph Laplacian	0.636 ± 0.127 [0.478, 0.794]	0.667 ± 0.071 [0.578, 0.756]	0.666 ± 0.056 [0.597, 0.735]	1.846 ± 0.799 [0.854, 2.838]
Graph Temporal	0.556 ± 0.102 [0.429, 0.682]	0.640 ± 0.048 [0.580, 0.699]	0.706 ± 0.046 [0.649, 0.763]	1.739 ± 1.228 [0.214, 3.263]
Sage-Voronoi	0.165 ± 0.027 [0.131, 0.198]	0.173 ± 0.009 [0.162, 0.184]	0.194 ± 0.021 [0.168, 0.220]	1.196 ± 0.847 [0.144, 2.248]
GAIN	0.165 ± 0.016 [0.145, 0.185]	0.299 ± 0.052 [0.235, 0.363]	0.693 ± 0.074 [0.602, 0.785]	2.969 ± 1.210 [1.466, 4.472]
GRIN	**0.110** ± 0.015 [0.092, 0.129]	**0.129** ± 0.012 [0.114, 0.143]	**0.151** ± 0.007 [0.143, 0.160]	1.782 ± 1.164 [0.337, 3.227]

Table S73: MAE for sensor A142, January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 1$	Block $B = 2$	Block $B = 4$	Block $B = 8$
Linear	70.30 ± 73.46 [−20.91, 161.51]	22.53 ± 11.87 [7.79, 37.27]	19.66 ± 4.74 [13.77, 25.55]	48.67 ± 71.58 [−40.21, 137.54]
Median	110.00 ± 83.74 [6.03, 213.97]	79.00 ± 58.32 [6.59, 151.41]	46.60 ± 15.48 [27.37, 65.83]	140.45 ± 86.01 [33.65, 247.25]
FFill + BFill	58.40 ± 69.34 [−27.70, 144.50]	32.80 ± 22.94 [4.31, 61.29]	34.70 ± 11.27 [20.71, 48.69]	56.00 ± 56.64 [−14.33, 126.33]
Rolling mean	63.55 ± 90.75 [−49.13, 176.23]	41.91 ± 17.40 [20.31, 63.51]	29.53 ± 5.98 [22.10, 36.96]	47.99 ± 52.43 [−17.11, 113.09]
KNN (scaled)	194.58 ± 85.19 [88.81, 300.35]	80.04 ± 51.49 [16.11, 143.97]	105.28 ± 46.68 [47.32, 163.25]	72.47 ± 52.68 [7.06, 137.89]
KNN (unscaled)	70.44 ± 81.68 [−30.98, 171.86]	27.50 ± 10.91 [13.95, 41.05]	26.16 ± 8.64 [15.43, 36.89]	34.23 ± 8.42 [23.77, 44.68]
Historical	88.62 ± 91.63 [−25.15, 202.39]	36.77 ± 15.97 [16.94, 56.60]	29.26 ± 14.84 [10.84, 47.69]	36.55 ± 14.93 [18.01, 55.10]
Graph Laplacian	83.31 ± 60.39 [8.33, 158.30]	33.24 ± 18.17 [10.68, 55.81]	51.68 ± 22.46 [23.80, 79.56]	56.06 ± 25.68 [24.17, 87.96]
Graph Temporal	71.90 ± 73.14 [−18.92, 162.72]	31.48 ± 18.36 [8.68, 54.27]	30.78 ± 14.22 [13.13, 48.44]	46.38 ± 16.68 [25.67, 67.09]
Sage-Voronoi	**55.59** ± 69.09 [−30.19, 141.38]	32.34 ± 18.10 [9.87, 54.81]	23.43 ± 5.55 [16.54, 30.32]	25.35 ± 19.78 [0.79, 49.92]
GAIN	79.97 ± 90.98 [−33.00, 192.94]	27.41 ± 21.25 [1.03, 53.80]	37.60 ± 19.13 [13.84, 61.35]	29.13 ± 24.14 [−0.85, 59.10]
GRIN	66.24 ± 87.01 [−41.80, 174.28]	**16.24** ± 7.66 [6.72, 25.75]	**16.24** ± 7.95 [6.37, 26.10]	**16.28** ± 12.03 [1.34, 31.22]

Table S74: MAE for sensor A142, January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 16$	Block $B = 32$	Block $B = 64$
Linear	31.35 ± 4.84 [25.34, 37.36]	40.62 ± 18.64 [17.48, 63.76]	94.26 ± 59.34 [20.58, 167.94]
Median	43.09 ± 17.11 [21.85, 64.33]	70.88 ± 31.71 [31.50, 110.26]	85.19 ± 36.67 [39.66, 130.72]
FFill + BFill	47.06 ± 28.58 [11.57, 82.55]	75.43 ± 48.10 [15.71, 135.16]	129.42 ± 91.39 [15.94, 242.90]
Rolling mean	31.92 ± 8.12 [21.84, 42.01]	42.43 ± 16.06 [22.49, 62.37]	96.06 ± 57.65 [24.48, 167.64]
KNN (scaled)	115.91 ± 27.04 [82.34, 149.48]	116.31 ± 36.01 [71.59, 161.02]	87.13 ± 36.90 [41.31, 132.95]
KNN (unscaled)	35.52 ± 12.19 [20.38, 50.66]	35.15 ± 10.36 [22.29, 48.01]	33.45 ± 5.36 [26.79, 40.11]
Historical	30.49 ± 14.85 [12.05, 48.93]	35.50 ± 4.82 [29.52, 41.48]	47.14 ± 14.41 [29.25, 65.03]
Graph Laplacian	64.01 ± 11.77 [49.39, 78.62]	62.07 ± 22.05 [34.69, 89.45]	54.88 ± 8.36 [44.50, 65.27]
Graph Temporal	60.48 ± 12.71 [44.69, 76.26]	46.01 ± 13.21 [29.61, 62.41]	45.73 ± 4.06 [40.69, 50.77]
Sage-Voronoi	39.02 ± 7.92 [29.19, 48.85]	45.94 ± 11.89 [31.17, 60.70]	48.36 ± 12.68 [32.62, 64.10]
GAIN	49.01 ± 28.85 [13.19, 84.83]	42.58 ± 15.69 [23.09, 62.06]	37.13 ± 11.90 [22.35, 51.91]
GRIN	**23.04** ± 7.51 [13.71, 32.36]	**25.58** ± 7.88 [15.79, 35.37]	**25.80** ± 11.30 [11.77, 39.83]

Table S75: MAE for sensor A142, January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 128$	Block $B = 144$
Linear	97.69 ± 21.64 [70.82, 124.56]	141.76 ± 40.99 [90.86, 192.66]
Median	107.13 ± 14.85 [88.70, 125.56]	102.05 ± 10.44 [89.09, 115.01]
FFill + BFill	124.96 ± 20.58 [99.41, 150.52]	141.00 ± 33.91 [98.89, 183.10]
Rolling mean	97.63 ± 21.27 [71.22, 124.04]	141.31 ± 41.34 [89.98, 192.64]
KNN (scaled)	104.97 ± 12.44 [89.52, 120.41]	107.61 ± 14.59 [89.49, 125.72]
KNN (unscaled)	33.86 ± 1.77 [31.66, 36.06]	32.27 ± 2.15 [29.60, 34.95]
Historical	46.92 ± 2.95 [43.25, 50.58]	44.05 ± 4.26 [38.76, 49.34]
Graph Laplacian	58.59 ± 6.42 [50.63, 66.56]	58.67 ± 6.44 [50.67, 66.67]
Graph Temporal	48.09 ± 6.02 [40.61, 55.56]	47.07 ± 2.88 [43.49, 50.66]
Sage-Voronoi	53.73 ± 10.16 [41.11, 66.34]	50.56 ± 8.87 [39.55, 61.57]
GAIN	35.53 ± 5.66 [28.51, 42.56]	34.66 ± 4.04 [29.65, 39.67]
GRIN	**31.02** ± 5.77 [23.85, 38.18]	**26.74** ± 1.62 [24.73, 28.75]

Table S76: MAE for sensor A142, January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	MCAR 10%	MCAR 30%	MCAR 50%	Synchronous $B = 144$
Linear	26.16 ± 1.90 [23.79, 28.52]	25.20 ± 0.83 [24.17, 26.24]	25.57 ± 0.75 [24.63, 26.50]	112.72 ± 27.12 [79.05, 146.40]
Median	102.15 ± 7.43 [92.93, 111.37]	101.29 ± 2.23 [98.52, 104.06]	100.11 ± 1.76 [97.93, 102.30]	101.49 ± 5.86 [94.21, 108.77]
FFill + BFill	32.36 ± 2.83 [28.84, 35.87]	31.30 ± 1.86 [28.99, 33.62]	33.45 ± 1.18 [31.99, 34.92]	128.00 ± 49.32 [66.75, 189.24]
Rolling mean	32.49 ± 2.78 [29.03, 35.94]	31.49 ± 2.33 [28.60, 34.38]	32.74 ± 0.90 [31.62, 33.85]	113.40 ± 27.78 [78.90, 147.90]
KNN (scaled)	101.22 ± 3.16 [97.30, 105.14]	96.21 ± 1.52 [94.32, 98.09]	85.42 ± 2.15 [82.75, 88.08]	106.75 ± 6.12 [99.15, 114.36]
KNN (unscaled)	34.24 ± 1.78 [32.03, 36.45]	33.87 ± 1.22 [32.36, 35.38]	35.04 ± 0.33 [34.63, 35.46]	106.75 ± 6.12 [99.15, 114.36]
Historical	44.24 ± 2.82 [40.73, 47.74]	46.04 ± 1.50 [44.18, 47.91]	46.07 ± 1.03 [44.79, 47.34]	**45.65** ± 15.50 [26.41, 64.90]
Graph Laplacian	58.09 ± 3.54 [53.69, 62.49]	58.33 ± 1.31 [56.70, 59.95]	60.93 ± 0.88 [59.83, 62.03]	106.75 ± 6.12 [99.15, 114.36]
Graph Temporal	41.10 ± 2.74 [37.69, 44.50]	40.92 ± 1.12 [39.53, 42.31]	43.13 ± 0.89 [42.03, 44.24]	163.51 ± 59.33 [89.84, 237.19]
Sage-Voronoi	28.49 ± 1.59 [26.52, 30.47]	29.43 ± 1.91 [27.06, 31.81]	31.53 ± 1.29 [29.93, 33.13]	113.25 ± 32.08 [73.42, 153.09]
GAIN	35.78 ± 5.58 [28.86, 42.71]	38.00 ± 3.38 [33.81, 42.20]	48.15 ± 3.04 [44.38, 51.93]	133.28 ± 33.92 [91.17, 175.40]
GRIN	**20.62** ± 1.83 [18.34, 22.90]	**20.02** ± 0.66 [19.20, 20.84]	**22.45** ± 0.86 [21.39, 23.52]	109.96 ± 8.48 [99.43, 120.48]

Table S77: RMSE for sensor A142, January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 1$	Block $B = 2$	Block $B = 4$	Block $B = 8$
Linear	70.30 ± 73.46 [−20.91, 161.51]	26.50 ± 14.64 [8.33, 44.68]	23.66 ± 6.79 [15.23, 32.10]	53.69 ± 72.03 [−35.74, 143.13]
Median	110.00 ± 83.74 [6.03, 213.97]	83.23 ± 57.47 [11.87, 154.59]	49.98 ± 15.89 [30.25, 69.71]	144.04 ± 82.58 [41.50, 246.58]
FFill + BFill	58.40 ± 69.34 [−27.70, 144.50]	38.51 ± 27.81 [3.97, 73.04]	37.80 ± 12.95 [21.72, 53.88]	60.45 ± 55.84 [−8.89, 129.79]
Rolling mean	63.55 ± 90.75 [−49.13, 176.23]	46.64 ± 21.20 [20.32, 72.96]	35.03 ± 6.64 [26.79, 43.27]	55.00 ± 57.36 [−16.23, 126.23]
KNN (scaled)	194.58 ± 85.19 [88.81, 300.35]	81.14 ± 51.65 [17.01, 145.27]	106.93 ± 45.77 [50.10, 163.75]	77.59 ± 51.76 [13.32, 141.86]
KNN (unscaled)	70.44 ± 81.68 [−30.98, 171.86]	30.85 ± 13.54 [14.04, 47.66]	29.29 ± 10.36 [16.43, 42.15]	40.50 ± 8.73 [29.67, 51.34]
Historical	88.62 ± 91.63 [−25.15, 202.39]	39.85 ± 17.25 [18.43, 61.28]	34.34 ± 14.83 [15.93, 52.75]	40.06 ± 14.52 [22.02, 58.09]
Graph Laplacian	83.31 ± 60.39 [8.33, 158.30]	36.93 ± 19.12 [13.20, 60.67]	53.70 ± 22.31 [26.00, 81.41]	59.59 ± 25.87 [27.47, 91.71]
Graph Temporal	71.90 ± 73.14 [−18.92, 162.72]	33.29 ± 20.01 [8.44, 58.13]	36.45 ± 13.16 [20.11, 52.80]	49.48 ± 18.42 [26.60, 72.35]
Sage-Voronoi	**55.59** ± 69.09 [−30.19, 141.38]	36.25 ± 23.07 [7.60, 64.89]	27.90 ± 5.38 [21.22, 34.58]	29.41 ± 20.95 [3.39, 55.42]
GAIN	79.97 ± 90.98 [−33.00, 192.94]	31.98 ± 26.02 [−0.33, 64.29]	39.85 ± 19.79 [15.27, 64.43]	33.53 ± 26.76 [0.31, 66.75]
GRIN	66.24 ± 87.01 [−41.80, 174.28]	**19.41** ± 10.93 [5.83, 32.98]	**18.00** ± 8.78 [7.09, 28.91]	**20.01** ± 14.62 [1.86, 38.17]

Table S78: RMSE for sensor A142, January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 16$	Block $B = 32$	Block $B = 64$
Linear	41.05 ± 11.00 [27.40, 54.71]	54.47 ± 21.84 [27.36, 81.59]	115.31 ± 70.28 [28.05, 202.58]
Median	56.04 ± 18.61 [32.93, 79.15]	87.52 ± 35.79 [43.09, 131.96]	103.31 ± 43.25 [49.61, 157.02]
FFill + BFill	57.24 ± 29.98 [20.02, 94.46]	91.04 ± 54.15 [23.80, 158.27]	147.16 ± 101.00 [21.75, 272.57]
Rolling mean	41.11 ± 13.95 [23.80, 58.43]	55.55 ± 19.73 [31.05, 80.05]	117.70 ± 69.61 [31.27, 204.13]
KNN (scaled)	124.04 ± 25.95 [91.82, 156.25]	127.02 ± 35.05 [83.49, 170.54]	101.97 ± 34.29 [59.39, 144.54]
KNN (unscaled)	46.69 ± 16.12 [26.67, 66.70]	45.25 ± 15.25 [26.32, 64.18]	43.50 ± 7.51 [34.18, 52.82]
Historical	38.99 ± 17.32 [17.49, 60.50]	48.31 ± 10.51 [35.26, 61.36]	57.78 ± 16.05 [37.85, 77.71]
Graph Laplacian	70.25 ± 11.21 [56.33, 84.17]	69.01 ± 21.20 [42.69, 95.33]	62.89 ± 8.83 [51.93, 73.85]
Graph Temporal	67.83 ± 13.76 [50.74, 84.92]	54.47 ± 15.00 [35.84, 73.10]	53.84 ± 6.39 [45.90, 61.78]
Sage-Voronoi	48.46 ± 7.75 [38.83, 58.08]	56.52 ± 10.01 [44.10, 68.94]	61.04 ± 15.73 [41.51, 80.57]
GAIN	60.17 ± 29.77 [23.20, 97.13]	54.82 ± 21.49 [28.14, 81.50]	49.12 ± 14.92 [30.60, 67.65]
GRIN	**32.45** ± 12.55 [16.87, 48.04]	**36.66** ± 13.91 [19.38, 53.93]	**32.81** ± 13.18 [16.44, 49.18]

Table S79: RMSE for sensor A142, January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 128$	Block $B = 144$
Linear	126.94 ± 27.54 [92.74, 161.13]	170.20 ± 39.36 [121.33, 219.06]
Median	132.34 ± 13.40 [115.69, 148.98]	127.30 ± 10.03 [114.84, 139.75]
FFill + BFill	153.28 ± 13.65 [136.33, 170.23]	165.80 ± 36.06 [121.03, 210.57]
Rolling mean	126.87 ± 27.16 [93.15, 160.59]	169.79 ± 39.62 [120.59, 218.99]
KNN (scaled)	125.38 ± 14.13 [107.84, 142.92]	127.19 ± 17.25 [105.78, 148.61]
KNN (unscaled)	48.52 ± 4.81 [42.54, 54.50]	46.30 ± 3.83 [41.54, 51.05]
Historical	64.40 ± 5.86 [57.13, 71.68]	60.80 ± 9.19 [49.40, 72.21]
Graph Laplacian	69.24 ± 7.23 [60.26, 78.22]	68.26 ± 7.22 [59.30, 77.23]
Graph Temporal	60.06 ± 6.08 [52.51, 67.61]	57.14 ± 4.08 [52.07, 62.20]
Sage-Voronoi	70.12 ± 12.87 [54.14, 86.09]	65.28 ± 10.49 [52.26, 78.30]
GAIN	51.37 ± 7.73 [41.78, 60.97]	49.42 ± 5.01 [43.20, 55.65]
GRIN	**45.92** ± 6.66 [37.65, 54.19]	**39.79** ± 2.56 [36.60, 42.97]

Table S80: RMSE for sensor A142, January 2023 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	MCAR 10%	MCAR 30%	MCAR 50%	Synchronous $B = 144$
Linear	38.10 ± 3.47 [33.80, 42.41]	36.85 ± 2.14 [34.19, 39.51]	36.61 ± 1.43 [34.84, 38.39]	148.87 ± 24.94 [117.91, 179.84]
Median	127.58 ± 7.89 [117.78, 137.38]	125.95 ± 2.30 [123.10, 128.80]	124.81 ± 1.79 [122.59, 127.04]	125.68 ± 7.68 [116.14, 135.22]
FFill + BFill	46.18 ± 6.02 [38.71, 53.65]	45.71 ± 2.97 [42.02, 49.39]	47.48 ± 2.13 [44.84, 50.13]	164.29 ± 47.52 [105.28, 223.29]
Rolling mean	44.50 ± 4.19 [39.30, 49.70]	44.44 ± 3.19 [40.48, 48.41]	46.05 ± 1.70 [43.94, 48.17]	149.33 ± 25.42 [117.76, 180.90]
KNN (scaled)	119.50 ± 4.82 [113.51, 125.48]	117.77 ± 1.75 [115.60, 119.94]	110.49 ± 2.71 [107.12, 113.86]	124.29 ± 4.61 [118.57, 130.01]
KNN (unscaled)	46.46 ± 3.45 [42.18, 50.74]	46.06 ± 2.38 [43.10, 49.01]	47.62 ± 0.88 [46.53, 48.71]	124.29 ± 4.61 [118.57, 130.01]
Historical	59.96 ± 4.51 [54.36, 65.57]	63.58 ± 2.72 [60.20, 66.96]	63.76 ± 1.61 [61.75, 65.76]	**60.69** ± 20.52 [35.21, 86.17]
Graph Laplacian	67.31 ± 2.80 [63.84, 70.79]	69.13 ± 1.57 [67.18, 71.08]	71.82 ± 0.76 [70.87, 72.77]	124.29 ± 4.61 [118.57, 130.01]
Graph Temporal	50.85 ± 2.36 [47.92, 53.77]	51.35 ± 2.13 [48.71, 53.99]	53.43 ± 0.68 [52.59, 54.28]	199.46 ± 61.02 [123.69, 275.22]
Sage-Voronoi	40.19 ± 3.05 [36.41, 43.98]	41.54 ± 2.65 [38.25, 44.83]	44.19 ± 2.39 [41.22, 47.15]	149.97 ± 34.76 [106.80, 193.13]
GAIN	49.25 ± 6.49 [41.18, 57.31]	48.96 ± 3.35 [44.80, 53.11]	59.37 ± 3.80 [54.65, 64.09]	172.83 ± 25.32 [141.39, 204.27]
GRIN	**32.01** ± 3.79 [27.31, 36.71]	**30.81** ± 1.86 [28.50, 33.12]	**32.96** ± 1.31 [31.33, 34.59]	127.85 ± 5.55 [120.95, 134.74]

Table S81: MRE for sensor A142, January 2023 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 1$	Block $B = 2$	Block $B = 4$	Block $B = 8$
Linear	0.172 ± 0.146 [−0.010, 0.353]	0.117 ± 0.094 [−0.000, 0.233]	0.075 ± 0.025 [0.043, 0.107]	0.276 ± 0.198 [0.030, 0.523]
Median	0.266 ± 0.160 [0.067, 0.466]	0.643 ± 0.798 [−0.348, 1.635]	0.176 ± 0.094 [0.059, 0.292]	2.901 ± 3.260 [−1.147, 6.948]
FFill + BFill	0.140 ± 0.140 [−0.034, 0.313]	0.182 ± 0.164 [−0.022, 0.386]	0.131 ± 0.059 [0.057, 0.205]	0.449 ± 0.312 [0.061, 0.837]
Rolling mean	0.151 ± 0.186 [−0.080, 0.381]	0.242 ± 0.199 [−0.006, 0.489]	0.116 ± 0.056 [0.047, 0.185]	0.335 ± 0.178 [0.114, 0.556]
KNN (scaled)	0.503 ± 0.117 [0.358, 0.648]	0.342 ± 0.118 [0.195, 0.489]	0.355 ± 0.122 [0.203, 0.506]	0.694 ± 0.483 [0.094, 1.294]
KNN (unscaled)	0.172 ± 0.159 [−0.025, 0.370]	0.162 ± 0.138 [−0.010, 0.333]	0.097 ± 0.034 [0.055, 0.140]	0.481 ± 0.432 [−0.056, 1.018]
Historical	0.207 ± 0.181 [−0.017, 0.432]	0.189 ± 0.090 [0.076, 0.301]	0.116 ± 0.096 [−0.003, 0.235]	0.525 ± 0.450 [−0.035, 1.084]
Graph Laplacian	0.218 ± 0.116 [0.074, 0.363]	0.196 ± 0.158 [0.000, 0.392]	0.178 ± 0.057 [0.108, 0.249]	0.848 ± 0.845 [−0.202, 1.897]
Graph Temporal	0.182 ± 0.143 [0.004, 0.360]	0.179 ± 0.138 [0.007, 0.351]	0.109 ± 0.033 [0.068, 0.151]	0.715 ± 0.720 [−0.179, 1.609]
Sage-Voronoi	**0.132** ± 0.135 [−0.037, 0.300]	0.185 ± 0.151 [−0.003, 0.373]	0.089 ± 0.029 [0.053, 0.125]	0.277 ± 0.217 [0.007, 0.547]
GAIN	0.202 ± 0.182 [−0.025, 0.428]	0.136 ± 0.142 [−0.040, 0.313]	0.129 ± 0.050 [0.067, 0.191]	0.216 ± 0.069 [0.130, 0.302]
GRIN	0.157 ± 0.171 [−0.056, 0.369]	**0.081** ± 0.054 [0.014, 0.148]	**0.063** ± 0.036 [0.018, 0.108]	**0.133** ± 0.051 [0.070, 0.196]

Table S82: MRE for sensor A142, January 2023 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 16$	Block $B = 32$	Block $B = 64$
Linear	0.104 ± 0.018 [0.081, 0.127]	0.176 ± 0.113 [0.036, 0.316]	1.037 ± 1.750 [−1.137, 3.210]
Median	0.133 ± 0.045 [0.077, 0.189]	0.570 ± 0.901 [−0.549, 1.688]	1.137 ± 1.329 [−0.513, 2.787]
FFill + BFill	0.162 ± 0.103 [0.034, 0.291]	0.659 ± 1.110 [−0.719, 2.037]	0.973 ± 1.208 [−0.526, 2.473]
Rolling mean	0.107 ± 0.029 [0.071, 0.143]	0.197 ± 0.154 [0.005, 0.388]	1.023 ± 1.733 [−1.128, 3.175]
KNN (scaled)	0.376 ± 0.063 [0.298, 0.454]	0.425 ± 0.045 [0.369, 0.481]	0.492 ± 0.292 [0.129, 0.855]
KNN (unscaled)	0.115 ± 0.036 [0.070, 0.160]	0.145 ± 0.056 [0.075, 0.215]	0.232 ± 0.145 [0.053, 0.412]
Historical	0.102 ± 0.058 [0.029, 0.174]	0.196 ± 0.204 [−0.057, 0.449]	0.303 ± 0.204 [0.050, 0.556]
Graph Laplacian	0.215 ± 0.037 [0.169, 0.262]	0.296 ± 0.184 [0.067, 0.524]	0.422 ± 0.316 [0.030, 0.813]
Graph Temporal	0.204 ± 0.046 [0.147, 0.262]	0.227 ± 0.161 [0.028, 0.427]	0.349 ± 0.255 [0.033, 0.666]
Sage-Voronoi	0.130 ± 0.031 [0.093, 0.168]	0.274 ± 0.296 [−0.094, 0.642]	0.333 ± 0.360 [−0.114, 0.779]
GAIN	0.156 ± 0.088 [0.046, 0.265]	0.151 ± 0.026 [0.119, 0.183]	0.173 ± 0.058 [0.100, 0.245]
GRIN	**0.073** ± 0.021 [0.047, 0.099]	**0.094** ± 0.022 [0.067, 0.121]	**0.124** ± 0.050 [0.062, 0.187]

Table S83: MRE for sensor A142, January 2023 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 128$	Block $B = 144$
Linear	1.547 ± 0.705 [0.672, 2.423]	1.272 ± 0.802 [0.277, 2.268]
Median	1.631 ± 0.464 [1.055, 2.207]	1.484 ± 0.385 [1.006, 1.962]
FFill + BFill	1.574 ± 0.615 [0.811, 2.338]	1.271 ± 0.766 [0.320, 2.222]
Rolling mean	1.538 ± 0.706 [0.661, 2.415]	1.270 ± 0.800 [0.276, 2.263]
KNN (scaled)	0.626 ± 0.124 [0.472, 0.779]	0.553 ± 0.072 [0.464, 0.642]
KNN (unscaled)	0.255 ± 0.043 [0.202, 0.308]	0.228 ± 0.037 [0.182, 0.275]
Historical	0.373 ± 0.050 [0.310, 0.435]	0.356 ± 0.036 [0.312, 0.401]
Graph Laplacian	0.539 ± 0.133 [0.374, 0.705]	0.508 ± 0.118 [0.361, 0.655]
Graph Temporal	0.465 ± 0.135 [0.298, 0.632]	0.430 ± 0.105 [0.300, 0.559]
Sage-Voronoi	0.462 ± 0.176 [0.244, 0.681]	0.379 ± 0.151 [0.192, 0.566]
GAIN	0.196 ± 0.040 [0.147, 0.245]	0.183 ± 0.028 [0.148, 0.218]
GRIN	**0.154** ± 0.022 [0.127, 0.181]	**0.135** ± 0.030 [0.098, 0.173]

Table S84: MRE for sensor A142, January 2023 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	MCAR 10%	MCAR 30%	MCAR 50%	Synchronous $B = 144$
Linear	0.184 ± 0.010 [0.171, 0.197]	0.147 ± 0.008 [0.137, 0.157]	0.148 ± 0.009 [0.136, 0.160]	1.693 ± 0.432 [1.156, 2.230]
Median	1.676 ± 0.311 [1.290, 2.062]	1.389 ± 0.099 [1.266, 1.512]	1.334 ± 0.075 [1.240, 1.428]	1.398 ± 0.380 [0.927, 1.869]
FFill + BFill	0.209 ± 0.020 [0.184, 0.234]	0.181 ± 0.017 [0.160, 0.202]	0.190 ± 0.009 [0.179, 0.200]	1.629 ± 0.805 [0.629, 2.629]
Rolling mean	0.213 ± 0.013 [0.197, 0.229]	0.186 ± 0.018 [0.163, 0.208]	0.185 ± 0.007 [0.176, 0.193]	1.694 ± 0.433 [1.157, 2.232]
KNN (scaled)	0.703 ± 0.089 [0.593, 0.814]	0.610 ± 0.048 [0.550, 0.669]	0.566 ± 0.043 [0.513, 0.620]	1.278 ± 0.319 [0.882, 1.674]
KNN (unscaled)	0.304 ± 0.049 [0.244, 0.365]	0.259 ± 0.024 [0.229, 0.290]	0.272 ± 0.021 [0.246, 0.298]	1.278 ± 0.319 [0.882, 1.674]
Historical	0.403 ± 0.055 [0.335, 0.472]	0.363 ± 0.023 [0.334, 0.391]	0.350 ± 0.020 [0.325, 0.375]	**0.365** ± 0.244 [0.062, 0.668]
Graph Laplacian	0.571 ± 0.098 [0.450, 0.692]	0.492 ± 0.033 [0.452, 0.533]	0.494 ± 0.026 [0.462, 0.526]	1.278 ± 0.319 [0.882, 1.674]
Graph Temporal	0.400 ± 0.066 [0.318, 0.482]	0.341 ± 0.022 [0.313, 0.368]	0.340 ± 0.022 [0.312, 0.367]	2.199 ± 0.645 [1.398, 3.000]
Sage-Voronoi	0.190 ± 0.025 [0.160, 0.221]	0.176 ± 0.017 [0.155, 0.196]	0.182 ± 0.005 [0.176, 0.187]	1.380 ± 0.566 [0.678, 2.083]
GAIN	0.226 ± 0.031 [0.188, 0.265]	0.293 ± 0.039 [0.244, 0.342]	0.497 ± 0.066 [0.415, 0.579]	2.041 ± 0.619 [1.273, 2.809]
GRIN	**0.114** ± 0.012 [0.099, 0.129]	**0.106** ± 0.009 [0.095, 0.117]	**0.126** ± 0.012 [0.111, 0.140]	1.223 ± 0.415 [0.708, 1.739]

Table S85: MAE for sensor A142, October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 1$	Block $B = 2$	Block $B = 4$	Block $B = 8$
Linear	19.80 ± 8.30 [9.49, 30.11]	21.87 ± 10.95 [8.27, 35.47]	35.81 ± 25.39 [4.28, 67.34]	**16.81** ± 6.90 [8.23, 25.38]
Median	136.20 ± 96.08 [16.90, 255.50]	148.20 ± 61.76 [71.52, 224.88]	43.15 ± 18.64 [20.01, 66.29]	93.20 ± 92.21 [−21.29, 207.69]
FFill + BFill	20.40 ± 14.47 [2.44, 38.36]	22.60 ± 14.40 [4.72, 40.48]	42.85 ± 16.43 [22.44, 63.26]	28.40 ± 18.72 [5.16, 51.64]
Rolling mean	**11.40** ± 6.76 [3.01, 19.79]	35.00 ± 21.90 [7.81, 62.19]	57.09 ± 24.92 [26.14, 88.04]	23.72 ± 9.50 [11.93, 35.52]
KNN (scaled)	40.60 ± 47.12 [−17.91, 99.11]	42.52 ± 29.40 [6.02, 79.02]	**24.73** ± 8.91 [13.66, 35.79]	30.91 ± 21.45 [4.27, 57.54]
KNN (unscaled)	16.08 ± 9.16 [4.70, 27.46]	43.64 ± 28.17 [8.67, 78.61]	32.26 ± 10.75 [18.91, 45.61]	35.92 ± 19.84 [11.29, 60.56]
Historical	48.11 ± 66.62 [−34.61, 130.83]	93.44 ± 44.10 [38.68, 148.20]	26.18 ± 9.35 [14.57, 37.78]	43.25 ± 45.38 [−13.10, 99.60]
Graph Laplacian	42.74 ± 25.57 [10.98, 74.49]	34.96 ± 18.11 [12.48, 57.44]	61.51 ± 25.10 [30.35, 92.68]	58.64 ± 39.56 [9.51, 107.76]
Graph Temporal	20.10 ± 18.73 [−3.15, 43.36]	28.19 ± 18.79 [4.86, 51.52]	57.65 ± 14.86 [39.19, 76.10]	55.31 ± 45.31 [−0.95, 111.58]
Sage-Voronoi	14.21 ± 9.79 [2.06, 26.36]	29.84 ± 18.04 [7.44, 52.24]	53.97 ± 21.86 [26.83, 81.11]	28.97 ± 18.91 [5.49, 52.45]
GAIN	32.07 ± 27.04 [−1.51, 65.64]	36.96 ± 43.15 [−16.63, 90.54]	71.19 ± 16.08 [51.23, 91.15]	59.59 ± 53.91 [−7.35, 126.53]
GRIN	15.40 ± 14.61 [−2.75, 33.55]	**18.15** ± 9.67 [6.14, 30.15]	30.48 ± 4.72 [24.61, 36.34]	26.98 ± 21.02 [0.88, 53.08]

Table S86: MAE for sensor A142, October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 16$	Block $B = 32$	Block $B = 64$
Linear	26.55 ± 13.94 [9.24, 43.85]	55.85 ± 29.34 [19.41, 92.28]	90.28 ± 24.82 [59.46, 121.10]
Median	115.83 ± 70.88 [27.82, 203.83]	99.17 ± 76.49 [4.20, 194.15]	83.08 ± 24.18 [53.05, 113.11]
FFill + BFill	49.55 ± 21.07 [23.39, 75.71]	92.47 ± 88.92 [−17.93, 202.88]	161.71 ± 83.10 [58.53, 264.89]
Rolling mean	28.16 ± 15.86 [8.47, 47.85]	58.67 ± 37.25 [12.42, 104.92]	91.41 ± 23.67 [62.02, 120.81]
KNN (scaled)	28.98 ± 20.20 [3.90, 54.07]	30.48 ± 13.61 [13.58, 47.38]	**35.28** ± 8.44 [24.80, 45.75]
KNN (unscaled)	27.05 ± 12.20 [11.91, 42.20]	34.34 ± 11.26 [20.36, 48.32]	46.77 ± 6.58 [38.60, 54.94]
Historical	68.36 ± 56.42 [−1.70, 138.41]	35.87 ± 13.68 [18.88, 52.86]	53.56 ± 15.61 [34.18, 72.94]
Graph Laplacian	33.73 ± 14.90 [15.22, 52.23]	51.12 ± 7.94 [41.26, 60.98]	51.64 ± 13.83 [34.47, 68.81]
Graph Temporal	30.62 ± 14.43 [12.69, 48.54]	42.48 ± 4.61 [36.76, 48.21]	46.20 ± 13.24 [29.76, 62.65]
Sage-Voronoi	30.10 ± 12.47 [14.61, 45.58]	42.31 ± 15.14 [23.51, 61.10]	59.27 ± 10.49 [46.25, 72.30]
GAIN	36.68 ± 7.18 [27.77, 45.60]	56.56 ± 33.41 [15.08, 98.04]	60.21 ± 17.95 [37.92, 82.50]
GRIN	**16.65** ± 8.47 [6.13, 27.17]	**26.91** ± 11.56 [12.55, 41.27]	39.56 ± 9.85 [27.33, 51.80]

Table S87: MAE for sensor A142, October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 128$	Block $B = 144$
Linear	139.72 ± 59.32 [66.06, 213.37]	136.05 ± 47.73 [76.79, 195.31]
Median	93.65 ± 20.25 [68.51, 118.80]	107.08 ± 13.81 [89.94, 124.22]
FFill + BFill	185.98 ± 34.20 [143.52, 228.44]	137.20 ± 44.89 [81.47, 192.94]
Rolling mean	140.36 ± 58.98 [67.12, 213.59]	136.10 ± 47.52 [77.09, 195.10]
KNN (scaled)	35.26 ± 9.30 [23.71, 46.80]	29.54 ± 4.04 [24.53, 34.55]
KNN (unscaled)	35.83 ± 4.88 [29.77, 41.90]	35.29 ± 2.87 [31.73, 38.85]
Historical	43.86 ± 17.02 [22.73, 65.00]	59.30 ± 25.78 [27.29, 91.31]
Graph Laplacian	48.47 ± 7.02 [39.76, 57.19]	48.08 ± 6.96 [39.43, 56.72]
Graph Temporal	45.81 ± 8.73 [34.97, 56.66]	42.72 ± 6.96 [34.09, 51.36]
Sage-Voronoi	48.64 ± 10.96 [35.02, 62.25]	44.95 ± 9.12 [33.63, 56.27]
GAIN	42.62 ± 10.15 [30.01, 55.22]	48.47 ± 5.62 [41.50, 55.45]
GRIN	**31.10** ± 7.82 [21.39, 40.82]	**28.99** ± 5.56 [22.09, 35.90]

Table S88: MAE for sensor A142, October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	MCAR 10%	MCAR 30%	MCAR 50%	Synchronous $B = 144$
Linear	26.34 ± 1.09 [24.99, 27.70]	25.79 ± 0.33 [25.38, 26.20]	26.68 ± 0.78 [25.71, 27.65]	113.28 ± 15.89 [93.55, 133.00]
Median	99.19 ± 2.70 [95.84, 102.55]	97.47 ± 0.99 [96.23, 98.70]	98.52 ± 1.21 [97.01, 100.03]	100.44 ± 16.20 [80.33, 120.55]
FFill + BFill	31.34 ± 1.05 [30.04, 32.65]	31.21 ± 0.23 [30.93, 31.49]	33.89 ± 1.20 [32.41, 35.38]	121.23 ± 20.69 [95.55, 146.92]
Rolling mean	30.40 ± 0.63 [29.61, 31.18]	30.86 ± 0.64 [30.06, 31.65]	32.22 ± 0.94 [31.05, 33.38]	113.59 ± 15.66 [94.14, 133.03]
KNN (scaled)	31.86 ± 1.38 [30.15, 33.56]	30.09 ± 0.73 [29.19, 30.99]	31.76 ± 0.48 [31.17, 32.36]	94.70 ± 13.95 [77.37, 112.02]
KNN (unscaled)	36.50 ± 2.51 [33.38, 39.62]	34.79 ± 0.76 [33.85, 35.74]	35.46 ± 0.23 [35.17, 35.75]	94.70 ± 13.95 [77.37, 112.02]
Historical	50.76 ± 1.37 [49.06, 52.45]	49.08 ± 1.39 [47.35, 50.81]	49.78 ± 0.97 [48.58, 50.99]	**55.84** ± 18.41 [32.98, 78.70]
Graph Laplacian	48.17 ± 3.10 [44.31, 52.02]	47.49 ± 1.75 [45.32, 49.67]	49.48 ± 0.51 [48.84, 50.11]	94.70 ± 13.95 [77.37, 112.02]
Graph Temporal	36.95 ± 1.94 [34.54, 39.37]	37.41 ± 0.92 [36.26, 38.56]	40.60 ± 0.27 [40.26, 40.94]	110.49 ± 20.37 [85.20, 135.79]
Sage-Voronoi	27.16 ± 0.93 [26.00, 28.32]	27.98 ± 0.80 [26.99, 28.97]	31.15 ± 0.70 [30.28, 32.02]	132.20 ± 36.42 [86.98, 177.41]
GAIN	40.57 ± 4.84 [34.56, 46.57]	51.38 ± 6.67 [43.09, 59.66]	120.88 ± 13.36 [104.28, 137.47]	527.42 ± 38.53 [479.58, 575.26]
GRIN	**20.71** ± 0.92 [19.57, 21.85]	**20.42** ± 0.50 [19.80, 21.03]	**22.85** ± 0.38 [22.37, 23.32]	97.04 ± 22.03 [69.69, 124.39]

Table S89: RMSE for sensor A142, October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 1$	Block $B = 2$	Block $B = 4$	Block $B = 8$
Linear	19.80 ± 8.30 [9.49, 30.11]	22.50 ± 11.16 [8.63, 36.36]	40.33 ± 24.62 [9.76, 70.89]	**24.00** ± 14.52 [5.97, 42.03]
Median	136.20 ± 96.08 [16.90, 255.50]	149.06 ± 61.32 [72.93, 225.20]	51.68 ± 19.13 [27.93, 75.43]	97.68 ± 90.35 [−14.51, 209.87]
FFill + BFill	20.40 ± 14.47 [2.44, 38.36]	23.96 ± 14.66 [5.76, 42.16]	51.41 ± 14.49 [33.43, 69.40]	35.81 ± 22.76 [7.55, 64.08]
Rolling mean	**11.40** ± 6.76 [3.01, 19.79]	37.07 ± 22.10 [9.63, 64.50]	66.70 ± 18.27 [44.00, 89.39]	31.76 ± 15.38 [12.66, 50.86]
KNN (scaled)	40.60 ± 47.12 [−17.91, 99.11]	45.91 ± 29.85 [8.85, 82.97]	**32.27** ± 13.01 [16.11, 48.42]	38.47 ± 23.66 [9.09, 67.85]
KNN (unscaled)	16.08 ± 9.16 [4.70, 27.46]	44.64 ± 29.01 [8.63, 80.66]	37.10 ± 12.51 [21.56, 52.63]	43.73 ± 23.84 [14.13, 73.34]
Historical	48.11 ± 66.62 [−34.61, 130.83]	94.19 ± 44.63 [38.77, 149.61]	33.10 ± 13.16 [16.77, 49.44]	49.64 ± 43.16 [−3.95, 103.24]
Graph Laplacian	42.74 ± 25.57 [10.98, 74.49]	36.91 ± 16.80 [16.04, 57.78]	69.58 ± 19.28 [45.65, 93.52]	63.37 ± 41.80 [11.47, 115.26]
Graph Temporal	20.10 ± 18.73 [−3.15, 43.36]	30.03 ± 17.67 [8.09, 51.98]	65.22 ± 9.96 [52.84, 77.59]	59.99 ± 47.56 [0.94, 119.03]
Sage-Voronoi	14.21 ± 9.79 [2.06, 26.36]	32.00 ± 18.45 [9.09, 54.91]	61.30 ± 17.71 [39.32, 83.29]	35.52 ± 25.65 [3.68, 67.36]
GAIN	32.07 ± 27.04 [−1.51, 65.64]	39.00 ± 42.19 [−13.39, 91.39]	76.39 ± 15.66 [56.95, 95.84]	63.83 ± 55.55 [−5.14, 132.80]
GRIN	15.40 ± 14.61 [−2.75, 33.55]	**19.91** ± 9.77 [7.77, 32.04]	36.93 ± 7.07 [28.15, 45.70]	32.96 ± 27.64 [−1.36, 67.28]

Table S90: RMSE for sensor A142, October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 16$	Block $B = 32$	Block $B = 64$
Linear	36.08 ± 19.47 [11.90, 60.25]	71.06 ± 36.03 [26.32, 115.79]	117.85 ± 35.00 [74.39, 161.32]
Median	122.57 ± 68.81 [37.13, 208.02]	113.09 ± 70.86 [25.10, 201.07]	108.92 ± 31.48 [69.83, 148.00]
FFill + BFill	60.37 ± 24.13 [30.41, 90.33]	107.35 ± 98.49 [−14.94, 229.64]	187.31 ± 90.96 [74.36, 300.25]
Rolling mean	36.08 ± 20.82 [10.22, 61.93]	73.02 ± 43.76 [18.69, 127.36]	120.18 ± 35.50 [76.10, 164.27]
KNN (scaled)	34.94 ± 21.75 [7.93, 61.95]	38.97 ± 18.35 [16.18, 61.76]	**54.06** ± 12.82 [38.13, 69.98]
KNN (unscaled)	33.23 ± 12.78 [17.36, 49.10]	43.72 ± 15.74 [24.18, 63.26]	68.12 ± 14.29 [50.37, 85.86]
Historical	73.59 ± 56.19 [3.81, 143.36]	46.82 ± 19.66 [22.41, 71.23]	73.84 ± 18.15 [51.30, 96.37]
Graph Laplacian	39.44 ± 18.19 [16.85, 62.03]	59.99 ± 7.15 [51.12, 68.87]	70.13 ± 18.18 [47.55, 92.70]
Graph Temporal	37.75 ± 18.55 [14.71, 60.78]	52.38 ± 4.90 [46.29, 58.46]	66.35 ± 17.78 [44.27, 88.43]
Sage-Voronoi	35.66 ± 13.47 [18.94, 52.38]	51.72 ± 18.68 [28.53, 74.91]	80.65 ± 14.27 [62.93, 98.37]
GAIN	45.89 ± 9.16 [34.51, 57.26]	66.97 ± 35.07 [23.42, 110.51]	81.74 ± 20.46 [56.33, 107.15]
GRIN	**23.84** ± 14.07 [6.37, 41.32]	**36.42** ± 17.41 [14.81, 58.04]	57.51 ± 14.72 [39.24, 75.79]

Table S91: RMSE for sensor A142, October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 128$	Block $B = 144$
Linear	162.21 ± 60.84 [86.67, 237.76]	167.37 ± 46.69 [109.39, 225.34]
Median	121.86 ± 22.83 [93.52, 150.21]	133.75 ± 11.56 [119.40, 148.11]
FFill + BFill	212.76 ± 27.86 [178.17, 247.36]	168.77 ± 43.29 [115.01, 222.52]
Rolling mean	162.93 ± 60.58 [87.72, 238.15]	167.40 ± 46.57 [109.58, 225.22]
KNN (scaled)	48.88 ± 9.27 [37.37, 60.39]	42.83 ± 6.85 [34.32, 51.34]
KNN (unscaled)	49.19 ± 3.36 [45.02, 53.36]	50.33 ± 6.75 [41.94, 58.71]
Historical	60.80 ± 21.35 [34.29, 87.32]	79.53 ± 32.43 [39.26, 119.80]
Graph Laplacian	61.10 ± 7.72 [51.52, 70.69]	59.96 ± 6.64 [51.72, 68.20]
Graph Temporal	60.07 ± 10.20 [47.41, 72.73]	55.27 ± 8.61 [44.58, 65.96]
Sage-Voronoi	63.16 ± 11.75 [48.58, 77.75]	62.20 ± 10.69 [48.93, 75.48]
GAIN	58.88 ± 9.92 [46.57, 71.20]	66.12 ± 7.06 [57.36, 74.88]
GRIN	**44.63** ± 6.18 [36.95, 52.30]	**42.60** ± 7.28 [33.56, 51.63]

Table S92: RMSE for sensor A142, October–December 2022 (vehicles per ten-minute interval). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	MCAR 10%	MCAR 30%	MCAR 50%	Synchronous $B = 144$
Linear	39.72 ± 2.81 [36.24, 43.20]	38.99 ± 0.92 [37.85, 40.13]	40.70 ± 1.27 [39.13, 42.27]	142.09 ± 20.80 [116.27, 167.91]
Median	127.07 ± 2.03 [124.55, 129.60]	125.09 ± 1.10 [123.73, 126.45]	126.43 ± 1.29 [124.83, 128.03]	126.16 ± 15.77 [106.57, 145.75]
FFill + BFill	46.36 ± 3.69 [41.77, 50.94]	46.94 ± 1.64 [44.91, 48.98]	50.99 ± 2.72 [47.62, 54.37]	149.78 ± 24.93 [118.83, 180.73]
Rolling mean	43.41 ± 1.80 [41.18, 45.65]	43.36 ± 1.28 [41.77, 44.96]	46.85 ± 1.92 [44.46, 49.23]	142.33 ± 20.74 [116.57, 168.09]
KNN (scaled)	44.87 ± 3.33 [40.73, 49.00]	41.93 ± 1.73 [39.79, 44.08]	45.21 ± 0.70 [44.34, 46.08]	111.97 ± 15.02 [93.33, 130.61]
KNN (unscaled)	51.74 ± 4.85 [45.71, 57.76]	48.80 ± 1.45 [46.99, 50.60]	51.30 ± 0.46 [50.73, 51.88]	111.97 ± 15.02 [93.33, 130.61]
Historical	71.93 ± 1.52 [70.04, 73.82]	69.08 ± 1.83 [66.81, 71.35]	70.47 ± 1.51 [68.60, 72.34]	**74.81** ± 23.49 [45.65, 103.98]
Graph Laplacian	60.93 ± 4.18 [55.74, 66.12]	59.21 ± 1.63 [57.18, 61.23]	62.45 ± 0.53 [61.80, 63.11]	111.97 ± 15.02 [93.33, 130.61]
Graph Temporal	50.36 ± 3.90 [45.52, 55.19]	49.27 ± 1.14 [47.86, 50.69]	53.93 ± 0.36 [53.48, 54.37]	133.50 ± 26.57 [100.51, 166.49]
Sage-Voronoi	39.28 ± 2.10 [36.67, 41.89]	38.74 ± 1.35 [37.06, 40.42]	44.36 ± 1.03 [43.08, 45.65]	161.96 ± 39.10 [113.41, 210.50]
GAIN	56.18 ± 5.98 [48.75, 63.60]	63.92 ± 6.91 [55.34, 72.50]	133.53 ± 13.01 [117.38, 149.67]	538.02 ± 36.67 [492.49, 583.54]
GRIN	**33.73** ± 2.91 [30.12, 37.34]	**31.34** ± 1.26 [29.77, 32.91]	**35.65** ± 0.74 [34.73, 36.58]	118.33 ± 27.48 [84.20, 152.45]

Table S93: MRE for sensor A142, October–December 2022 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 1$	Block $B = 2$	Block $B = 4$	Block $B = 8$
Linear	0.157 ± 0.072 [0.067, 0.246]	0.212 ± 0.216 [−0.056, 0.481]	0.115 ± 0.078 [0.018, 0.211]	0.154 ± 0.204 [−0.099, 0.407]
Median	1.872 ± 1.991 [−0.600, 4.345]	2.449 ± 3.861 [−2.345, 7.243]	0.135 ± 0.050 [0.073, 0.197]	2.214 ± 4.503 [−3.377, 7.805]
FFill + BFill	0.139 ± 0.121 [−0.011, 0.289]	0.179 ± 0.099 [0.056, 0.303]	0.137 ± 0.055 [0.068, 0.205]	0.170 ± 0.123 [0.017, 0.323]
Rolling mean	**0.080** ± 0.045 [0.023, 0.136]	0.270 ± 0.136 [0.101, 0.439]	0.181 ± 0.073 [0.090, 0.271]	0.163 ± 0.138 [−0.008, 0.334]
KNN (scaled)	0.250 ± 0.174 [0.033, 0.466]	0.290 ± 0.148 [0.106, 0.474]	**0.078** ± 0.038 [0.031, 0.126]	0.196 ± 0.150 [0.009, 0.382]
KNN (unscaled)	0.150 ± 0.112 [0.011, 0.290]	0.267 ± 0.087 [0.159, 0.374]	0.104 ± 0.038 [0.056, 0.151]	0.292 ± 0.342 [−0.132, 0.716]
Historical	0.418 ± 0.648 [−0.386, 1.223]	0.726 ± 0.376 [0.258, 1.193]	0.081 ± 0.027 [0.047, 0.114]	0.344 ± 0.364 [−0.108, 0.796]
Graph Laplacian	0.348 ± 0.282 [−0.002, 0.697]	0.618 ± 1.041 [−0.674, 1.910]	0.191 ± 0.089 [0.081, 0.302]	0.671 ± 1.075 [−0.663, 2.005]
Graph Temporal	0.191 ± 0.234 [−0.099, 0.481]	0.456 ± 0.776 [−0.508, 1.420]	0.179 ± 0.057 [0.108, 0.250]	0.562 ± 0.845 [−0.488, 1.611]
Sage-Voronoi	0.131 ± 0.100 [0.007, 0.256]	0.214 ± 0.101 [0.088, 0.340]	0.167 ± 0.068 [0.083, 0.251]	0.192 ± 0.168 [−0.017, 0.401]
GAIN	0.297 ± 0.257 [−0.022, 0.615]	0.181 ± 0.080 [0.081, 0.280]	0.218 ± 0.036 [0.174, 0.262]	0.254 ± 0.131 [0.092, 0.416]
GRIN	0.104 ± 0.132 [−0.060, 0.267]	**0.139** ± 0.073 [0.049, 0.229]	0.093 ± 0.018 [0.071, 0.116]	**0.128** ± 0.054 [0.062, 0.195]

Table S94: MRE for sensor A142, October–December 2022 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 16$	Block $B = 32$	Block $B = 64$
Linear	0.152 ± 0.052 [0.087, 0.217]	0.388 ± 0.488 [−0.218, 0.994]	0.368 ± 0.088 [0.258, 0.477]
Median	1.320 ± 1.554 [−0.609, 3.250]	1.335 ± 2.195 [−1.390, 4.061]	0.815 ± 0.576 [0.101, 1.530]
FFill + BFill	0.382 ± 0.232 [0.093, 0.670]	0.545 ± 0.576 [−0.170, 1.260]	0.588 ± 0.248 [0.280, 0.895]
Rolling mean	0.171 ± 0.057 [0.100, 0.241]	0.411 ± 0.501 [−0.211, 1.033]	0.351 ± 0.064 [0.271, 0.431]
KNN (scaled)	0.170 ± 0.065 [0.089, 0.250]	0.136 ± 0.049 [0.075, 0.196]	**0.154** ± 0.032 [0.114, 0.195]
KNN (unscaled)	0.168 ± 0.044 [0.114, 0.223]	0.177 ± 0.114 [0.035, 0.319]	0.186 ± 0.023 [0.157, 0.214]
Historical	0.458 ± 0.446 [−0.095, 1.012]	0.180 ± 0.132 [0.016, 0.345]	0.231 ± 0.076 [0.137, 0.325]
Graph Laplacian	0.285 ± 0.272 [−0.053, 0.623]	0.406 ± 0.487 [−0.199, 1.010]	0.275 ± 0.133 [0.110, 0.440]
Graph Temporal	0.257 ± 0.257 [−0.062, 0.576]	0.329 ± 0.390 [−0.155, 0.813]	0.235 ± 0.102 [0.109, 0.362]
Sage-Voronoi	0.214 ± 0.101 [0.089, 0.339]	0.226 ± 0.157 [0.032, 0.421]	0.241 ± 0.027 [0.208, 0.275]
GAIN	0.244 ± 0.133 [0.078, 0.410]	0.210 ± 0.070 [0.123, 0.297]	0.238 ± 0.076 [0.144, 0.333]
GRIN	**0.094** ± 0.025 [0.064, 0.125]	**0.107** ± 0.020 [0.082, 0.132]	0.155 ± 0.032 [0.115, 0.195]

Table S95: MRE for sensor A142, October–December 2022 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	Block $B = 128$	Block $B = 144$
Linear	0.920 ± 0.636 [0.131, 1.710]	1.005 ± 0.579 [0.286, 1.725]
Median	1.261 ± 0.534 [0.598, 1.923]	1.350 ± 0.266 [1.020, 1.681]
FFill + BFill	1.388 ± 0.936 [0.226, 2.550]	1.015 ± 0.581 [0.293, 1.737]
Rolling mean	0.921 ± 0.642 [0.124, 1.718]	1.005 ± 0.580 [0.285, 1.725]
KNN (scaled)	0.173 ± 0.040 [0.124, 0.223]	0.162 ± 0.041 [0.111, 0.214]
KNN (unscaled)	0.190 ± 0.039 [0.143, 0.238]	0.196 ± 0.038 [0.149, 0.243]
Historical	0.280 ± 0.200 [0.031, 0.528]	0.471 ± 0.343 [0.045, 0.897]
Graph Laplacian	0.364 ± 0.089 [0.254, 0.474]	0.371 ± 0.073 [0.281, 0.461]
Graph Temporal	0.320 ± 0.074 [0.228, 0.411]	0.314 ± 0.044 [0.259, 0.368]
Sage-Voronoi	0.267 ± 0.053 [0.201, 0.332]	0.260 ± 0.063 [0.182, 0.339]
GAIN	0.194 ± 0.062 [0.117, 0.271]	0.228 ± 0.029 [0.192, 0.265]
GRIN	**0.140** ± 0.034 [0.098, 0.183]	**0.137** ± 0.027 [0.103, 0.171]

Table S96: MRE for sensor A142, October–December 2022 (dimensionless ratio). Each cell gives mean ± sample SD, followed by the 95% confidence interval; $n = 5$ seed runs. Bold means identify the lowest unrounded mean within each column, not statistical significance.

Method	MCAR 10%	MCAR 30%	MCAR 50%	Synchronous $B = 144$
Linear	0.140 ± 0.001 [0.138, 0.141]	0.142 ± 0.007 [0.134, 0.151]	0.145 ± 0.006 [0.138, 0.153]	0.952 ± 0.610 [0.194, 1.710]
Median	1.213 ± 0.055 [1.144, 1.281]	1.204 ± 0.022 [1.177, 1.231]	1.216 ± 0.034 [1.174, 1.259]	1.174 ± 0.502 [0.551, 1.798]
FFill + BFill	0.166 ± 0.003 [0.162, 0.170]	0.175 ± 0.011 [0.161, 0.189]	0.187 ± 0.010 [0.175, 0.199]	0.996 ± 0.625 [0.220, 1.773]
Rolling mean	0.170 ± 0.006 [0.162, 0.178]	0.176 ± 0.007 [0.167, 0.185]	0.180 ± 0.007 [0.172, 0.188]	0.954 ± 0.611 [0.195, 1.713]
KNN (scaled)	0.166 ± 0.003 [0.162, 0.169]	0.158 ± 0.003 [0.154, 0.162]	0.162 ± 0.003 [0.158, 0.167]	0.998 ± 0.430 [0.465, 1.531]
KNN (unscaled)	0.191 ± 0.007 [0.182, 0.200]	0.186 ± 0.006 [0.179, 0.194]	0.182 ± 0.003 [0.178, 0.185]	0.998 ± 0.430 [0.465, 1.531]
Historical	0.340 ± 0.024 [0.310, 0.370]	0.337 ± 0.017 [0.315, 0.358]	0.335 ± 0.015 [0.317, 0.354]	**0.425** ± 0.267 [0.094, 0.756]
Graph Laplacian	0.344 ± 0.029 [0.308, 0.380]	0.347 ± 0.023 [0.318, 0.375]	0.360 ± 0.018 [0.338, 0.382]	0.998 ± 0.430 [0.465, 1.531]
Graph Temporal	0.246 ± 0.009 [0.235, 0.258]	0.254 ± 0.006 [0.247, 0.262]	0.271 ± 0.006 [0.263, 0.279]	0.968 ± 0.508 [0.337, 1.598]
Sage-Voronoi	0.151 ± 0.009 [0.140, 0.162]	0.161 ± 0.007 [0.152, 0.170]	0.178 ± 0.009 [0.167, 0.189]	0.816 ± 0.393 [0.328, 1.304]
GAIN	0.192 ± 0.019 [0.168, 0.215]	0.303 ± 0.043 [0.250, 0.356]	0.783 ± 0.075 [0.690, 0.877]	4.495 ± 1.381 [2.780, 6.209]
GRIN	**0.099** ± 0.004 [0.094, 0.103]	**0.101** ± 0.003 [0.097, 0.105]	**0.114** ± 0.004 [0.109, 0.119]	0.975 ± 0.424 [0.449, 1.501]

Additional analyses: supporting tables

Tables A1–A3 cover persistent stream loss; A4–A9 graph controls; A10–A12 common-k KNN scaling; A13–A24 count/degree strata for $B = 144$ and MCAR 30%; A25–A30 daily sensor-count sums; A31 plain Sage-Voronoi forecasting. Every metric cell gives mean, sample SD and an unadjusted run-level 95% Student-t CI from five seeds. MAE/RMSE use vehicles per 10-minute interval, except daily sums, which use aggregated sensor counts; MRE is a dimensionless ratio on nonzero targets. Full auxiliary summaries for all completed scenarios are in `additional_analyses_p03.csv`; per-seed training records and rounding differences are supplied separately. No method is omitted from the primary-model comparisons. Low/high count thresholds are per-sensor training quartiles; degree groups use the road graph. Daily sums score affected test days only and can include partial boundary days. Negative CI endpoints, if present, are retained as calculated and do not imply negative errors.

Table A1: MAE under persistent loss of sensor streams. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	Jan 25% streams	Jan 50% streams	Oct–Dec 25% streams	Oct–Dec 50% streams
Linear	114.98 ± 17.12 [93.72, 136.23]	115.77 ± 3.20 [111.79, 119.74]	107.72 ± 17.43 [86.07, 129.37]	111.34 ± 4.14 [106.20, 116.48]
Median	114.98 ± 17.12 [93.72, 136.23]	115.77 ± 3.20 [111.79, 119.74]	107.72 ± 17.43 [86.07, 129.37]	111.34 ± 4.14 [106.20, 116.48]
FFill + BFill	114.98 ± 17.12 [93.72, 136.23]	115.77 ± 3.20 [111.79, 119.74]	107.72 ± 17.43 [86.07, 129.37]	111.34 ± 4.14 [106.20, 116.48]
Rolling mean	114.98 ± 17.12 [93.72, 136.23]	115.77 ± 3.20 [111.79, 119.74]	107.72 ± 17.43 [86.07, 129.37]	111.34 ± 4.14 [106.20, 116.48]
KNN (scaled)	101.43 ± 41.09 [50.41, 152.46]	90.03 ± 36.07 [45.25, 134.82]	37.83 ± 17.57 [16.01, 59.65]	41.45 ± 3.47 [37.14, 45.76]
KNN (unscaled)	40.73 ± 17.45 [19.06, 62.40]	51.45 ± 2.04 [48.92, 53.99]	41.30 ± 16.31 [21.04, 61.56]	42.91 ± 2.44 [39.87, 45.94]
Historical	56.80 ± 18.22 [34.18, 79.43]	65.17 ± 5.06 [58.88, 71.45]	57.14 ± 17.00 [36.03, 78.25]	60.65 ± 3.75 [55.99, 65.31]
Graph Laplacian	67.03 ± 33.89 [24.95, 109.11]	78.51 ± 18.38 [55.68, 101.34]	44.54 ± 21.79 [17.48, 71.60]	51.53 ± 2.52 [48.40, 54.66]
Graph Temporal	83.67 ± 36.20 [38.72, 128.61]	80.69 ± 21.61 [53.85, 107.52]	46.23 ± 20.41 [20.89, 71.56]	52.23 ± 2.69 [48.88, 55.57]
Sage-Voronoi	60.22 ± 29.52 [23.57, 96.88]	74.12 ± 7.48 [64.83, 83.41]	48.27 ± 21.91 [21.06, 75.48]	58.79 ± 6.46 [50.77, 66.80]
GAIN	37.90 ± 16.02 [18.00, 57.80]	58.94 ± 3.49 [54.61, 63.27]	39.57 ± 17.19 [18.23, 60.91]	69.30 ± 2.84 [65.78, 72.82]
GRIN	40.24 ± 15.27 [21.28, 59.20]	51.42 ± 5.08 [45.12, 57.72]	36.23 ± 16.78 [15.39, 57.07]	44.14 ± 3.54 [39.74, 48.54]

Table A2: RMSE under persistent loss of sensor streams. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	Jan 25% streams	Jan 50% streams	Oct–Dec 25% streams	Oct–Dec 50% streams
Linear	192.51 ± 27.32 [158.59, 226.43]	220.44 ± 35.02 [176.96, 263.92]	191.38 ± 43.45 [137.43, 245.34]	208.06 ± 11.78 [193.44, 222.69]
Median	192.51 ± 27.32 [158.59, 226.43]	220.44 ± 35.02 [176.96, 263.92]	191.38 ± 43.45 [137.43, 245.34]	208.06 ± 11.78 [193.44, 222.69]
FFill + BFill	192.51 ± 27.32 [158.59, 226.43]	220.44 ± 35.02 [176.96, 263.92]	191.38 ± 43.45 [137.43, 245.34]	208.06 ± 11.78 [193.44, 222.69]
Rolling mean	192.51 ± 27.32 [158.59, 226.43]	220.44 ± 35.02 [176.96, 263.92]	191.38 ± 43.45 [137.43, 245.34]	208.06 ± 11.78 [193.44, 222.69]
KNN (scaled)	194.01 ± 62.69 [116.18, 271.85]	221.71 ± 32.26 [181.66, 261.76]	108.50 ± 64.91 [27.91, 189.09]	141.00 ± 10.53 [127.93, 154.07]
KNN (unscaled)	99.29 ± 55.13 [30.84, 167.74]	187.68 ± 48.36 [127.63, 247.73]	114.17 ± 63.94 [34.77, 193.57]	144.55 ± 10.99 [130.90, 158.20]
Historical	126.63 ± 52.72 [61.17, 192.09]	198.42 ± 44.64 [142.99, 253.84]	130.95 ± 48.10 [71.23, 190.67]	150.03 ± 12.53 [134.47, 165.59]
Graph Laplacian	152.73 ± 107.43 [19.34, 286.12]	260.95 ± 44.05 [206.26, 315.65]	118.44 ± 80.98 [17.89, 218.99]	172.44 ± 10.38 [159.55, 185.33]
Graph Temporal	208.31 ± 92.64 [93.28, 323.34]	278.77 ± 70.13 [191.70, 365.85]	121.89 ± 74.83 [28.98, 214.80]	171.18 ± 10.15 [158.57, 183.78]
Sage-Voronoi	135.77 ± 91.60 [22.03, 249.50]	230.23 ± 38.12 [182.90, 277.56]	126.15 ± 79.00 [28.06, 224.24]	181.16 ± 12.05 [166.19, 196.13]
GAIN	93.99 ± 53.08 [28.09, 159.90]	170.54 ± 48.88 [109.85, 231.23]	110.57 ± 69.27 [24.56, 196.59]	161.52 ± 5.80 [154.33, 168.72]
GRIN	103.82 ± 52.37 [38.79, 168.85]	177.87 ± 43.94 [123.31, 232.42]	108.96 ± 72.16 [19.36, 198.57]	147.91 ± 17.83 [125.77, 170.05]

Table A3: MRE under persistent loss of sensor streams. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	Jan 25% streams	Jan 50% streams	Oct–Dec 25% streams	Oct–Dec 50% streams
Linear	2.295 ± 0.119 [2.147, 2.443]	2.273 ± 0.031 [2.235, 2.311]	2.403 ± 0.056 [2.334, 2.472]	2.366 ± 0.074 [2.275, 2.458]
Median	2.295 ± 0.119 [2.147, 2.443]	2.273 ± 0.031 [2.235, 2.311]	2.403 ± 0.056 [2.334, 2.472]	2.366 ± 0.074 [2.275, 2.458]
FFill + BFill	2.295 ± 0.119 [2.147, 2.443]	2.273 ± 0.031 [2.235, 2.311]	2.403 ± 0.056 [2.334, 2.472]	2.366 ± 0.074 [2.275, 2.458]
Rolling mean	2.295 ± 0.119 [2.147, 2.443]	2.273 ± 0.031 [2.235, 2.311]	2.403 ± 0.056 [2.334, 2.472]	2.366 ± 0.074 [2.275, 2.458]
KNN (scaled)	1.167 ± 0.145 [0.988, 1.347]	0.969 ± 0.582 [0.246, 1.691]	0.402 ± 0.112 [0.264, 0.541]	0.388 ± 0.091 [0.275, 0.502]
KNN (unscaled)	0.531 ± 0.078 [0.434, 0.627]	0.635 ± 0.303 [0.259, 1.012]	0.469 ± 0.101 [0.344, 0.594]	0.411 ± 0.099 [0.287, 0.534]
Historical	0.546 ± 0.057 [0.476, 0.617]	0.563 ± 0.018 [0.541, 0.585]	0.690 ± 0.118 [0.543, 0.836]	0.681 ± 0.097 [0.560, 0.801]
Graph Laplacian	1.038 ± 0.234 [0.747, 1.329]	0.902 ± 0.401 [0.404, 1.399]	0.518 ± 0.056 [0.449, 0.588]	0.513 ± 0.049 [0.452, 0.573]
Graph Temporal	1.204 ± 0.417 [0.686, 1.722]	1.061 ± 0.632 [0.276, 1.846]	0.562 ± 0.082 [0.460, 0.663]	0.549 ± 0.052 [0.484, 0.613]
Sage-Voronoi	0.908 ± 0.277 [0.564, 1.252]	0.882 ± 0.319 [0.486, 1.278]	0.424 ± 0.052 [0.358, 0.489]	0.452 ± 0.035 [0.408, 0.496]
GAIN	0.508 ± 0.088 [0.399, 0.617]	0.965 ± 0.129 [0.804, 1.126]	0.443 ± 0.080 [0.343, 0.543]	1.066 ± 0.098 [0.944, 1.188]
GRIN	0.514 ± 0.103 [0.386, 0.641]	0.590 ± 0.103 [0.462, 0.719]	0.348 ± 0.050 [0.286, 0.411]	0.459 ± 0.068 [0.374, 0.544]

Table A4: Graph Temporal sensitivity controls, January 2023, MAE. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	$B = 16$	$B = 144$	MCAR 30%
Road, standardized	61.16 ± 1.87 [58.83, 63.48]	72.02 ± 4.78 [66.09, 77.96]	53.79 ± 0.45 [53.23, 54.34]
Road, raw counts	141.52 ± 9.21 [130.08, 152.96]	147.89 ± 4.16 [142.72, 153.05]	106.69 ± 0.49 [106.08, 107.29]
No spatial edges	34.77 ± 7.82 [25.06, 44.48]	133.05 ± 5.42 [126.32, 139.79]	23.16 ± 0.19 [22.93, 23.40]
Training correlation	38.89 ± 2.04 [36.36, 41.43]	42.24 ± 4.19 [37.04, 47.44]	36.44 ± 0.38 [35.96, 36.91]

Table A5: Graph Temporal sensitivity controls, January 2023, RMSE. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	$B = 16$	$B = 144$	MCAR 30%
Road, standardized	182.17 ± 20.91 [156.20, 208.14]	240.03 ± 26.62 [206.97, 273.08]	173.76 ± 1.50 [171.90, 175.62]
Road, raw counts	247.24 ± 31.93 [207.59, 286.89]	271.82 ± 19.22 [247.96, 295.67]	203.05 ± 1.56 [201.12, 204.99]
No spatial edges	117.12 ± 55.76 [47.88, 186.36]	263.30 ± 20.72 [237.57, 289.02]	77.64 ± 1.42 [75.87, 79.41]
Training correlation	127.12 ± 34.78 [83.93, 170.30]	161.02 ± 35.90 [116.44, 205.59]	142.71 ± 3.39 [138.49, 146.92]

Table A6: Graph Temporal sensitivity controls, January 2023, MRE. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	$B = 16$	$B = 144$	MCAR 30%
Road, standardized	0.879 ± 0.495 [0.265, 1.493]	1.050 ± 0.158 [0.853, 1.247]	0.750 ± 0.031 [0.711, 0.788]
Road, raw counts	6.589 ± 1.182 [5.121, 8.057]	7.450 ± 0.737 [6.535, 8.365]	4.953 ± 0.073 [4.863, 5.043]
No spatial edges	0.374 ± 0.060 [0.300, 0.448]	2.913 ± 0.286 [2.558, 3.268]	0.262 ± 0.003 [0.258, 0.266]
Training correlation	0.582 ± 0.256 [0.265, 0.900]	0.597 ± 0.064 [0.517, 0.676]	0.525 ± 0.014 [0.508, 0.543]

Table A7: Graph Temporal sensitivity controls, October–December 2022, MAE. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	$B = 16$	$B = 144$	MCAR 30%
Road, standardized	48.14 ± 5.03 [41.90, 54.38]	44.67 ± 3.18 [40.72, 48.62]	35.37 ± 0.16 [35.17, 35.57]
Road, raw counts	130.75 ± 8.45 [120.26, 141.24]	133.32 ± 6.49 [125.26, 141.38]	97.55 ± 0.16 [97.35, 97.76]
No spatial edges	29.12 ± 2.21 [26.38, 31.86]	115.76 ± 6.37 [107.85, 123.68]	20.58 ± 0.14 [20.41, 20.75]
Training correlation	41.82 ± 4.61 [36.09, 47.54]	38.54 ± 4.57 [32.87, 44.21]	30.75 ± 0.22 [30.48, 31.02]

Table A8: Graph Temporal sensitivity controls, October–December 2022, RMSE. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	$B = 16$	$B = 144$	MCAR 30%
Road, standardized	177.38 ± 55.64 [108.29, 246.47]	133.66 ± 21.55 [106.90, 160.42]	104.23 ± 1.62 [102.22, 106.23]
Road, raw counts	261.77 ± 39.41 [212.83, 310.71]	242.93 ± 21.96 [215.66, 270.19]	182.93 ± 1.26 [181.36, 184.50]
No spatial edges	86.28 ± 17.56 [64.48, 108.07]	208.23 ± 21.84 [181.11, 235.36]	60.51 ± 1.16 [59.06, 61.95]
Training correlation	155.65 ± 49.06 [94.73, 216.57]	124.18 ± 27.44 [90.11, 158.25]	84.63 ± 1.81 [82.38, 86.88]

Table A9: Graph Temporal sensitivity controls, October–December 2022, MRE. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	$B = 16$	$B = 144$	MCAR 30%
Road, standardized	0.552 ± 0.189 [0.317, 0.787]	0.620 ± 0.089 [0.509, 0.731]	0.457 ± 0.003 [0.454, 0.461]
Road, raw counts	4.886 ± 0.677 [4.044, 5.727]	6.021 ± 0.319 [5.625, 6.417]	4.240 ± 0.048 [4.180, 4.300]
No spatial edges	0.371 ± 0.052 [0.306, 0.436]	2.537 ± 0.109 [2.402, 2.673]	0.251 ± 0.001 [0.249, 0.252]
Training correlation	0.752 ± 0.730 [−0.154, 1.658]	0.585 ± 0.246 [0.280, 0.890]	0.414 ± 0.005 [0.407, 0.421]

Table A10: KNN scaling control at common $k = 10$, MAE. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	Jan $B = 144$	Jan MCAR 30%	Oct–Dec $B = 144$	Oct–Dec MCAR 30%
KNN, standardized	113.60 ± 5.43 [106.85, 120.34]	104.12 ± 1.46 [102.30, 105.93]	35.87 ± 1.81 [33.63, 38.11]	36.78 ± 0.19 [36.55, 37.02]
KNN, unscaled	41.52 ± 3.35 [37.37, 45.68]	41.89 ± 0.22 [41.61, 42.16]	38.87 ± 1.84 [36.58, 41.15]	39.13 ± 0.12 [38.98, 39.29]

Table A11: KNN scaling control at common $k = 10$, RMSE. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	Jan $B = 144$	Jan MCAR 30%	Oct–Dec $B = 144$	Oct–Dec MCAR 30%
KNN, standardized	237.08 ± 24.33 [206.87, 267.29]	227.19 ± 1.50 [225.33, 229.06]	112.74 ± 6.58 [104.57, 120.92]	123.63 ± 1.28 [122.05, 125.22]
KNN, unscaled	162.90 ± 31.08 [124.31, 201.49]	169.71 ± 2.82 [166.21, 173.22]	114.80 ± 6.69 [106.50, 123.11]	125.07 ± 1.30 [123.46, 126.69]

Table A12: KNN scaling control at common $k = 10$, MRE. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	Jan $B = 144$	Jan MCAR 30%	Oct–Dec $B = 144$	Oct–Dec MCAR 30%
KNN, standardized	1.051 ± 0.068 [0.966, 1.136]	1.045 ± 0.024 [1.015, 1.074]	0.491 ± 0.125 [0.335, 0.646]	0.457 ± 0.008 [0.447, 0.467]
KNN, unscaled	0.448 ± 0.034 [0.406, 0.491]	0.451 ± 0.012 [0.437, 0.466]	0.582 ± 0.141 [0.407, 0.757]	0.512 ± 0.009 [0.500, 0.524]

Table A13: January 2023, $B = 144$, MAE by count and road-graph degree strata. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	Low count	High count	Degree 0–2	Degree ≥ 3
Linear	129.15 ± 12.72 [113.36, 144.94]	134.22 ± 13.09 [117.97, 150.47]	87.87 ± 10.17 [75.24, 100.51]	162.12 ± 5.22 [155.64, 168.61]
Median	108.45 ± 2.55 [105.29, 111.62]	154.48 ± 10.02 [142.04, 166.92]	75.89 ± 2.79 [72.42, 79.35]	136.81 ± 6.77 [128.40, 145.21]
FFill + BFill	128.71 ± 11.64 [114.26, 143.17]	144.35 ± 10.77 [130.98, 157.73]	88.10 ± 8.27 [77.84, 98.37]	173.53 ± 10.92 [159.98, 187.08]
Rolling mean	129.13 ± 12.61 [113.47, 144.78]	134.33 ± 13.03 [118.15, 150.52]	87.71 ± 10.25 [74.99, 100.44]	162.42 ± 5.31 [155.82, 169.01]
KNN (scaled)	39.93 ± 3.90 [35.09, 44.77]	189.84 ± 11.67 [175.36, 204.33]	74.00 ± 4.83 [68.01, 80.00]	137.19 ± 7.11 [128.36, 146.02]
KNN (unscaled)	28.31 ± 3.85 [23.53, 33.09]	52.90 ± 8.39 [42.48, 63.32]	33.89 ± 3.01 [30.16, 37.63]	47.95 ± 5.44 [41.19, 54.72]
Historical	33.57 ± 3.67 [29.01, 38.12]	83.06 ± 9.86 [70.82, 95.31]	45.56 ± 2.76 [42.13, 48.99]	67.43 ± 6.11 [59.84, 75.02]
Graph Laplacian	54.09 ± 3.01 [50.35, 57.83]	66.93 ± 9.68 [54.91, 78.95]	64.72 ± 5.57 [57.81, 71.63]	67.45 ± 5.30 [60.87, 74.04]
Graph Temporal	59.63 ± 6.45 [51.61, 67.64]	77.39 ± 8.52 [66.81, 87.98]	64.13 ± 3.85 [59.35, 68.91]	76.73 ± 5.54 [69.85, 83.60]
Sage-Voronoi	48.83 ± 7.43 [39.61, 58.06]	58.60 ± 8.71 [47.79, 69.41]	55.63 ± 4.35 [50.23, 61.04]	62.91 ± 3.86 [58.12, 67.70]
GAIN	18.12 ± 1.66 [16.07, 20.18]	58.97 ± 10.74 [45.64, 72.31]	29.65 ± 2.64 [26.36, 32.93]	46.72 ± 6.49 [38.66, 54.79]
GRIN	22.89 ± 2.68 [19.57, 26.22]	56.50 ± 9.24 [45.03, 67.96]	29.01 ± 2.10 [26.40, 31.62]	46.99 ± 6.66 [38.71, 55.26]

Table A14: January 2023, $B = 144$, RMSE by count and road-graph degree strata. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	Low count	High count	Degree 0–2	Degree ≥ 3
Linear	249.86 ± 31.56 [210.67, 289.04]	299.75 ± 35.86 [255.22, 344.28]	197.58 ± 57.78 [125.84, 269.32]	297.74 ± 22.91 [269.30, 326.18]
Median	167.53 ± 7.38 [158.37, 176.70]	301.48 ± 42.66 [248.52, 354.45]	164.78 ± 17.99 [142.45, 187.11]	244.87 ± 36.05 [200.10, 289.63]
FFill + BFill	254.95 ± 38.45 [207.22, 302.69]	333.95 ± 51.05 [270.55, 397.34]	193.82 ± 45.53 [137.29, 250.36]	342.28 ± 52.07 [277.62, 406.93]
Rolling mean	249.84 ± 31.53 [210.70, 288.99]	300.43 ± 35.98 [255.76, 345.10]	197.03 ± 57.72 [125.35, 268.70]	298.69 ± 23.53 [269.47, 327.91]
KNN (scaled)	99.08 ± 11.03 [85.39, 112.77]	345.68 ± 36.60 [300.23, 391.12]	174.19 ± 21.57 [147.41, 200.98]	266.91 ± 33.74 [225.03, 308.80]
KNN (unscaled)	112.27 ± 21.52 [85.55, 138.98]	223.36 ± 53.90 [156.44, 290.28]	131.53 ± 18.77 [108.23, 154.84]	178.47 ± 46.19 [121.12, 235.82]
Historical	127.16 ± 13.92 [109.88, 144.44]	242.32 ± 52.00 [177.75, 306.88]	158.46 ± 17.73 [136.45, 180.48]	189.71 ± 45.65 [133.03, 246.39]
Graph Laplacian	189.36 ± 17.88 [167.16, 211.57]	233.80 ± 54.26 [166.43, 301.17]	252.93 ± 40.08 [203.16, 302.70]	191.85 ± 41.59 [140.21, 243.49]
Graph Temporal	212.06 ± 22.79 [183.76, 240.35]	264.79 ± 45.74 [208.00, 321.59]	254.63 ± 20.27 [229.47, 279.80]	229.84 ± 38.58 [181.94, 277.74]
Sage-Voronoi	165.69 ± 15.07 [146.97, 184.41]	223.55 ± 55.30 [154.89, 292.21]	215.71 ± 32.91 [174.85, 256.57]	188.25 ± 36.72 [142.65, 233.85]
GAIN	62.39 ± 10.33 [49.56, 75.22]	239.55 ± 55.44 [170.71, 308.39]	125.00 ± 21.02 [98.90, 151.10]	179.85 ± 46.47 [122.14, 237.55]
GRIN	80.64 ± 15.56 [61.32, 99.96]	232.63 ± 54.66 [164.75, 300.50]	117.38 ± 17.03 [96.24, 138.52]	180.09 ± 47.77 [120.79, 239.40]

Table A15: January 2023, $B = 144$, MRE by count and road-graph degree strata. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	Low count	High count	Degree 0–2	Degree ≥ 3
Linear	10.958 ± 1.427 [9.186, 12.731]	0.512 ± 0.034 [0.469, 0.554]	3.224 ± 0.447 [2.668, 3.779]	2.774 ± 0.251 [2.462, 3.085]
Median	8.649 ± 0.343 [8.223, 9.076]	0.567 ± 0.005 [0.561, 0.574]	2.431 ± 0.079 [2.333, 2.530]	2.178 ± 0.071 [2.090, 2.267]
FFill + BFill	10.669 ± 1.203 [9.176, 12.163]	0.551 ± 0.046 [0.493, 0.608]	3.182 ± 0.383 [2.706, 3.658]	2.769 ± 0.185 [2.539, 2.999]
Rolling mean	10.952 ± 1.424 [9.184, 12.720]	0.512 ± 0.034 [0.470, 0.554]	3.221 ± 0.446 [2.666, 3.775]	2.774 ± 0.249 [2.464, 3.084]
KNN (scaled)	2.891 ± 0.302 [2.516, 3.265]	0.660 ± 0.013 [0.644, 0.676]	1.164 ± 0.078 [1.067, 1.262]	0.987 ± 0.068 [0.902, 1.071]
KNN (unscaled)	1.187 ± 0.173 [0.972, 1.402]	0.216 ± 0.005 [0.211, 0.222]	0.602 ± 0.086 [0.495, 0.709]	0.376 ± 0.020 [0.351, 0.402]
Historical	1.452 ± 0.092 [1.338, 1.567]	0.288 ± 0.005 [0.282, 0.293]	0.672 ± 0.048 [0.613, 0.732]	0.484 ± 0.021 [0.459, 0.510]
Graph Laplacian	2.874 ± 0.292 [2.511, 3.237]	0.258 ± 0.008 [0.249, 0.268]	1.265 ± 0.132 [1.101, 1.430]	0.695 ± 0.027 [0.661, 0.728]
Graph Temporal	3.406 ± 0.608 [2.651, 4.160]	0.313 ± 0.009 [0.302, 0.325]	1.661 ± 0.336 [1.244, 2.078]	0.706 ± 0.066 [0.624, 0.789]
Sage-Voronoi	2.420 ± 0.587 [1.691, 3.148]	0.232 ± 0.020 [0.207, 0.256]	1.053 ± 0.178 [0.833, 1.274]	0.651 ± 0.118 [0.505, 0.797]
GAIN	0.844 ± 0.170 [0.633, 1.055]	0.218 ± 0.008 [0.208, 0.227]	0.470 ± 0.076 [0.375, 0.565]	0.293 ± 0.025 [0.262, 0.324]
GRIN	1.219 ± 0.133 [1.053, 1.384]	0.217 ± 0.014 [0.200, 0.235]	0.624 ± 0.071 [0.537, 0.712]	0.337 ± 0.031 [0.298, 0.376]

Table A16: January 2023, MCAR 30%, MAE by count and road-graph degree strata. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	Low count	High count	Degree 0–2	Degree ≥ 3
Linear	9.19 ± 0.28 [8.85, 9.54]	31.67 ± 0.42 [31.15, 32.19]	17.40 ± 0.36 [16.94, 17.85]	27.07 ± 0.16 [26.88, 27.27]
Median	109.30 ± 1.35 [107.62, 110.97]	150.21 ± 0.65 [149.40, 151.02]	72.42 ± 0.37 [71.96, 72.87]	135.85 ± 0.97 [134.65, 137.06]
FFill + BFill	11.46 ± 0.55 [10.77, 12.15]	39.06 ± 0.84 [38.01, 40.11]	21.86 ± 0.43 [21.33, 22.39]	34.51 ± 0.34 [34.09, 34.93]
Rolling mean	12.51 ± 0.40 [12.01, 13.01]	38.77 ± 0.59 [38.03, 39.50]	22.13 ± 0.50 [21.51, 22.75]	36.04 ± 0.21 [35.78, 36.29]
KNN (scaled)	43.70 ± 1.26 [42.13, 45.26]	166.96 ± 2.70 [163.62, 170.31]	67.10 ± 0.96 [65.91, 68.29]	126.25 ± 2.09 [123.65, 128.85]
KNN (unscaled)	30.87 ± 1.09 [29.51, 32.23]	51.44 ± 0.75 [50.51, 52.38]	33.39 ± 0.78 [32.42, 34.36]	48.95 ± 0.46 [48.38, 49.52]
Historical	35.37 ± 1.03 [34.09, 36.65]	80.61 ± 0.51 [79.97, 81.25]	44.53 ± 0.84 [43.49, 45.58]	68.21 ± 0.58 [67.49, 68.92]
Graph Laplacian	52.49 ± 1.75 [50.31, 54.66]	65.34 ± 1.05 [64.03, 66.64]	63.00 ± 0.66 [62.17, 63.82]	68.46 ± 0.61 [67.71, 69.22]
Graph Temporal	42.49 ± 1.01 [41.24, 43.75]	58.77 ± 1.32 [57.13, 60.41]	39.67 ± 0.71 [38.79, 40.56]	62.23 ± 0.90 [61.11, 63.34]
Sage-Voronoi	15.23 ± 0.95 [14.05, 16.42]	38.64 ± 1.18 [37.17, 40.11]	25.07 ± 1.13 [23.68, 26.47]	34.67 ± 0.59 [33.93, 35.40]
GAIN	27.79 ± 1.39 [26.07, 29.52]	56.98 ± 1.40 [55.23, 58.72]	29.14 ± 0.91 [28.01, 30.27]	50.78 ± 1.11 [49.39, 52.16]
GRIN	9.21 ± 0.51 [8.58, 9.85]	35.68 ± 1.10 [34.31, 37.05]	15.98 ± 0.46 [15.41, 16.54]	28.49 ± 0.48 [27.89, 29.08]

Table A17: January 2023, MCAR 30%, RMSE by count and road-graph degree strata. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	Low count	High count	Degree 0–2	Degree ≥ 3
Linear	38.99 ± 2.07 [36.43, 41.56]	107.26 ± 1.94 [104.85, 109.68]	69.40 ± 3.34 [65.25, 73.54]	82.68 ± 0.59 [81.95, 83.41]
Median	171.10 ± 2.49 [168.01, 174.20]	298.90 ± 3.56 [294.48, 303.32]	145.43 ± 3.96 [140.52, 150.35]	250.39 ± 2.72 [247.02, 253.77]
FFill + BFill	56.98 ± 4.08 [51.92, 62.04]	133.74 ± 6.35 [125.86, 141.62]	89.43 ± 4.27 [84.13, 94.73]	107.34 ± 4.47 [101.79, 112.89]
Rolling mean	59.62 ± 3.18 [55.67, 63.57]	135.08 ± 5.60 [128.13, 142.03]	88.82 ± 6.35 [80.93, 96.70]	109.88 ± 2.71 [106.52, 113.24]
KNN (scaled)	112.19 ± 2.57 [109.00, 115.39]	327.87 ± 2.83 [324.36, 331.39]	155.48 ± 3.48 [151.16, 159.81]	260.79 ± 3.17 [256.86, 264.73]
KNN (unscaled)	123.26 ± 3.50 [118.91, 127.61]	233.18 ± 5.58 [226.25, 240.11]	129.22 ± 5.19 [122.76, 135.67]	191.52 ± 3.70 [186.92, 196.12]
Historical	135.18 ± 4.40 [129.71, 140.64]	248.44 ± 4.93 [242.32, 254.56]	151.15 ± 5.58 [144.22, 158.08]	202.19 ± 3.58 [197.76, 206.63]
Graph Laplacian	177.73 ± 8.62 [167.02, 188.43]	241.53 ± 4.99 [235.34, 247.72]	250.30 ± 3.62 [245.81, 254.79]	204.18 ± 3.62 [199.69, 208.67]
Graph Temporal	137.88 ± 3.25 [133.85, 141.91]	210.72 ± 4.29 [205.39, 216.05]	142.61 ± 4.21 [137.38, 147.83]	189.93 ± 3.26 [185.88, 193.98]
Sage-Voronoi	60.31 ± 5.24 [53.81, 66.82]	157.20 ± 14.03 [139.77, 174.62]	96.36 ± 5.04 [90.10, 102.62]	125.27 ± 9.83 [113.07, 137.48]
GAIN	82.17 ± 3.76 [77.51, 86.84]	238.49 ± 5.62 [231.51, 245.47]	101.60 ± 4.55 [95.95, 107.26]	190.63 ± 4.21 [185.41, 195.85]
GRIN	32.39 ± 4.67 [26.59, 38.20]	160.30 ± 12.17 [145.18, 175.41]	66.49 ± 3.98 [61.55, 71.43]	124.34 ± 8.79 [113.43, 135.25]

Table A18: January 2023, MCAR 30%, MRE by count and road-graph degree strata. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	Low count	High count	Degree 0–2	Degree ≥ 3
Linear	0.552 ± 0.009 [0.541, 0.562]	0.178 ± 0.002 [0.175, 0.181]	0.308 ± 0.004 [0.303, 0.313]	0.242 ± 0.005 [0.235, 0.248]
Median	8.810 ± 0.107 [8.677, 8.943]	0.563 ± 0.001 [0.562, 0.565]	2.341 ± 0.045 [2.286, 2.397]	2.228 ± 0.016 [2.207, 2.248]
FFill + BFill	0.647 ± 0.017 [0.626, 0.668]	0.213 ± 0.004 [0.209, 0.217]	0.362 ± 0.003 [0.357, 0.366]	0.294 ± 0.007 [0.285, 0.303]
Rolling mean	0.655 ± 0.013 [0.638, 0.671]	0.194 ± 0.002 [0.191, 0.197]	0.343 ± 0.002 [0.340, 0.346]	0.301 ± 0.006 [0.293, 0.308]
KNN (scaled)	3.049 ± 0.088 [2.940, 3.158]	0.596 ± 0.008 [0.586, 0.606]	1.137 ± 0.036 [1.091, 1.182]	0.993 ± 0.020 [0.967, 1.018]
KNN (unscaled)	1.251 ± 0.064 [1.172, 1.330]	0.211 ± 0.002 [0.208, 0.214]	0.583 ± 0.023 [0.554, 0.612]	0.404 ± 0.012 [0.390, 0.419]
Historical	1.414 ± 0.025 [1.383, 1.445]	0.284 ± 0.001 [0.283, 0.285]	0.625 ± 0.011 [0.611, 0.639]	0.490 ± 0.005 [0.484, 0.496]
Graph Laplacian	2.813 ± 0.081 [2.712, 2.914]	0.256 ± 0.000 [0.255, 0.256]	1.181 ± 0.045 [1.126, 1.237]	0.717 ± 0.007 [0.709, 0.725]
Graph Temporal	2.356 ± 0.153 [2.165, 2.546]	0.248 ± 0.002 [0.246, 0.251]	1.074 ± 0.097 [0.954, 1.194]	0.567 ± 0.014 [0.549, 0.585]
Sage-Voronoi	0.698 ± 0.054 [0.631, 0.765]	0.164 ± 0.003 [0.161, 0.167]	0.363 ± 0.021 [0.336, 0.389]	0.274 ± 0.007 [0.266, 0.282]
GAIN	1.558 ± 0.160 [1.359, 1.757]	0.218 ± 0.002 [0.215, 0.220]	0.642 ± 0.049 [0.582, 0.703]	0.463 ± 0.035 [0.420, 0.506]
GRIN	0.552 ± 0.037 [0.506, 0.598]	0.152 ± 0.006 [0.145, 0.159]	0.312 ± 0.019 [0.288, 0.336]	0.203 ± 0.004 [0.198, 0.208]

Table A19: October–December 2022, $B = 144$, MAE by count and road-graph degree strata. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	Low count	High count	Degree 0–2	Degree ≥ 3
Linear	105.65 ± 9.76 [93.53, 117.77]	126.14 ± 9.67 [114.14, 138.14]	82.83 ± 12.29 [67.57, 98.08]	137.36 ± 4.42 [131.87, 142.85]
Median	113.24 ± 1.21 [111.75, 114.74]	150.25 ± 11.22 [136.31, 164.19]	74.32 ± 9.13 [62.99, 85.65]	119.92 ± 1.19 [118.45, 121.40]
FFill + BFill	106.13 ± 9.29 [94.60, 117.67]	130.70 ± 14.74 [112.40, 149.00]	83.08 ± 12.26 [67.86, 98.31]	140.07 ± 6.20 [132.38, 147.77]
Rolling mean	105.69 ± 9.74 [93.60, 117.78]	126.09 ± 9.63 [114.13, 138.06]	82.77 ± 12.28 [67.53, 98.02]	137.41 ± 4.45 [131.88, 142.94]
KNN (scaled)	18.29 ± 2.26 [15.48, 21.10]	63.58 ± 7.54 [54.22, 72.95]	30.59 ± 4.99 [24.40, 36.78]	38.99 ± 0.95 [37.82, 40.17]
KNN (unscaled)	20.35 ± 2.87 [16.78, 23.91]	64.38 ± 7.50 [55.07, 73.69]	34.28 ± 4.65 [28.51, 40.05]	43.11 ± 2.02 [40.60, 45.62]
Historical	32.08 ± 1.79 [29.86, 34.30]	74.55 ± 10.73 [61.22, 87.88]	39.88 ± 6.52 [31.78, 47.97]	59.64 ± 2.27 [56.82, 62.45]
Graph Laplacian	20.20 ± 1.72 [18.06, 22.33]	71.75 ± 11.99 [56.86, 86.65]	39.81 ± 8.28 [29.53, 50.08]	46.59 ± 0.55 [45.91, 47.27]
Graph Temporal	19.27 ± 2.03 [16.75, 21.78]	72.76 ± 11.50 [58.48, 87.04]	40.55 ± 9.26 [29.05, 52.04]	47.10 ± 1.18 [45.64, 48.57]
Sage-Voronoi	15.20 ± 2.02 [12.70, 17.70]	65.63 ± 9.92 [53.31, 77.95]	39.58 ± 6.36 [31.68, 47.49]	44.42 ± 2.33 [41.52, 47.31]
GAIN	13.55 ± 4.89 [7.47, 19.62]	76.18 ± 11.78 [61.55, 90.81]	34.49 ± 9.89 [22.22, 46.77]	42.25 ± 1.02 [40.99, 43.52]
GRIN	9.59 ± 1.62 [7.57, 11.60]	62.63 ± 11.54 [48.31, 76.96]	28.80 ± 8.29 [18.51, 39.09]	33.10 ± 0.97 [31.90, 34.30]

Table A20: October–December 2022, $B = 144$, RMSE by count and road-graph degree strata. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	Low count	High count	Degree 0–2	Degree ≥ 3
Linear	201.37 ± 19.72 [176.88, 225.86]	247.61 ± 26.46 [214.75, 280.46]	177.53 ± 51.59 [113.47, 241.58]	227.74 ± 9.88 [215.47, 240.01]
Median	172.09 ± 2.29 [169.25, 174.94]	265.55 ± 39.38 [216.65, 314.44]	166.32 ± 45.07 [110.36, 222.29]	183.57 ± 2.43 [180.56, 186.58]
FFill + BFill	200.29 ± 20.43 [174.92, 225.65]	252.74 ± 41.51 [201.20, 304.28]	175.55 ± 51.39 [111.74, 239.36]	230.87 ± 12.86 [214.91, 246.84]
Rolling mean	201.31 ± 19.57 [177.01, 225.61]	247.57 ± 26.45 [214.73, 280.41]	177.34 ± 51.48 [113.43, 241.26]	227.76 ± 9.80 [215.60, 239.93]
KNN (scaled)	78.86 ± 9.73 [66.78, 90.95]	180.42 ± 25.00 [149.38, 211.47]	117.55 ± 17.43 [95.90, 139.19]	109.13 ± 8.66 [98.38, 119.88]
KNN (unscaled)	90.05 ± 15.64 [70.64, 109.47]	181.02 ± 26.38 [148.27, 213.77]	124.32 ± 17.05 [103.14, 145.49]	115.99 ± 10.58 [102.86, 129.12]
Historical	91.73 ± 9.89 [79.45, 104.01]	177.91 ± 34.10 [135.56, 220.25]	120.76 ± 27.52 [86.59, 154.94]	121.05 ± 6.98 [112.38, 129.72]
Graph Laplacian	43.37 ± 5.44 [36.62, 50.12]	218.53 ± 54.97 [150.28, 286.78]	161.35 ± 52.35 [96.34, 226.36]	112.57 ± 8.94 [101.48, 123.67]
Graph Temporal	45.16 ± 8.38 [34.75, 55.57]	216.62 ± 52.66 [151.23, 282.00]	159.28 ± 50.83 [96.16, 222.40]	112.78 ± 10.46 [99.80, 125.77]
Sage-Voronoi	46.08 ± 12.17 [30.97, 61.18]	201.44 ± 47.25 [142.77, 260.11]	157.53 ± 41.84 [105.57, 209.48]	106.81 ± 10.25 [94.09, 119.54]
GAIN	42.53 ± 29.89 [5.42, 79.65]	222.35 ± 44.50 [167.09, 277.61]	135.57 ± 58.92 [62.41, 208.74]	113.95 ± 5.36 [107.29, 120.61]
GRIN	28.00 ± 5.07 [21.71, 34.30]	186.91 ± 40.22 [136.96, 236.85]	118.72 ± 45.80 [61.85, 175.59]	90.23 ± 11.46 [76.01, 104.46]

Table A21: October–December 2022, $B = 144$, MRE by count and road-graph degree strata. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	Low count	High count	Degree 0–2	Degree ≥ 3
Linear	8.470 ± 0.543 [7.796, 9.144]	0.455 ± 0.021 [0.429, 0.481]	2.634 ± 0.336 [2.217, 3.051]	2.508 ± 0.122 [2.356, 2.659]
Median	8.770 ± 0.301 [8.397, 9.143]	0.513 ± 0.007 [0.505, 0.521]	2.428 ± 0.100 [2.304, 2.551]	2.460 ± 0.049 [2.400, 2.521]
FFill + BFill	8.232 ± 0.532 [7.571, 8.893]	0.467 ± 0.023 [0.438, 0.496]	2.557 ± 0.309 [2.173, 2.941]	2.467 ± 0.141 [2.291, 2.642]
Rolling mean	8.463 ± 0.543 [7.789, 9.138]	0.455 ± 0.021 [0.429, 0.481]	2.631 ± 0.338 [2.212, 3.050]	2.507 ± 0.122 [2.356, 2.658]
KNN (scaled)	1.391 ± 0.541 [0.719, 2.063]	0.197 ± 0.010 [0.184, 0.210]	0.542 ± 0.159 [0.345, 0.739]	0.463 ± 0.118 [0.316, 0.610]
KNN (unscaled)	1.677 ± 0.710 [0.794, 2.559]	0.217 ± 0.013 [0.201, 0.233]	0.619 ± 0.149 [0.434, 0.803]	0.571 ± 0.199 [0.323, 0.818]
Historical	2.071 ± 0.223 [1.794, 2.347]	0.228 ± 0.008 [0.218, 0.238]	0.735 ± 0.156 [0.542, 0.929]	0.756 ± 0.031 [0.717, 0.795]
Graph Laplacian	1.617 ± 0.278 [1.272, 1.962]	0.217 ± 0.013 [0.201, 0.233]	0.772 ± 0.199 [0.525, 1.019]	0.491 ± 0.009 [0.479, 0.502]
Graph Temporal	1.708 ± 0.355 [1.268, 2.149]	0.230 ± 0.010 [0.217, 0.242]	0.820 ± 0.269 [0.486, 1.154]	0.514 ± 0.013 [0.498, 0.529]
Sage-Voronoi	0.953 ± 0.139 [0.781, 1.125]	0.213 ± 0.012 [0.198, 0.229]	0.589 ± 0.129 [0.429, 0.749]	0.361 ± 0.025 [0.329, 0.392]
GAIN	1.128 ± 0.804 [0.129, 2.126]	0.229 ± 0.009 [0.218, 0.240]	0.508 ± 0.170 [0.297, 0.720]	0.405 ± 0.240 [0.106, 0.703]
GRIN	0.776 ± 0.219 [0.504, 1.049]	0.205 ± 0.018 [0.183, 0.227]	0.485 ± 0.167 [0.278, 0.692]	0.260 ± 0.009 [0.249, 0.272]

Table A22: October–December 2022, MCAR 30%, MAE by count and road-graph degree strata. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	Low count	High count	Degree 0–2	Degree ≥ 3
Linear	7.95 ± 0.15 [7.76, 8.14]	32.14 ± 0.40 [31.65, 32.63]	17.93 ± 0.13 [17.76, 18.10]	22.63 ± 0.19 [22.39, 22.87]
Median	113.98 ± 0.47 [113.40, 114.56]	149.75 ± 0.91 [148.63, 150.88]	71.92 ± 0.45 [71.37, 72.48]	119.65 ± 0.60 [118.91, 120.39]
FFill + BFill	9.58 ± 0.13 [9.42, 9.73]	39.06 ± 0.63 [38.27, 39.84]	22.01 ± 0.25 [21.70, 22.32]	28.14 ± 0.22 [27.87, 28.42]
Rolling mean	10.72 ± 0.21 [10.46, 10.98]	38.05 ± 0.60 [37.31, 38.79]	21.73 ± 0.18 [21.50, 21.96]	29.06 ± 0.25 [28.74, 29.37]
KNN (scaled)	18.56 ± 0.28 [18.21, 18.90]	66.98 ± 0.96 [65.79, 68.16]	30.15 ± 0.30 [29.77, 30.52]	40.71 ± 0.44 [40.16, 41.26]
KNN (unscaled)	20.01 ± 0.48 [19.41, 20.62]	68.11 ± 0.84 [67.07, 69.15]	33.14 ± 0.33 [32.73, 33.55]	44.04 ± 0.35 [43.61, 44.47]
Historical	32.73 ± 0.26 [32.41, 33.06]	76.66 ± 0.95 [75.48, 77.84]	40.27 ± 0.14 [40.10, 40.45]	63.00 ± 0.45 [62.44, 63.56]
Graph Laplacian	19.77 ± 0.42 [19.25, 20.29]	74.07 ± 0.90 [72.95, 75.18]	38.90 ± 0.46 [38.32, 39.47]	46.96 ± 0.36 [46.51, 47.41]
Graph Temporal	14.98 ± 0.10 [14.86, 15.10]	57.67 ± 0.89 [56.56, 58.77]	26.79 ± 0.20 [26.55, 27.03]	40.44 ± 0.35 [40.01, 40.87]
Sage-Voronoi	8.99 ± 0.26 [8.67, 9.31]	35.26 ± 0.68 [34.41, 36.10]	20.07 ± 0.27 [19.74, 20.40]	24.84 ± 0.20 [24.58, 25.09]
GAIN	20.97 ± 0.88 [19.88, 22.05]	71.29 ± 1.15 [69.86, 72.72]	33.43 ± 0.39 [32.95, 33.90]	46.40 ± 0.49 [45.79, 47.00]
GRIN	6.15 ± 0.08 [6.05, 6.25]	30.57 ± 0.45 [30.01, 31.14]	15.98 ± 0.08 [15.88, 16.08]	18.98 ± 0.15 [18.79, 19.16]

Table A23: October–December 2022, MCAR 30%, RMSE by count and road-graph degree strata. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	Low count	High count	Degree 0–2	Degree ≥ 3
Linear	34.25 ± 1.89 [31.91, 36.59]	93.76 ± 1.74 [91.60, 95.92]	68.46 ± 0.87 [67.38, 69.55]	56.02 ± 1.42 [54.26, 57.78]
Median	173.61 ± 0.75 [172.68, 174.55]	282.24 ± 2.06 [279.68, 284.80]	163.64 ± 2.28 [160.81, 166.47]	193.95 ± 2.15 [191.28, 196.62]
FFill + BFill	46.65 ± 2.00 [44.16, 49.13]	115.36 ± 3.12 [111.49, 119.23]	86.76 ± 2.12 [84.13, 89.39]	71.86 ± 1.65 [69.81, 73.91]
Rolling mean	52.34 ± 2.53 [49.20, 55.48]	114.38 ± 2.91 [110.77, 117.99]	83.97 ± 1.74 [81.81, 86.13]	74.43 ± 1.63 [72.41, 76.45]
KNN (scaled)	86.30 ± 1.87 [83.97, 88.62]	206.56 ± 3.51 [202.20, 210.92]	118.04 ± 1.91 [115.68, 120.41]	126.80 ± 2.97 [123.11, 130.48]
KNN (unscaled)	91.76 ± 3.08 [87.94, 95.58]	209.12 ± 2.78 [205.67, 212.57]	123.48 ± 2.30 [120.63, 126.33]	131.30 ± 2.63 [128.03, 134.57]
Historical	95.28 ± 1.03 [94.00, 96.56]	203.19 ± 3.35 [199.02, 207.35]	119.75 ± 1.28 [118.17, 121.34]	139.93 ± 2.54 [136.78, 143.09]
Graph Laplacian	41.23 ± 0.86 [40.16, 42.29]	240.15 ± 2.09 [237.55, 242.75]	164.32 ± 2.82 [160.82, 167.82]	126.07 ± 2.54 [122.92, 129.22]
Graph Temporal	34.95 ± 0.78 [33.98, 35.92]	173.04 ± 4.39 [167.59, 178.49]	100.09 ± 1.56 [98.16, 102.02]	106.58 ± 2.99 [102.86, 110.30]
Sage-Voronoi	37.12 ± 1.96 [34.69, 39.56]	114.52 ± 3.43 [110.27, 118.78]	79.57 ± 1.77 [77.37, 81.76]	65.59 ± 1.96 [63.15, 68.03]
GAIN	57.76 ± 3.85 [52.98, 62.54]	224.94 ± 3.36 [220.77, 229.12]	128.78 ± 3.24 [124.75, 132.81]	126.19 ± 2.43 [123.17, 129.21]
GRIN	27.84 ± 1.55 [25.92, 29.76]	97.91 ± 2.98 [94.21, 101.60]	68.01 ± 0.73 [67.10, 68.92]	51.53 ± 1.70 [49.43, 53.64]

Table A24: October–December 2022, MCAR 30%, MRE by count and road-graph degree strata. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	Low count	High count	Degree 0–2	Degree ≥ 3
Linear	0.511 ± 0.005 [0.505, 0.517]	0.169 ± 0.001 [0.167, 0.170]	0.302 ± 0.004 [0.297, 0.307]	0.229 ± 0.003 [0.225, 0.233]
Median	8.597 ± 0.036 [8.553, 8.642]	0.515 ± 0.001 [0.514, 0.516]	2.452 ± 0.002 [2.450, 2.455]	2.350 ± 0.020 [2.326, 2.374]
FFill + BFill	0.589 ± 0.007 [0.580, 0.597]	0.201 ± 0.003 [0.197, 0.204]	0.352 ± 0.005 [0.347, 0.358]	0.273 ± 0.004 [0.269, 0.278]
Rolling mean	0.592 ± 0.008 [0.582, 0.602]	0.179 ± 0.002 [0.177, 0.181]	0.329 ± 0.004 [0.324, 0.333]	0.276 ± 0.004 [0.271, 0.280]
KNN (scaled)	1.255 ± 0.033 [1.214, 1.295]	0.204 ± 0.001 [0.202, 0.205]	0.481 ± 0.010 [0.468, 0.493]	0.444 ± 0.015 [0.425, 0.463]
KNN (unscaled)	1.404 ± 0.058 [1.332, 1.476]	0.224 ± 0.002 [0.222, 0.226]	0.548 ± 0.012 [0.533, 0.563]	0.503 ± 0.021 [0.477, 0.529]
Historical	1.904 ± 0.019 [1.880, 1.928]	0.230 ± 0.001 [0.228, 0.231]	0.707 ± 0.008 [0.697, 0.717]	0.718 ± 0.010 [0.706, 0.730]
Graph Laplacian	1.469 ± 0.014 [1.451, 1.486]	0.223 ± 0.003 [0.219, 0.226]	0.674 ± 0.009 [0.664, 0.685]	0.481 ± 0.004 [0.476, 0.486]
Graph Temporal	1.185 ± 0.008 [1.175, 1.196]	0.194 ± 0.001 [0.193, 0.196]	0.516 ± 0.003 [0.513, 0.520]	0.425 ± 0.004 [0.421, 0.430]
Sage-Voronoi	0.518 ± 0.020 [0.494, 0.542]	0.146 ± 0.002 [0.144, 0.148]	0.292 ± 0.004 [0.287, 0.297]	0.235 ± 0.006 [0.228, 0.243]
GAIN	1.489 ± 0.073 [1.399, 1.579]	0.212 ± 0.001 [0.210, 0.214]	0.616 ± 0.017 [0.595, 0.637]	0.510 ± 0.024 [0.481, 0.539]
GRIN	0.443 ± 0.010 [0.430, 0.456]	0.142 ± 0.009 [0.131, 0.153]	0.272 ± 0.004 [0.267, 0.277]	0.179 ± 0.003 [0.176, 0.183]

Table A25: January 2023, MAE of reconstructed daily sensor-count sums on affected test days. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	$B = 144$	MCAR 30%	Synchronous
Linear	72410.75 ± 12109.05 [57375.38, 87446.11]	6150.56 ± 2243.87 [3364.43, 8936.69]	980293.78 ± 377426.58 [511656.79, 1448930.77]
Median	90035.00 ± 16381.39 [69694.82, 110375.18]	231525.20 ± 2903.38 [227920.17, 235130.23]	611806.00 ± 163725.07 [408514.47, 815097.53]
FFill + BFill	85656.88 ± 18957.11 [62118.52, 109195.24]	8610.08 ± 1657.37 [6552.18, 10667.98]	972934.90 ± 366138.17 [518314.31, 1427555.49]
Rolling mean	72274.58 ± 12233.86 [57084.23, 87464.93]	9196.60 ± 2676.48 [5873.31, 12519.90]	980797.96 ± 372104.43 [518769.29, 1442826.64]
KNN (scaled)	190782.77 ± 15735.16 [171244.99, 210320.55]	446441.30 ± 10826.67 [432998.21, 459884.38]	589644.89 ± 154911.41 [397296.97, 781992.81]
KNN (unscaled)	31607.26 ± 9718.70 [19539.90, 43674.62]	15783.40 ± 2244.63 [12996.33, 18570.47]	589644.89 ± 154911.41 [397296.97, 781992.81]
Historical	73200.83 ± 15691.08 [53717.78, 92683.88]	198564.57 ± 2480.62 [195484.47, 201644.66]	451834.57 ± 100964.69 [326470.35, 577198.80]
Graph Laplacian	85428.42 ± 8061.82 [75418.35, 95438.48]	212077.53 ± 5574.20 [205156.24, 218998.82]	589644.89 ± 154911.41 [397296.97, 781992.81]
Graph Temporal	80436.01 ± 15084.37 [61706.29, 99165.74]	122172.38 ± 2556.79 [118997.70, 125347.05]	1453677.99 ± 738136.06 [537161.02, 2370194.96]
Sage-Voronoi	54436.33 ± 22194.49 [26878.23, 81994.43]	16735.64 ± 5408.87 [10019.65, 23451.63]	945616.77 ± 491318.39 [335564.41, 1555669.12]
GAIN	39654.53 ± 7559.95 [30267.62, 49041.45]	44242.60 ± 7305.22 [35171.98, 53313.23]	962114.41 ± 465157.59 [384544.98, 1539683.84]
GRIN	31783.75 ± 6919.57 [23191.97, 40375.53]	42622.32 ± 7061.68 [33854.09, 51390.56]	839838.44 ± 321189.96 [441028.42, 1238648.45]

Table A26: January 2023, RMSE of reconstructed daily sensor-count sums on affected test days. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	$B = 144$	MCAR 30%	Synchronous
Linear	93735.94 ± 25103.74 [62565.54, 124906.35]	7565.38 ± 2288.36 [4724.00, 10406.75]	1060912.23 ± 344025.04 [633748.73, 1488075.73]
Median	109975.87 ± 22153.24 [82468.99, 137482.76]	252456.67 ± 3676.35 [247891.87, 257021.46]	691354.84 ± 182989.29 [464143.63, 918566.06]
FFill + BFill	114645.91 ± 39533.28 [65558.86, 163732.96]	10368.82 ± 1634.50 [8339.31, 12398.32]	1113379.52 ± 313875.24 [723651.93, 1503107.11]
Rolling mean	93662.62 ± 25340.16 [62198.66, 125126.59]	10891.94 ± 2575.31 [7694.27, 14089.61]	1062969.39 ± 336871.97 [644687.59, 1481251.19]
KNN (scaled)	228627.08 ± 20218.51 [203522.48, 253731.68]	485432.27 ± 10598.40 [472272.61, 498591.92]	654996.70 ± 160460.72 [455758.40, 854235.01]
KNN (unscaled)	45674.69 ± 15208.28 [26791.12, 64558.26]	20444.83 ± 3757.69 [15779.04, 25110.62]	654996.70 ± 160460.72 [455758.40, 854235.01]
Historical	90330.33 ± 22764.42 [62064.57, 118596.08]	216997.29 ± 2943.84 [213342.03, 220652.54]	517418.71 ± 166511.62 [310667.23, 724170.19]
Graph Laplacian	111593.30 ± 12648.95 [95887.56, 127299.05]	221781.34 ± 6144.90 [214151.44, 229411.24]	654996.70 ± 160460.72 [455758.40, 854235.01]
Graph Temporal	106584.35 ± 16174.92 [86500.54, 126668.17]	128068.48 ± 2290.52 [125224.42, 130912.54]	1519204.64 ± 727950.07 [615335.24, 2423074.04]
Sage-Voronoi	76745.87 ± 27492.24 [42609.74, 110881.99]	19582.00 ± 5656.03 [12559.11, 26604.89]	1077640.27 ± 516068.67 [436856.39, 1718424.16]
GAIN	56597.96 ± 11846.24 [41888.92, 71307.01]	51489.16 ± 8047.30 [41497.12, 61481.21]	1056563.77 ± 453209.88 [493829.38, 1619298.16]
GRIN	45683.62 ± 10595.07 [32528.10, 58839.14]	46143.35 ± 7370.48 [36991.70, 55295.01]	953917.01 ± 384726.11 [476216.45, 1431617.56]

Table A27: January 2023, MRE of reconstructed daily sensor-count sums on affected test days. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	$B = 144$	MCAR 30%	Synchronous
Linear	0.019 ± 0.004 [0.013, 0.024]	0.002 ± 0.001 [0.001, 0.003]	0.225 ± 0.076 [0.131, 0.319]
Median	0.022 ± 0.004 [0.017, 0.027]	0.058 ± 0.001 [0.057, 0.059]	0.140 ± 0.033 [0.098, 0.181]
FFill + BFill	0.022 ± 0.007 [0.014, 0.031]	0.003 ± 0.001 [0.002, 0.004]	0.236 ± 0.092 [0.121, 0.350]
Rolling mean	0.019 ± 0.004 [0.013, 0.024]	0.003 ± 0.001 [0.002, 0.005]	0.226 ± 0.075 [0.132, 0.319]
KNN (scaled)	0.044 ± 0.003 [0.040, 0.048]	0.108 ± 0.003 [0.104, 0.112]	0.137 ± 0.037 [0.090, 0.183]
KNN (unscaled)	0.008 ± 0.003 [0.004, 0.011]	0.004 ± 0.000 [0.004, 0.005]	0.137 ± 0.037 [0.090, 0.183]
Historical	0.018 ± 0.004 [0.013, 0.023]	0.055 ± 0.000 [0.054, 0.055]	0.115 ± 0.054 [0.047, 0.182]
Graph Laplacian	0.020 ± 0.002 [0.018, 0.023]	0.054 ± 0.001 [0.052, 0.055]	0.137 ± 0.037 [0.090, 0.183]
Graph Temporal	0.019 ± 0.004 [0.014, 0.024]	0.031 ± 0.001 [0.030, 0.032]	0.322 ± 0.153 [0.132, 0.511]
Sage-Voronoi	0.014 ± 0.006 [0.006, 0.022]	0.006 ± 0.002 [0.004, 0.008]	0.207 ± 0.109 [0.072, 0.343]
GAIN	0.010 ± 0.001 [0.008, 0.011]	0.011 ± 0.002 [0.009, 0.013]	0.244 ± 0.181 [0.020, 0.469]
GRIN	0.008 ± 0.001 [0.006, 0.009]	0.011 ± 0.002 [0.009, 0.013]	0.187 ± 0.063 [0.108, 0.265]

Table A28: October–December 2022, MAE of reconstructed daily sensor-count sums on affected test days. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	$B = 144$	MCAR 30%	Synchronous
Linear	31034.47 ± 8046.66 [21043.22, 41025.72]	4149.24 ± 662.27 [3326.93, 4971.56]	928936.29 ± 470488.66 [344747.46, 1513125.12]
Median	27838.31 ± 4548.63 [22190.44, 33486.17]	193375.25 ± 2431.43 [190356.23, 196394.26]	391730.10 ± 291623.22 [29632.05, 753828.15]
FFill + BFill	32613.12 ± 7468.78 [23339.40, 41886.84]	5690.29 ± 1070.17 [4361.50, 7019.08]	964772.10 ± 547238.63 [285285.59, 1644258.61]
Rolling mean	30995.58 ± 7975.92 [21092.16, 40898.99]	5960.51 ± 1020.42 [4693.48, 7227.53]	930489.38 ± 470877.39 [345817.88, 1515160.88]
KNN (scaled)	12934.76 ± 1929.70 [10538.72, 15330.80]	58304.09 ± 1738.37 [56145.61, 60462.56]	395205.05 ± 290778.10 [34156.36, 756253.74]
KNN (unscaled)	13671.03 ± 2226.58 [10906.36, 16435.70]	33660.05 ± 1392.31 [31931.26, 35388.83]	395205.05 ± 290778.10 [34156.36, 756253.74]
Historical	25138.25 ± 2588.35 [21924.38, 28352.12]	198701.36 ± 1566.27 [196756.58, 200646.14]	337485.08 ± 280791.67 [−11163.83, 686133.98]
Graph Laplacian	14703.42 ± 3090.71 [10865.79, 18541.05]	27298.32 ± 2212.11 [24551.62, 30045.02]	395205.05 ± 290778.10 [34156.36, 756253.74]
Graph Temporal	15291.87 ± 3150.52 [11379.98, 19203.76]	18969.44 ± 2183.87 [16257.82, 21681.07]	877365.88 ± 466034.82 [298707.22, 1456024.53]
Sage-Voronoi	14930.71 ± 2077.24 [12351.47, 17509.94]	11986.24 ± 2893.52 [8393.46, 15579.02]	1198269.84 ± 494366.23 [584433.08, 1812106.59]
GAIN	19546.61 ± 3036.68 [15776.07, 23317.14]	33139.96 ± 1489.54 [31290.46, 34989.47]	1993002.23 ± 395703.37 [1501671.59, 2484332.86]
GRIN	12752.65 ± 3329.79 [8618.17, 16887.14]	20186.61 ± 2476.21 [17111.99, 23261.23]	380694.76 ± 116215.08 [236394.68, 524994.84]

Table A29: October–December 2022, RMSE of reconstructed daily sensor-count sums on affected test days. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	$B = 144$	MCAR 30%	Synchronous
Linear	42371.99 ± 12266.90 [27140.61, 57603.36]	5220.02 ± 758.17 [4278.62, 6161.41]	1151177.89 ± 631814.50 [366676.57, 1935679.21]
Median	37182.80 ± 5998.52 [29734.65, 44630.94]	228507.00 ± 3142.43 [224605.15, 232408.84]	429435.01 ± 330486.13 [19082.28, 839787.74]
FFill + BFill	45373.38 ± 11657.47 [30898.72, 59848.04]	7169.11 ± 907.52 [6042.27, 8295.95]	1197030.00 ± 691932.87 [337881.87, 2056178.14]
Rolling mean	42324.26 ± 12206.92 [27167.37, 57481.16]	7366.62 ± 1233.95 [5834.46, 8898.77]	1152280.61 ± 630178.26 [369810.95, 1934750.27]
KNN (scaled)	23804.14 ± 2738.26 [20404.14, 27204.13]	70492.53 ± 2513.60 [67371.48, 73613.58]	432017.13 ± 329251.82 [23197.00, 840837.25]
KNN (unscaled)	24235.77 ± 3901.40 [19391.54, 29079.99]	44152.82 ± 3043.37 [40373.98, 47931.66]	432017.13 ± 329251.82 [23197.00, 840837.25]
Historical	32296.63 ± 3429.63 [28038.18, 36555.08]	231819.60 ± 1790.85 [229595.97, 234043.23]	379035.11 ± 273528.22 [39404.97, 718665.25]
Graph Laplacian	24835.53 ± 6201.29 [17135.62, 32535.45]	35485.53 ± 2436.84 [32459.79, 38511.27]	432017.13 ± 329251.82 [23197.00, 840837.25]
Graph Temporal	25373.72 ± 6073.65 [17832.28, 32915.16]	24406.78 ± 3891.38 [19575.00, 29238.57]	1097689.10 ± 612649.02 [336984.87, 1858393.34]
Sage-Voronoi	21867.12 ± 4852.48 [15841.98, 27892.27]	14572.64 ± 3140.81 [10672.81, 18472.47]	1514135.61 ± 687397.74 [660618.59, 2367652.63]
GAIN	31231.74 ± 5273.92 [24683.30, 37780.18]	44286.82 ± 2021.76 [41776.47, 46797.16]	2305153.39 ± 407108.01 [1799662.03, 2810644.75]
GRIN	21933.16 ± 5011.53 [15710.52, 28155.81]	22050.23 ± 2362.88 [19116.33, 24984.13]	442367.46 ± 144468.27 [262986.41, 621748.51]

Table A30: October–December 2022, MRE of reconstructed daily sensor-count sums on affected test days. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	$B = 144$	MCAR 30%	Synchronous
Linear	0.008 ± 0.002 [0.005, 0.010]	0.001 ± 0.000 [0.001, 0.001]	0.259 ± 0.117 [0.113, 0.404]
Median	0.008 ± 0.001 [0.006, 0.009]	0.057 ± 0.001 [0.057, 0.058]	0.124 ± 0.123 [−0.029, 0.277]
FFill + BFill	0.008 ± 0.002 [0.006, 0.011]	0.002 ± 0.000 [0.001, 0.002]	0.267 ± 0.135 [0.099, 0.435]
Rolling mean	0.008 ± 0.002 [0.005, 0.010]	0.002 ± 0.000 [0.001, 0.002]	0.259 ± 0.117 [0.113, 0.405]
KNN (scaled)	0.003 ± 0.000 [0.003, 0.004]	0.014 ± 0.000 [0.014, 0.015]	0.125 ± 0.124 [−0.028, 0.279]
KNN (unscaled)	0.004 ± 0.001 [0.003, 0.004]	0.008 ± 0.000 [0.008, 0.009]	0.125 ± 0.124 [−0.028, 0.279]
Historical	0.007 ± 0.001 [0.006, 0.008]	0.060 ± 0.000 [0.060, 0.061]	0.110 ± 0.128 [−0.048, 0.269]
Graph Laplacian	0.004 ± 0.001 [0.003, 0.005]	0.007 ± 0.001 [0.006, 0.008]	0.125 ± 0.124 [−0.028, 0.279]
Graph Temporal	0.004 ± 0.001 [0.003, 0.005]	0.005 ± 0.001 [0.004, 0.006]	0.245 ± 0.117 [0.100, 0.390]
Sage-Voronoi	0.004 ± 0.001 [0.003, 0.005]	0.003 ± 0.001 [0.002, 0.004]	0.320 ± 0.089 [0.209, 0.431]
GAIN	0.005 ± 0.001 [0.004, 0.006]	0.010 ± 0.000 [0.009, 0.010]	0.594 ± 0.283 [0.242, 0.945]
GRIN	0.003 ± 0.001 [0.002, 0.004]	0.005 ± 0.001 [0.004, 0.006]	0.110 ± 0.055 [0.041, 0.178]

Table A31: Plain one-step (10-minute) Sage-Voronoi forecasting with observed past inputs. MAE and RMSE are in vehicles per 10-minute interval; MRE is dimensionless. Cells give mean ± sample SD and [95% Student-t CI] across five seed runs.

Method / variant	MAE	RMSE	MRE
January 2023	28.24 ± 0.37 [27.78, 28.69]	105.57 ± 4.79 [99.63, 111.51]	0.288 ± 0.009 [0.277, 0.299]
October–December 2022	21.58 ± 0.19 [21.35, 21.81]	66.06 ± 0.47 [65.48, 66.65]	0.244 ± 0.004 [0.239, 0.250]